

(12) **United States Patent**
Meyers et al.

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(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **METHOD FOR GENERATING NFTS IN RESPONSE TO IN-APP ISSUANCE REQUESTS**

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(73) Assignee: **Galiant Arts, LLC**, Austin, TX (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/929,744**

(22) Filed: **Oct. 29, 2024**

(65) **Prior Publication Data**

US 2025/0050227 A1 Feb. 13, 2025

Related U.S. Application Data

(63) Continuation of application No. 18/886,252, filed on Sep. 16, 2024, which is a continuation of application (Continued)

(51) **Int. Cl.**

A63F 13/00 (2014.01)

A63F 13/35 (2014.01)

(Continued)

(52) **U.S. Cl.**

CPC **A63F 13/73** (2014.09); **A63F 13/35** (2014.09); **G06F 21/10** (2013.01); **H04L 9/50** (2022.05)

(58) **Field of Classification Search**

CPC .. H04L 9/50; A63F 13/69; A63F 13/71; A63F 13/77

See application file for complete search history.

(56)

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Primary Examiner — David L Lewis

Assistant Examiner — Eric M Thomas

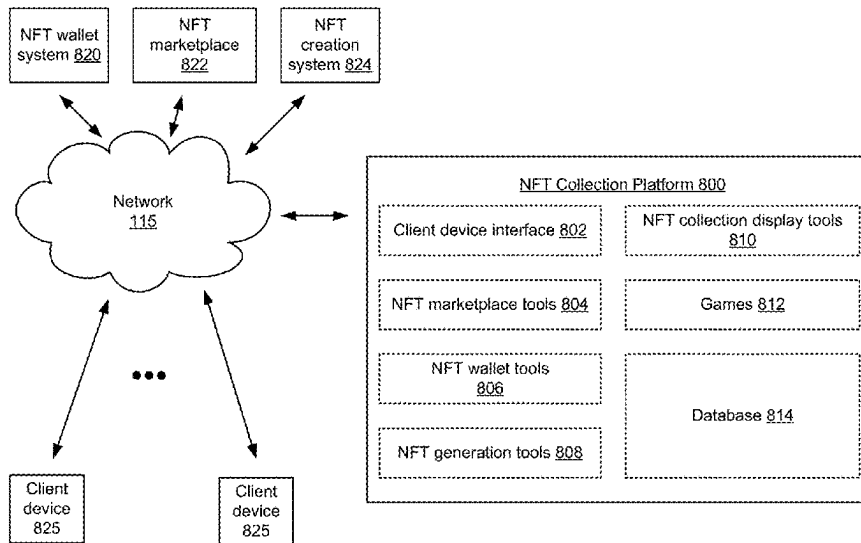
(74) *Attorney, Agent, or Firm* — Katz Ruby & Carle LLP; Bruce E. Stuckman

(57)

ABSTRACT

An NFT system operates by receiving an application launch request for a game application having a game non-fungible token (NFT) associated therewith; generating, via the game application and in response to the application launch request, launch metadata; sending the launch metadata to an NFT wallet, the NFT wallet containing a private key associated with the game NFT; receiving, from the NFT wallet, a launch response that is generated based on the private key; generating, via the game application and based on the launch response, an ownership check; sending the ownership check to a blockchain system that stores the game NFT; receiving, from the blockchain system, an ownership validation response; and enabling, via the game application, play of the game when the ownership validation response indicates ownership of the game NFT is verified.

20 Claims, 52 Drawing Sheets



Related U.S. Application Data

No. 17/661,362, filed on Apr. 29, 2022, now Pat. No. 12,121,820, which is a continuation-in-part of application No. 17/656,084, filed on Mar. 23, 2022, now Pat. No. 11,786,820.

- (60) Provisional application No. 63/362,581, filed on Apr. 6, 2022, provisional application No. 63/310,376, filed on Feb. 15, 2022, provisional application No. 63/309,382, filed on Feb. 11, 2022, provisional application No. 63/308,546, filed on Feb. 10, 2022, provisional application No. 63/307,349, filed on Feb. 7, 2022, provisional application No. 63/306,405, filed on Feb. 3, 2022, provisional application No. 63/306,412, filed on Feb. 3, 2022, provisional application No. 63/305,559, filed on Feb. 1, 2022, provisional application No. 63/305,505, filed on Feb. 1, 2022, provisional application No. 63/302,757, filed on Jan. 25, 2022, provisional application No. 63/302,768, filed on Jan. 25, 2022, provisional application No. 63/297,394, filed on Jan. 7, 2022, provisional application No. 63/266,478, filed on Jan. 6, 2022, provisional application No. 63/265,754, filed on Dec. 20, 2021, provisional application No. 63/264,811, filed on Dec. 2, 2021, provisional application No. 63/263,844, filed on Nov. 10, 2021, provisional application No. 63/262,858, filed on Oct. 21, 2021, provisional application No. 63/262,762, filed on Oct. 20, 2021, provisional application No. 63/262,536, filed on Oct. 14, 2021.

(51) **Int. Cl.**

A63F 13/73 (2014.01)
G06F 21/10 (2013.01)
H04L 9/00 (2022.01)

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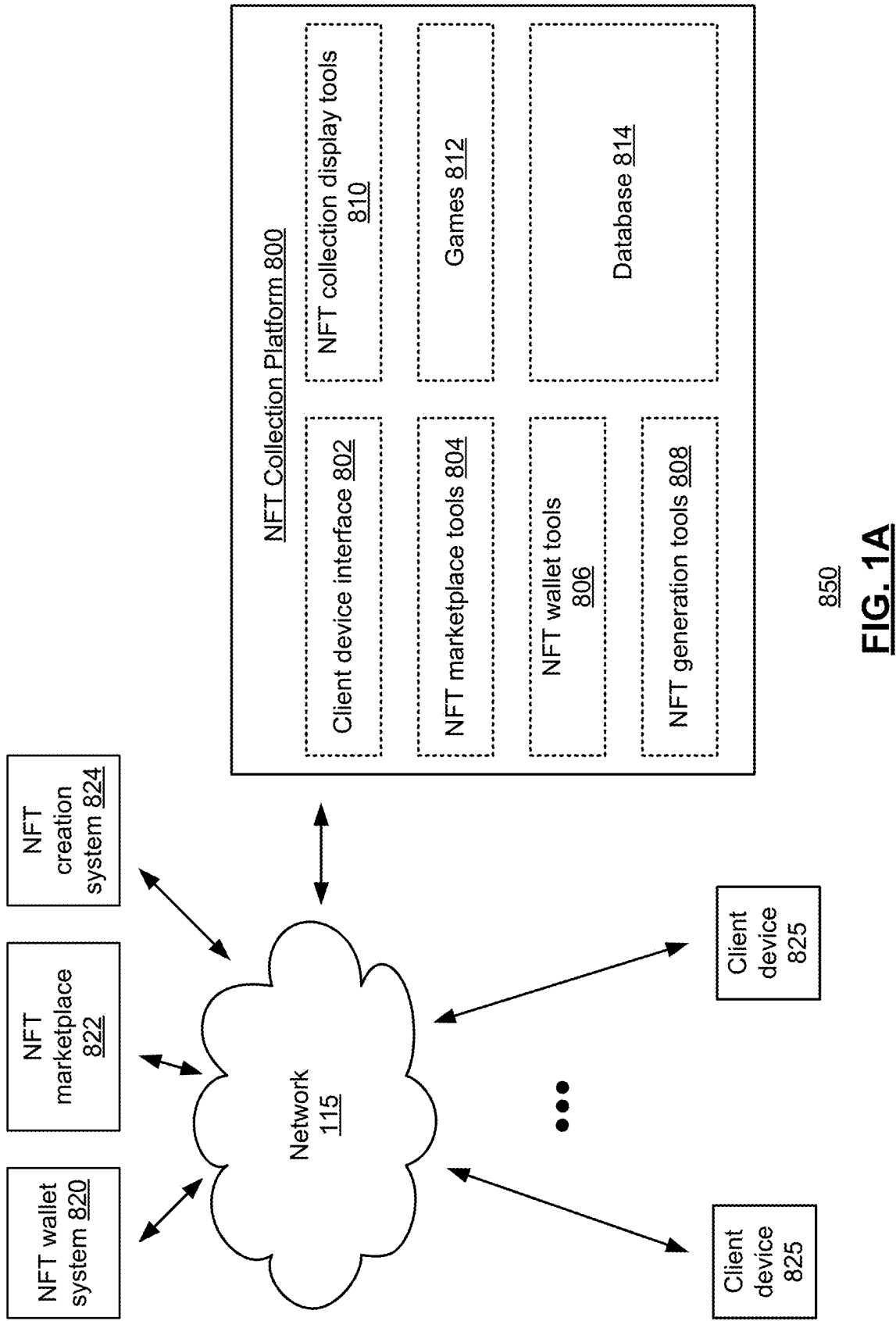
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 WO WO2023225437 11/2023
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 WO WO2024015704 1/2024
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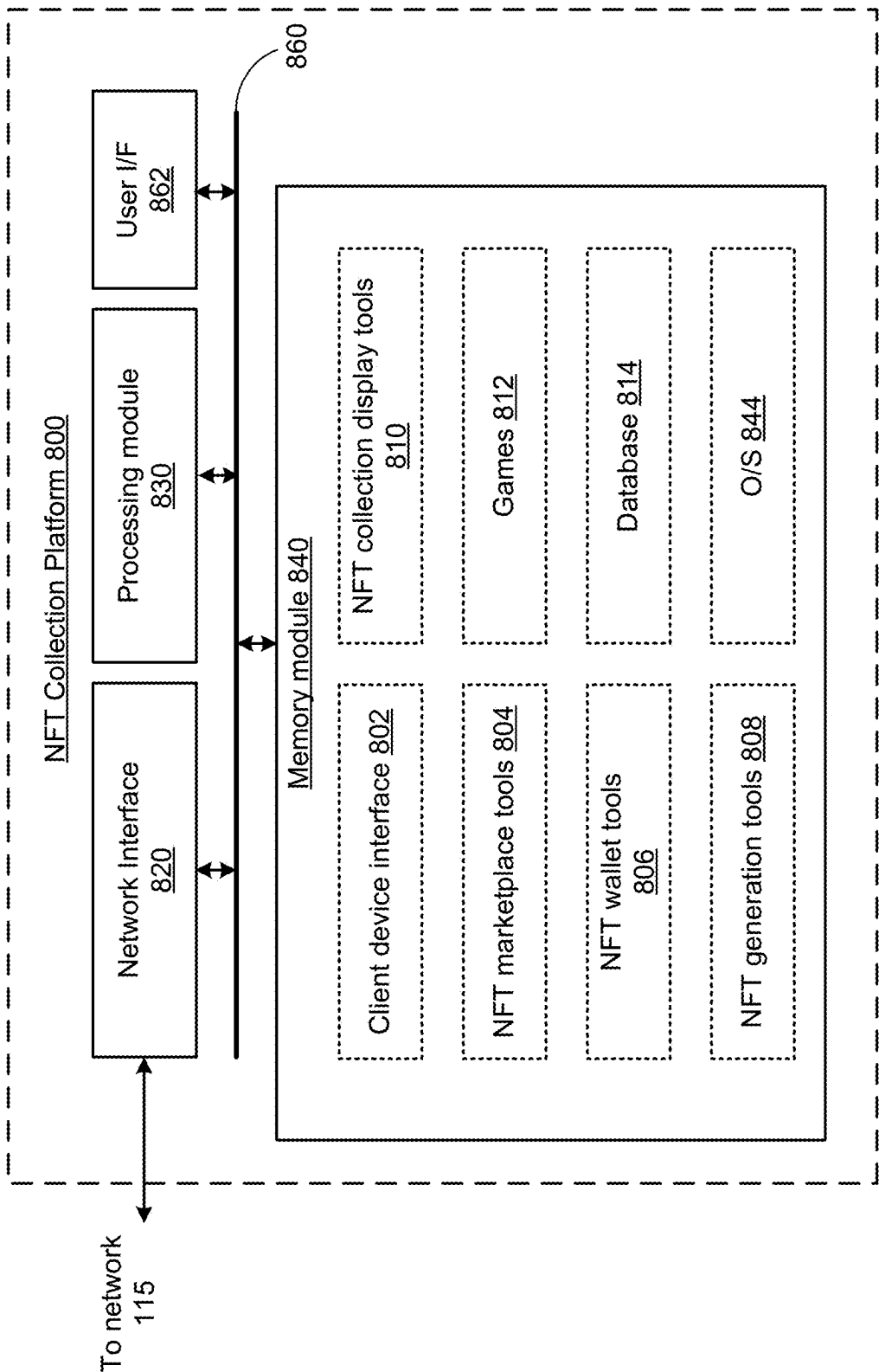


FIG. 1B

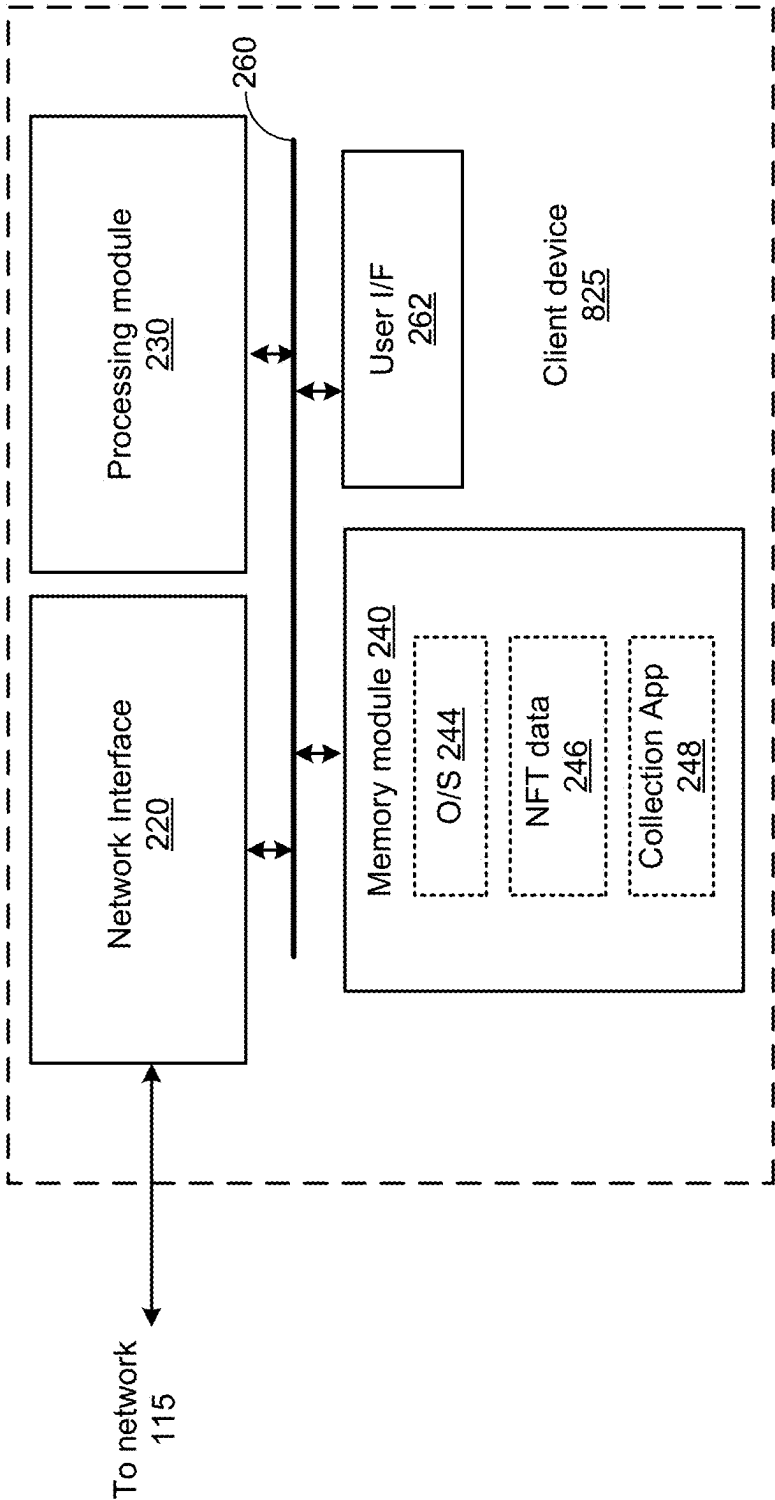
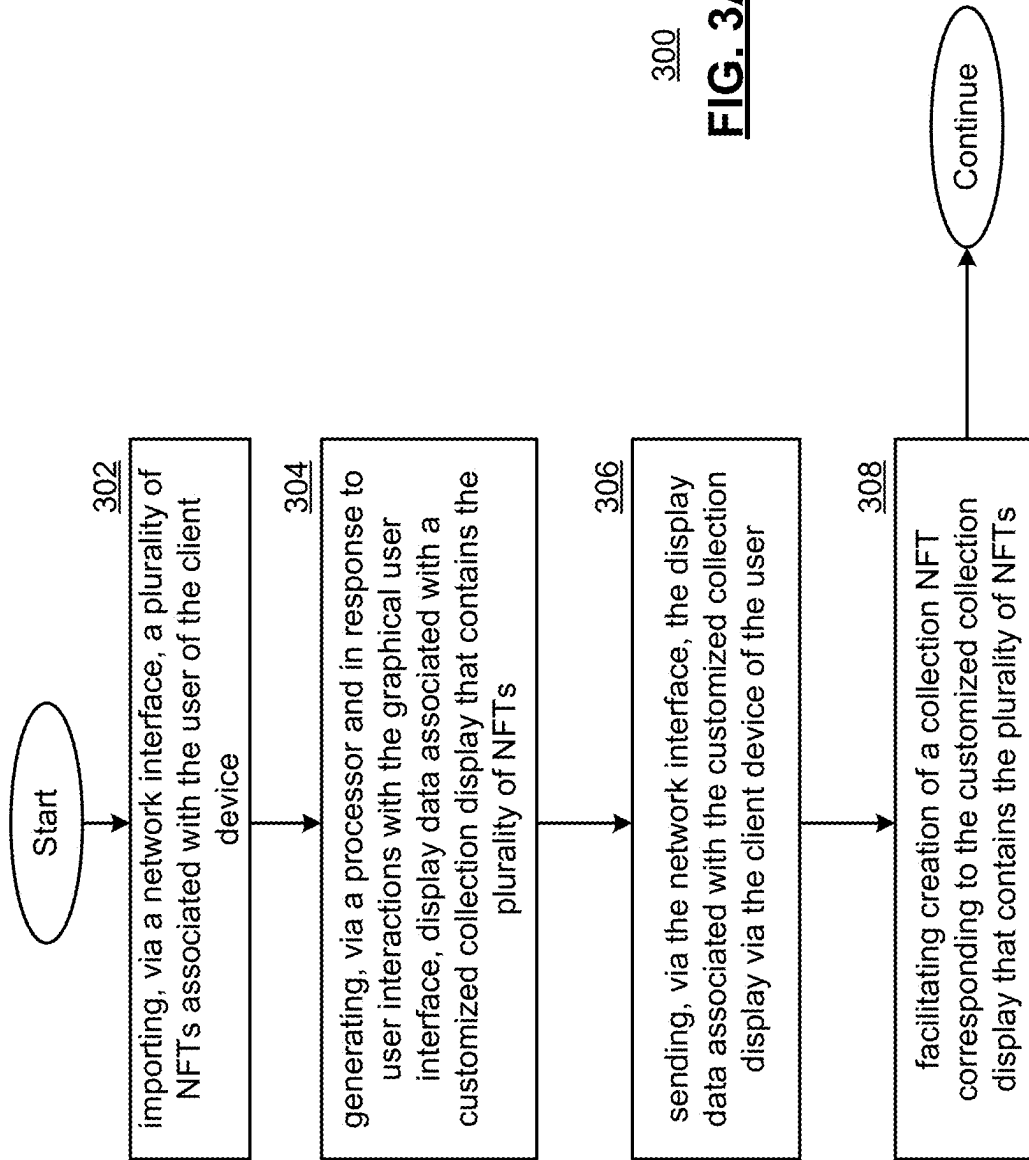


FIG. 2



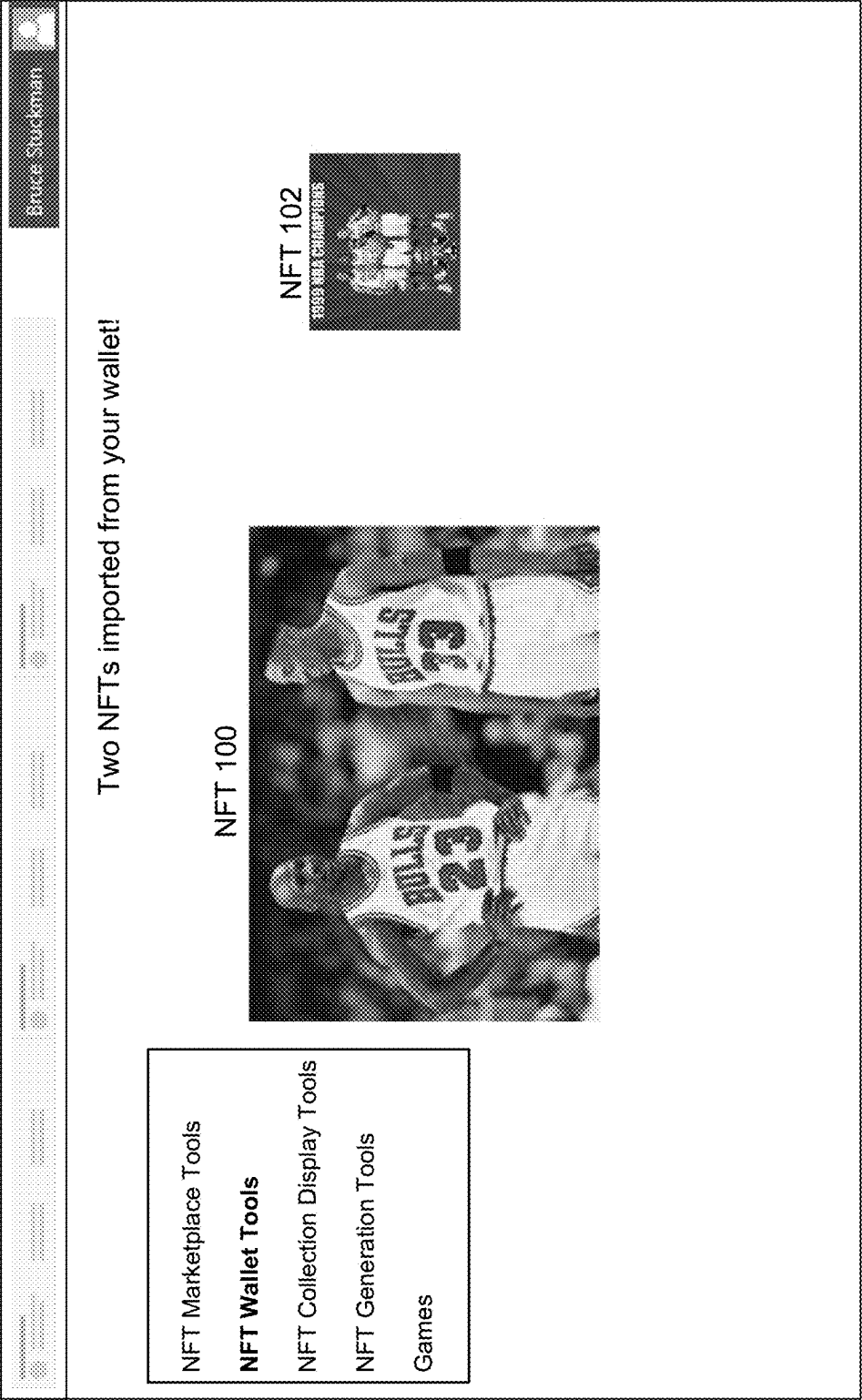


FIG. 3B



FIG. 3C

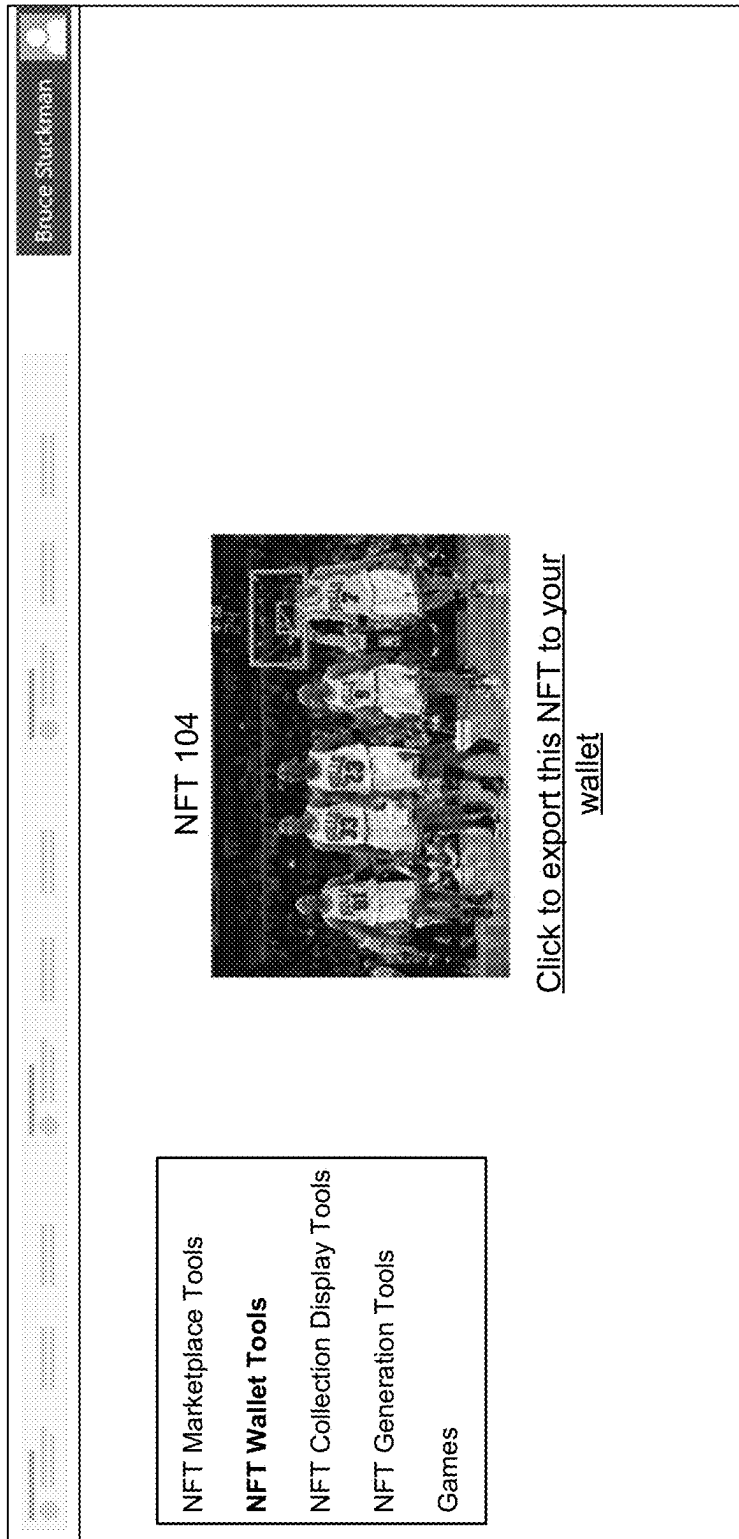


FIG. 3D

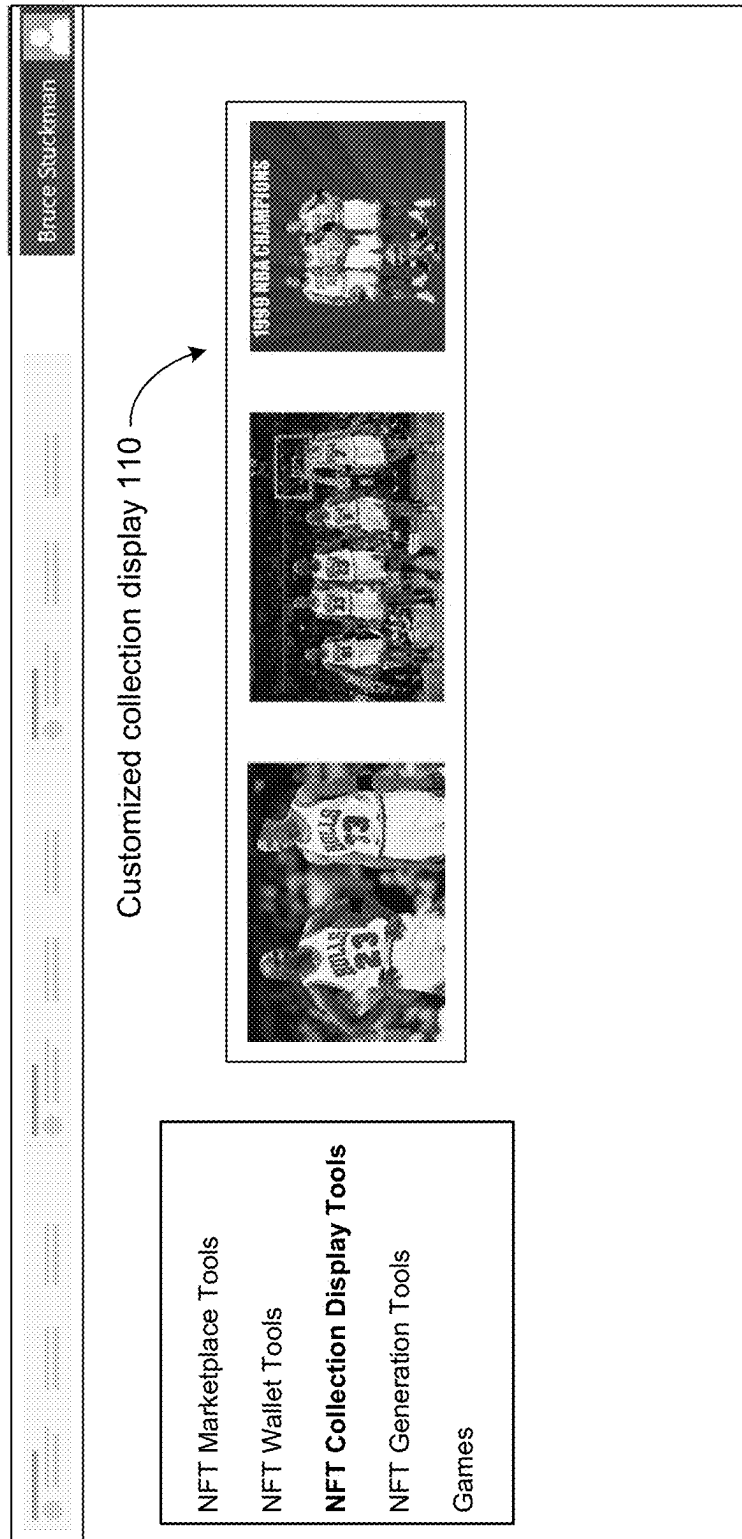


FIG. 3E

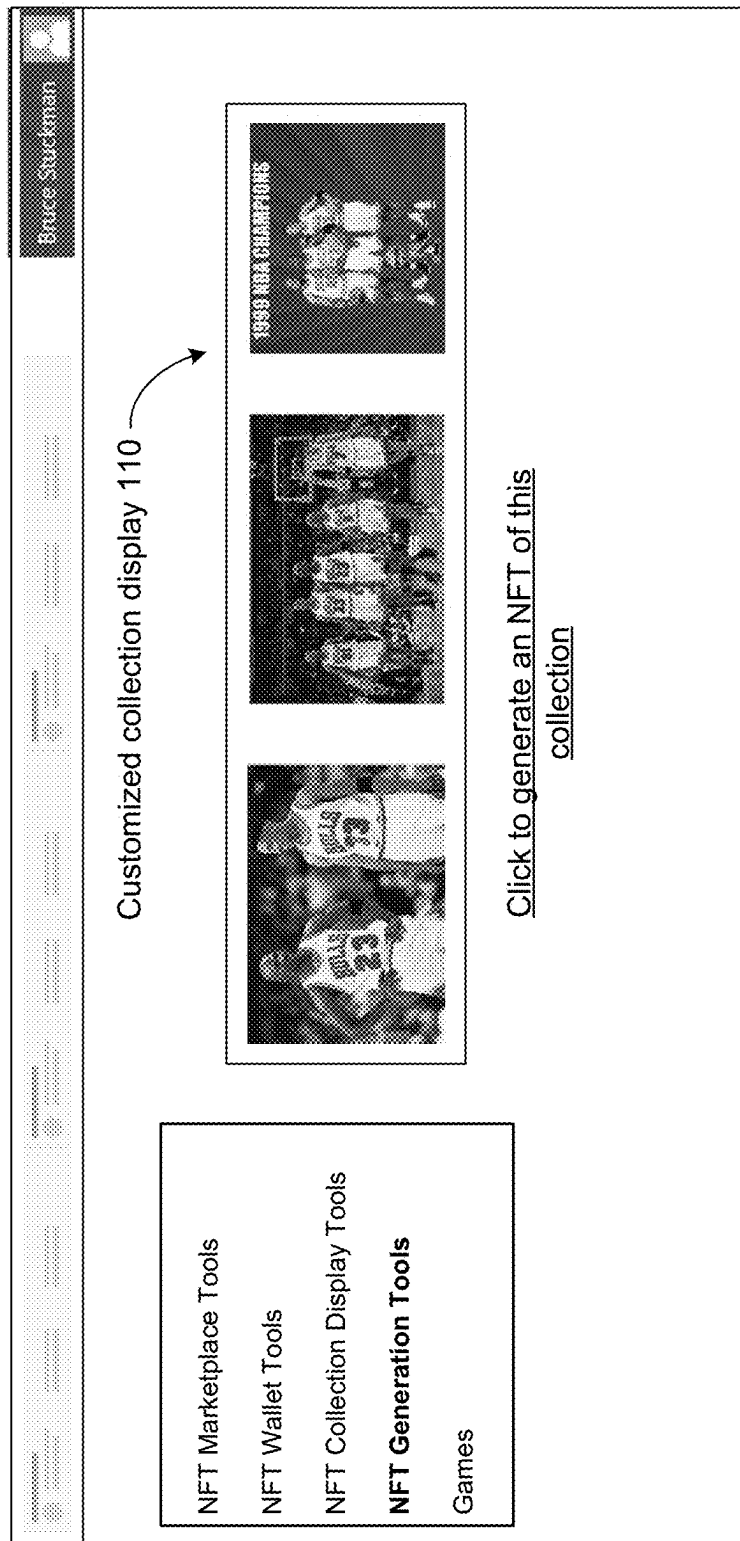


FIG. 3F

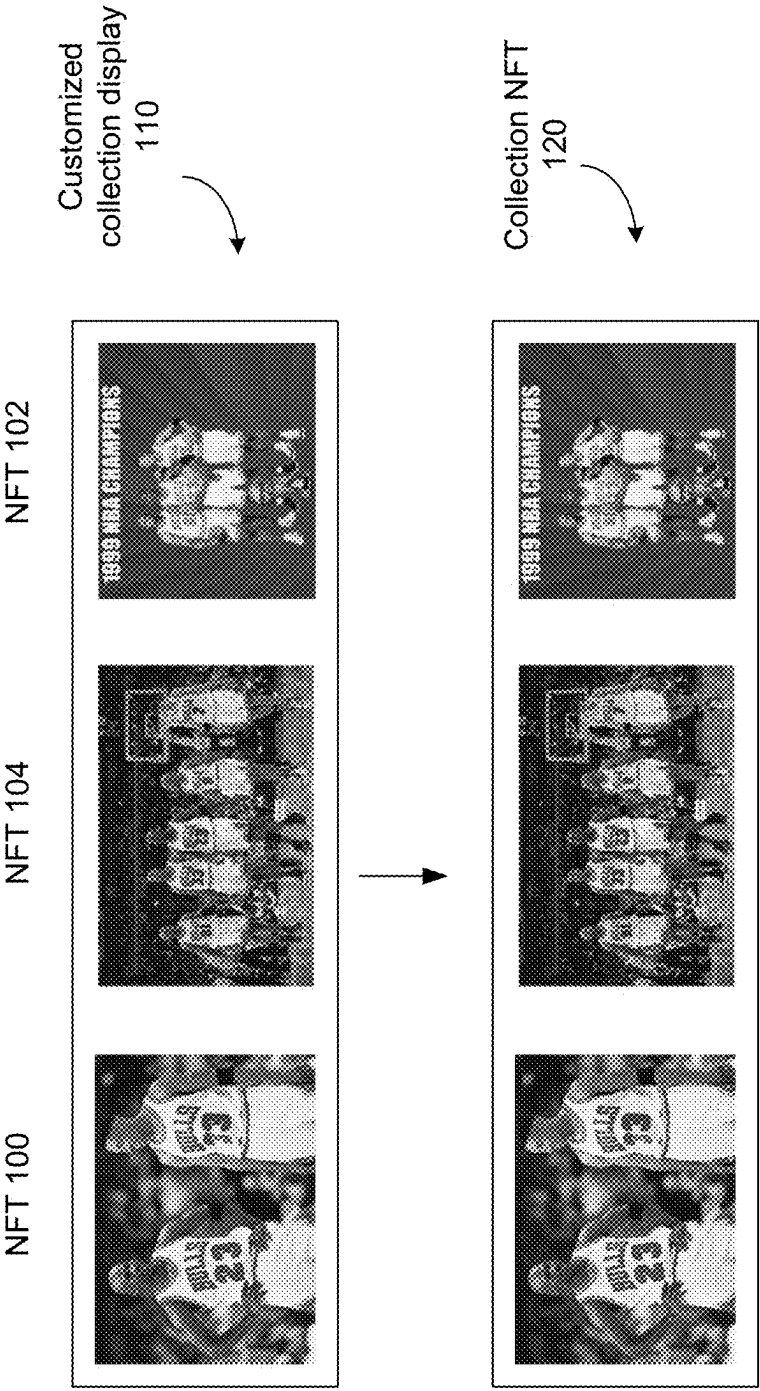
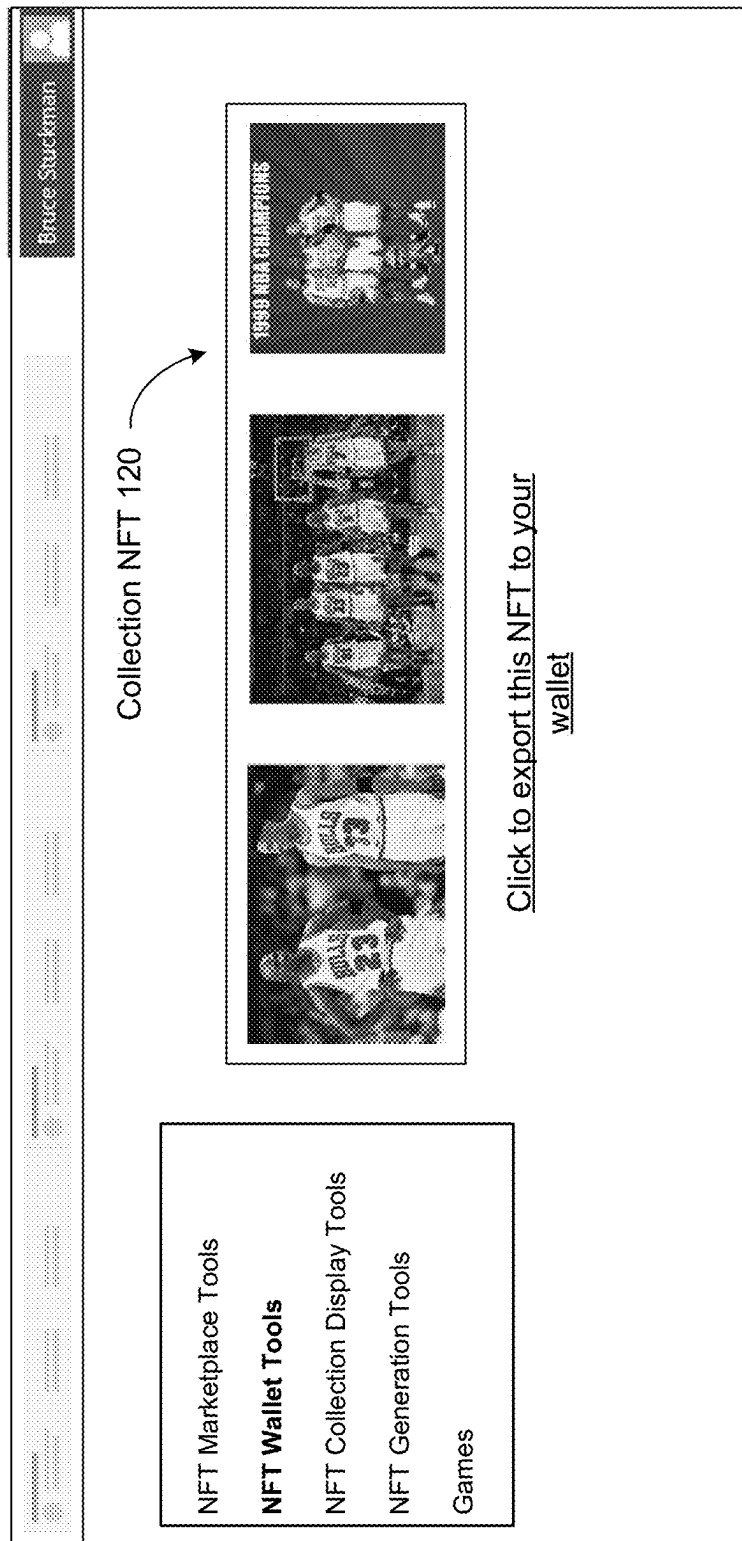
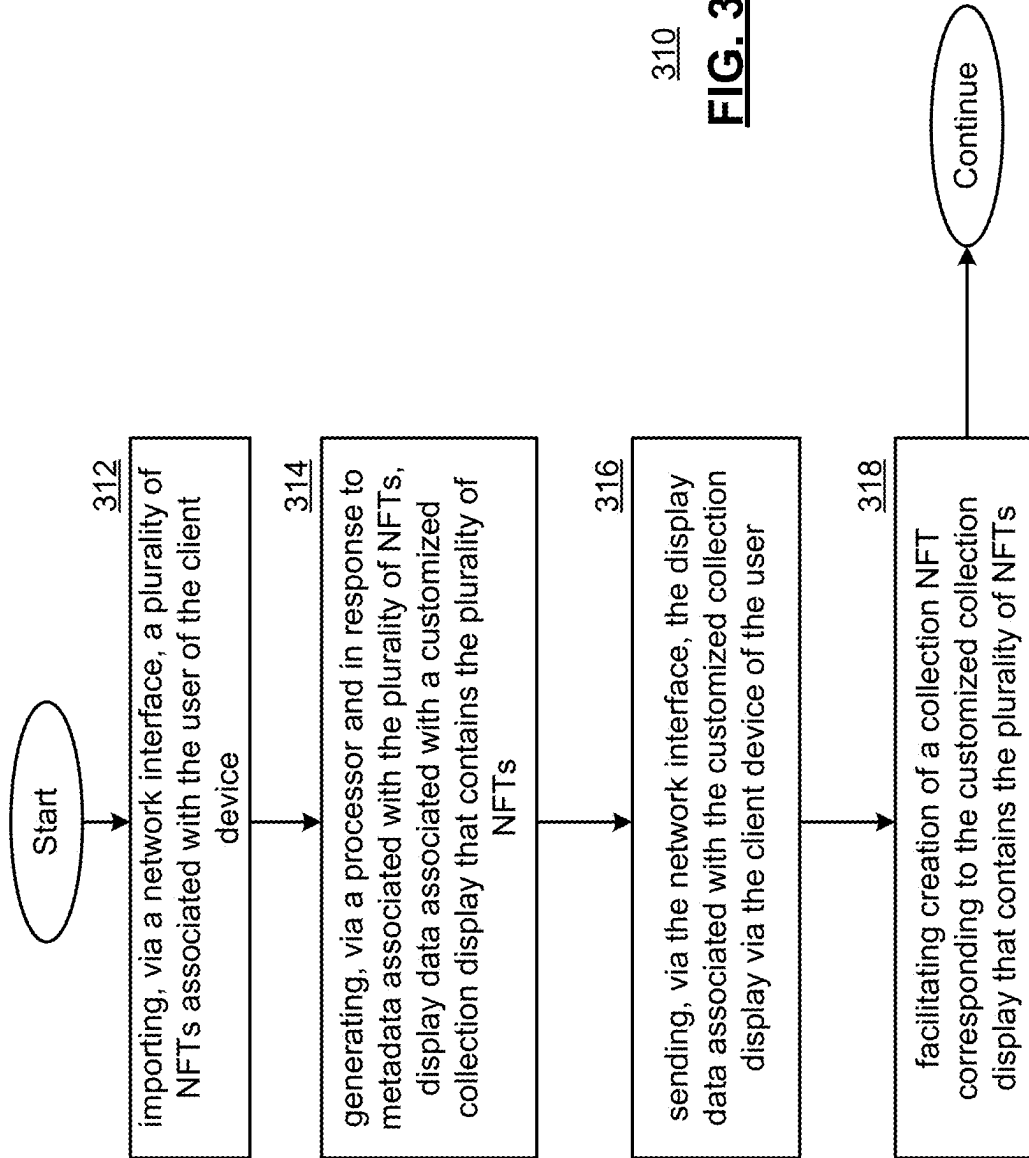
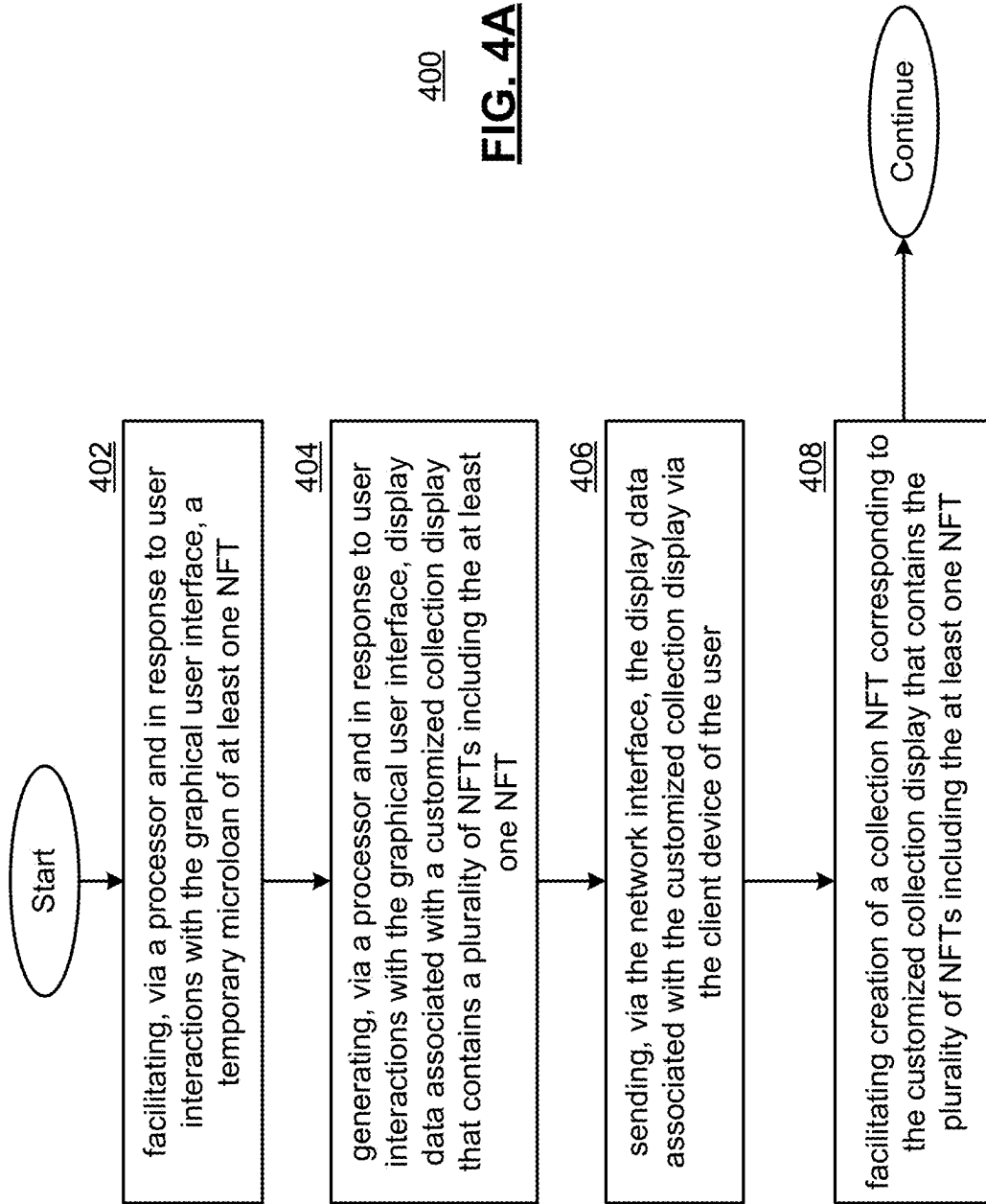


FIG. 3G

**FIG. 3H**

310**FIG. 31**



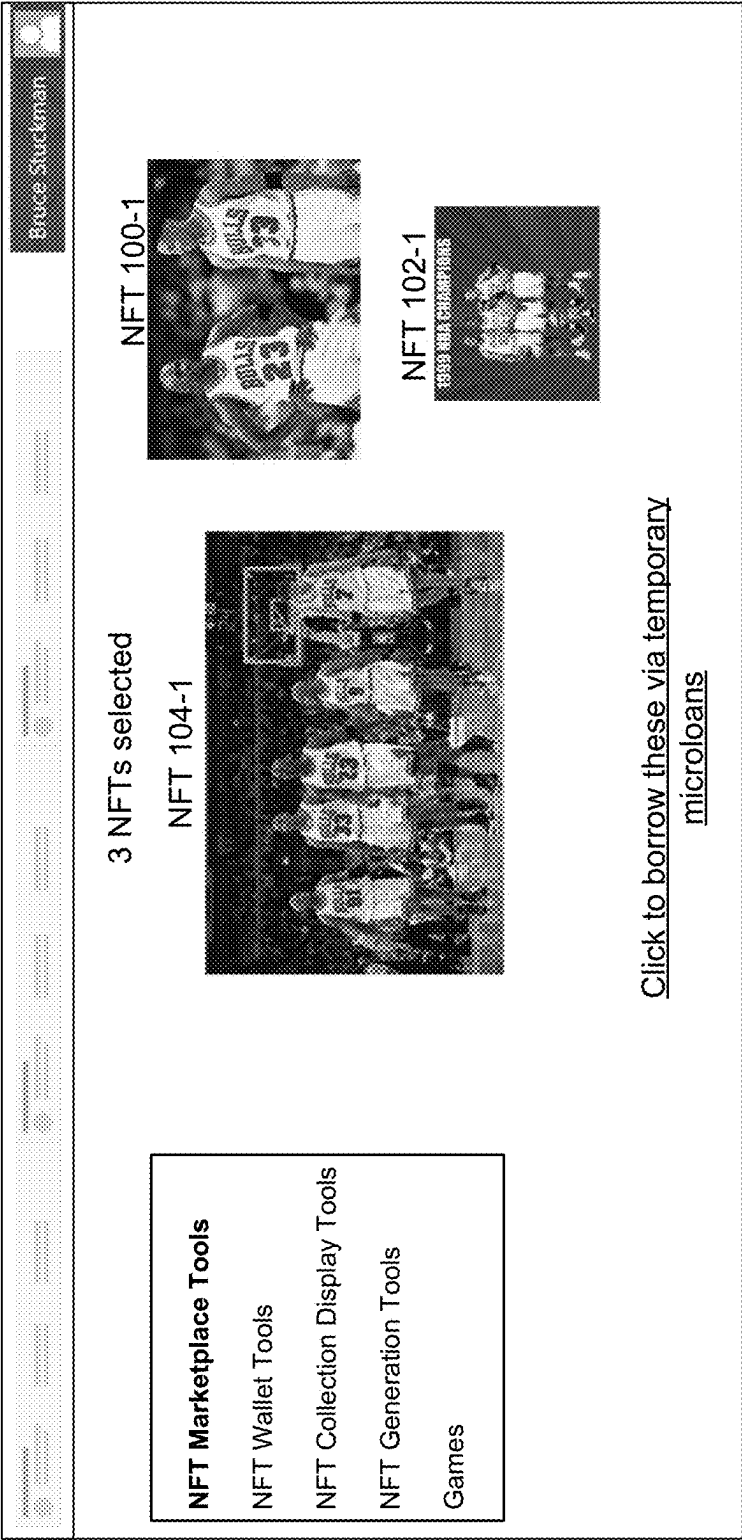


FIG. 4B

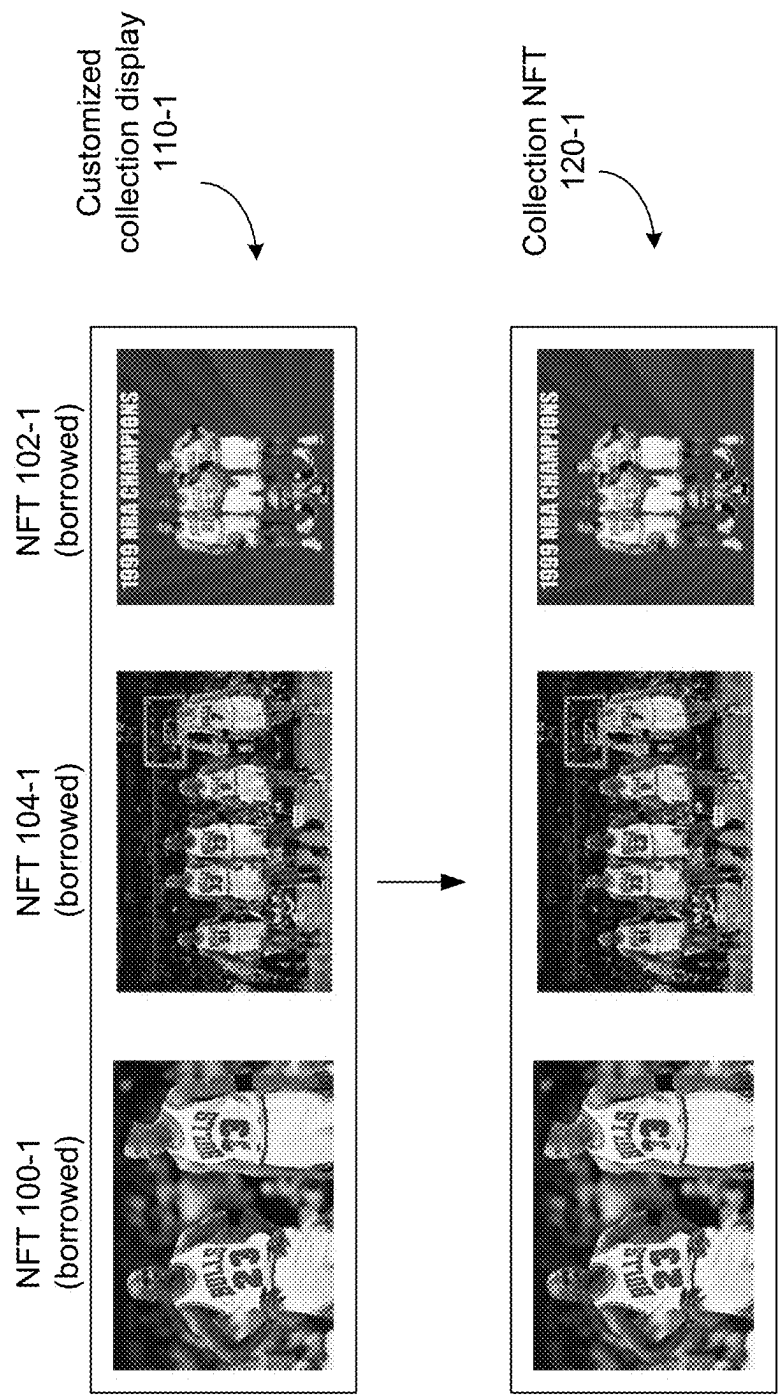
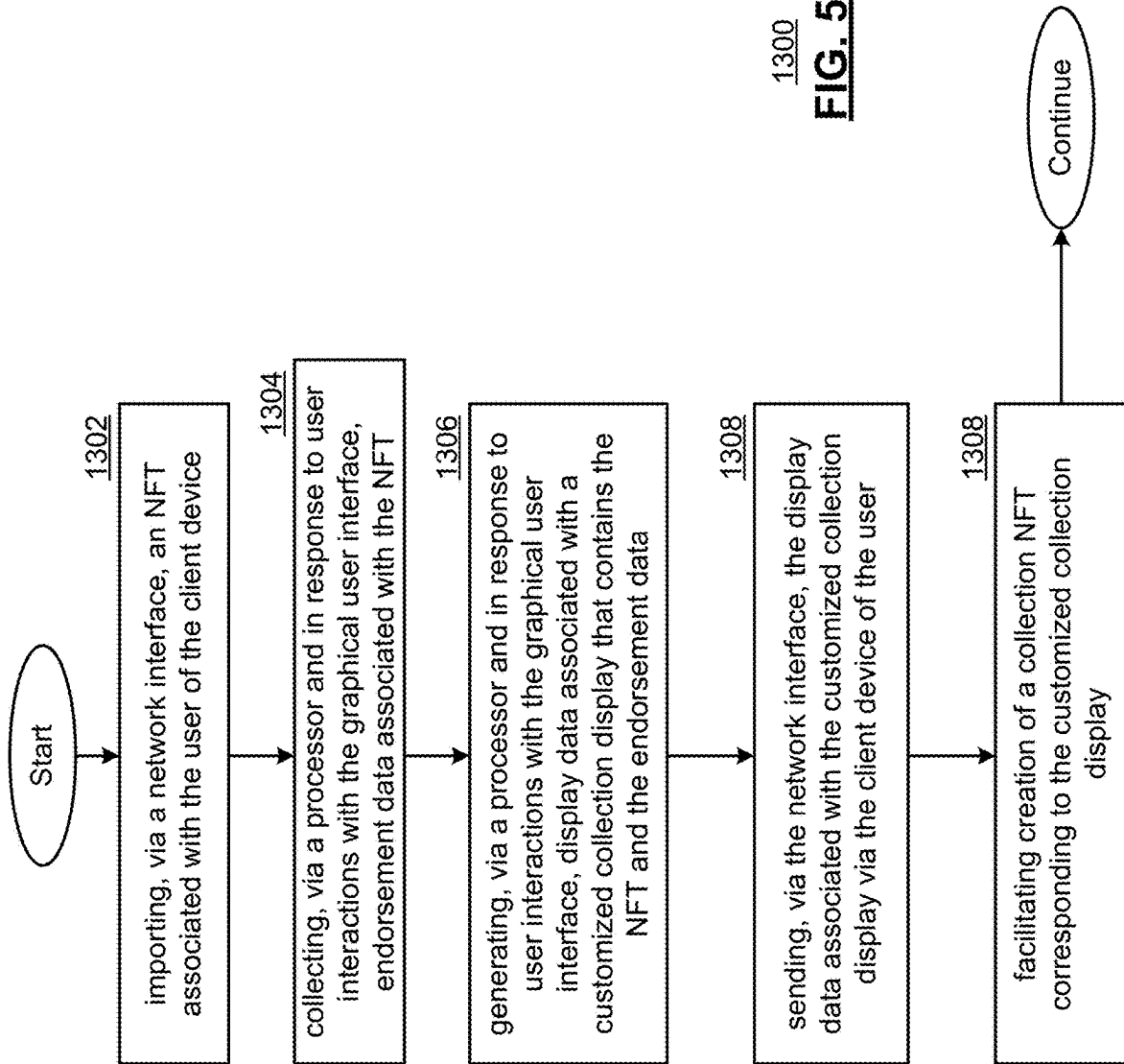


FIG. 4C



1300
FIG. 5A



FIG. 5B

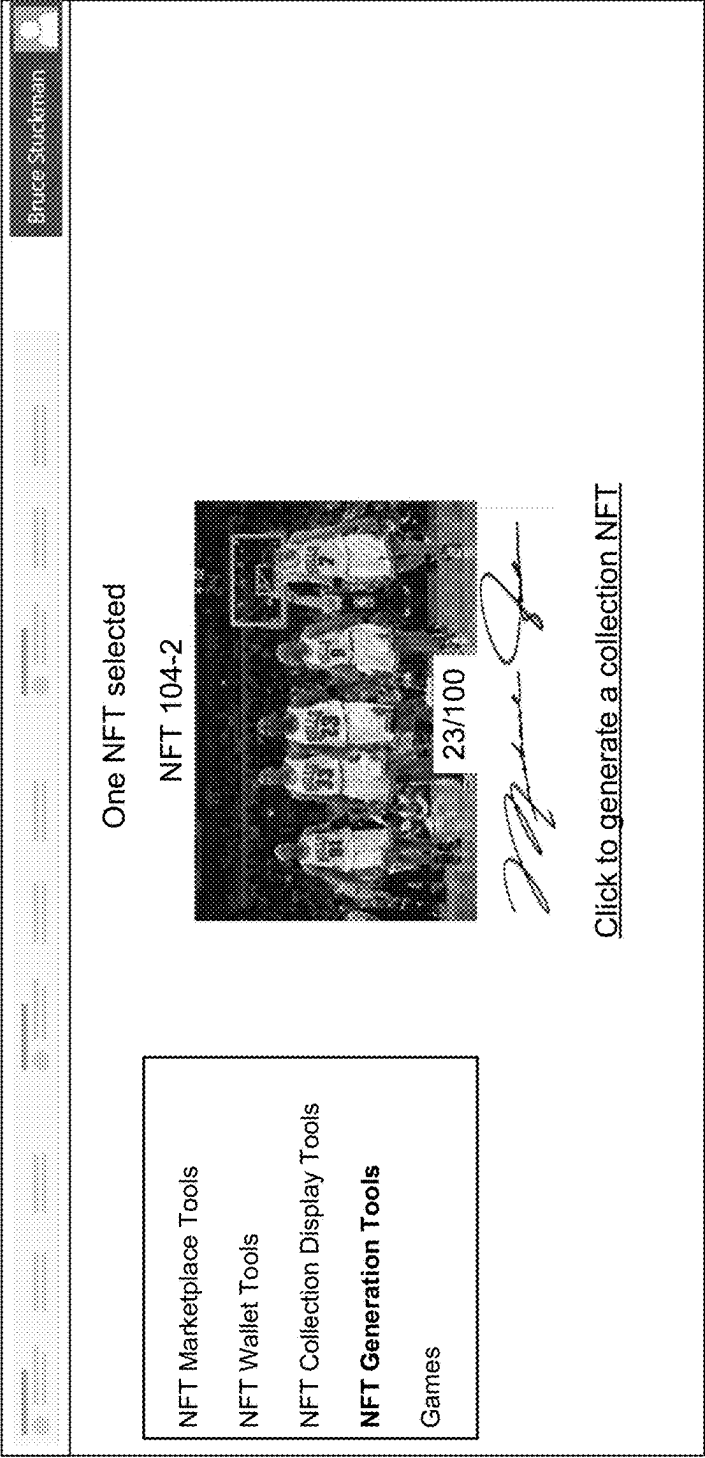


FIG. 5C

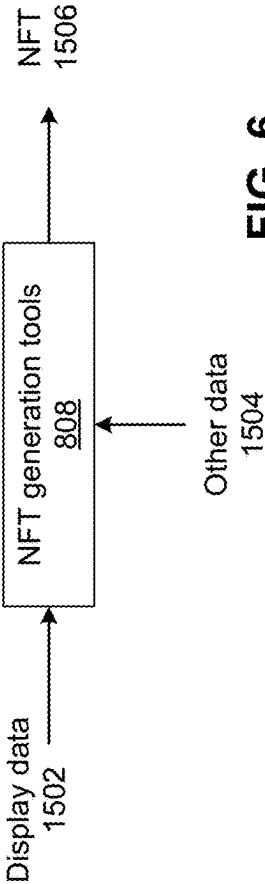
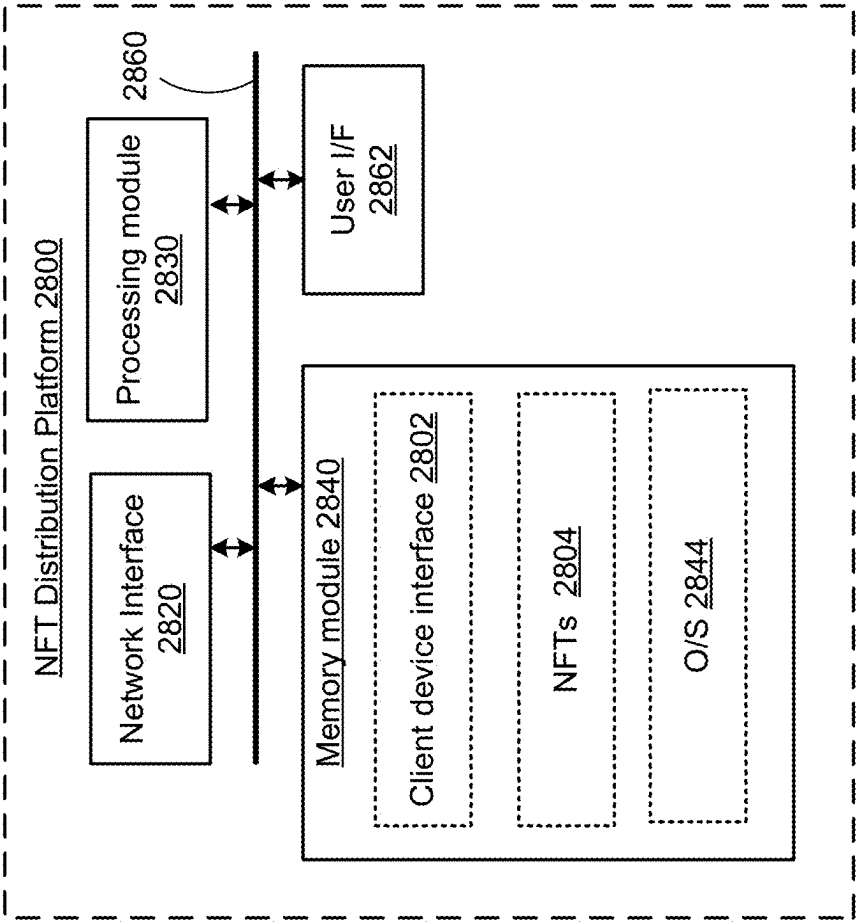
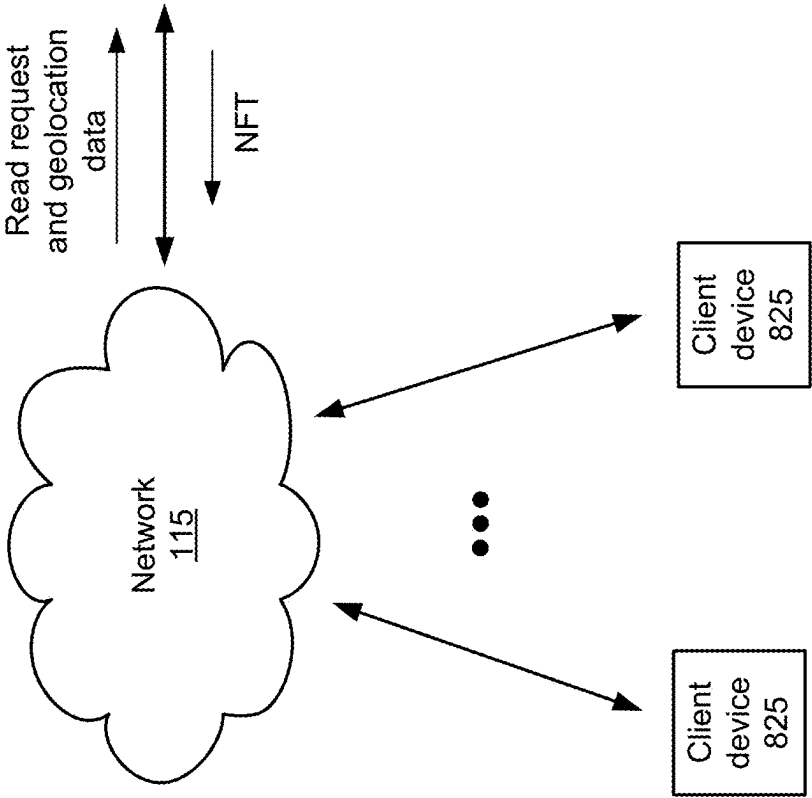


FIG. 6



2850

FIG. 7



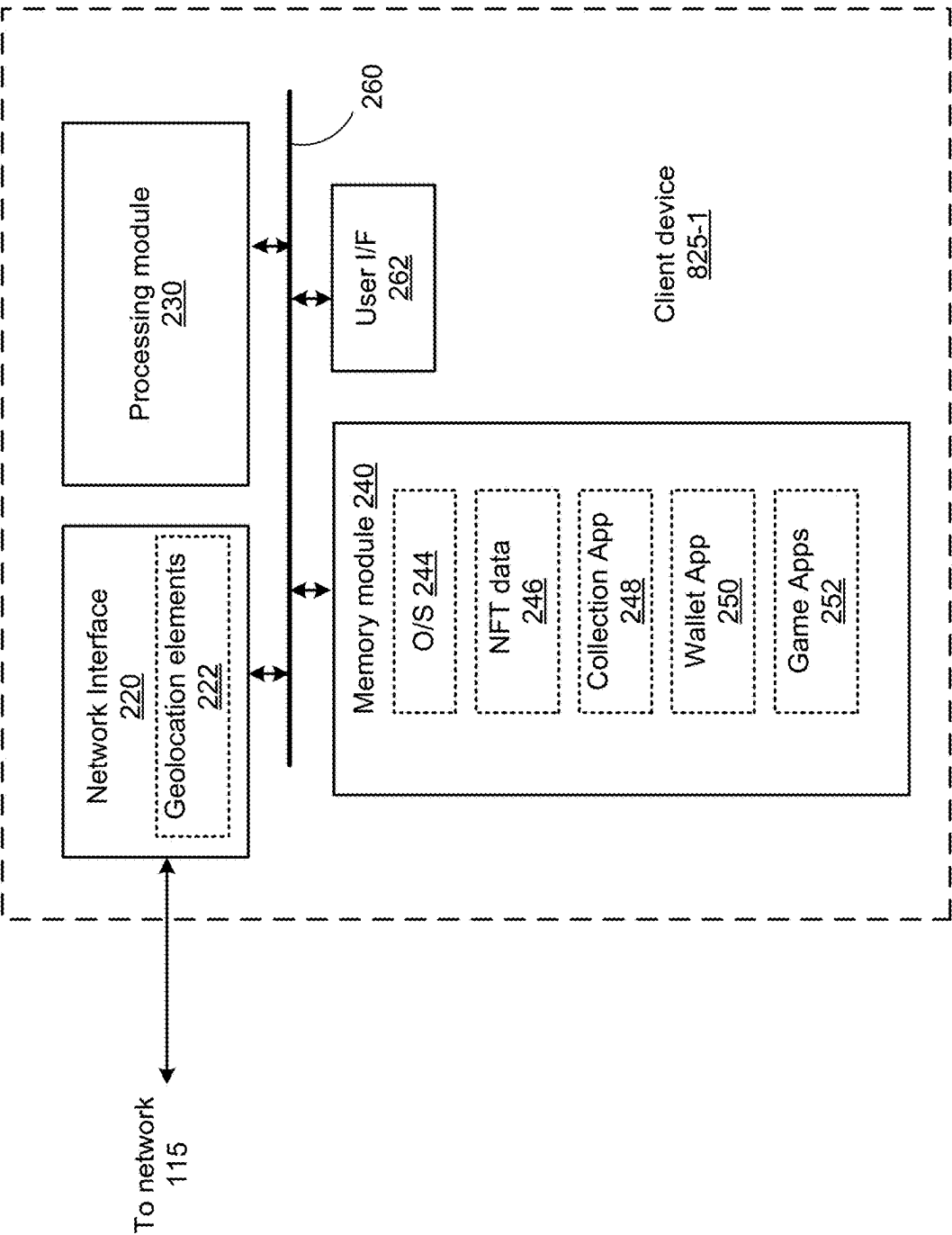


FIG. 8

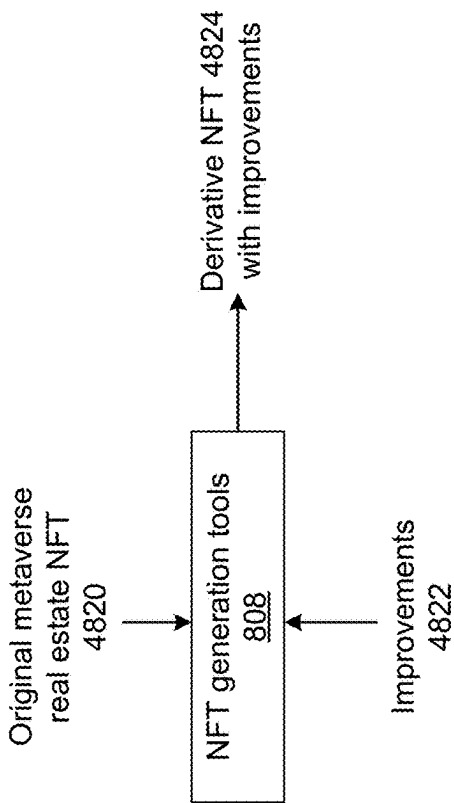


FIG. 9A

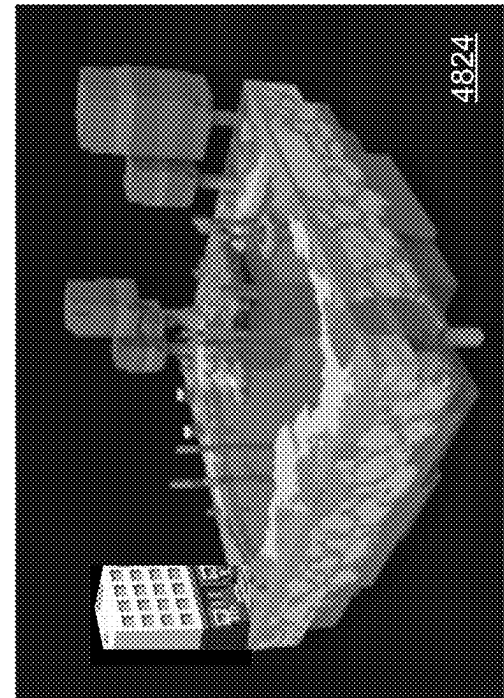
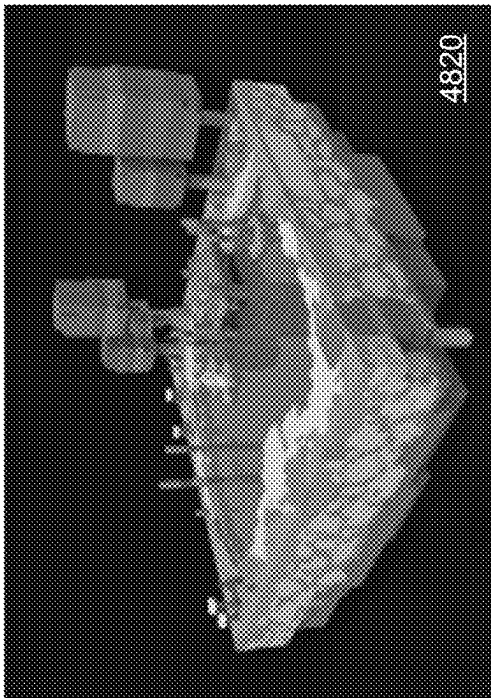


FIG. 9D

FIG. 9B

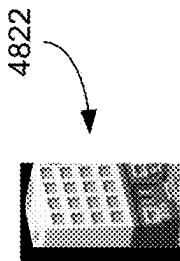
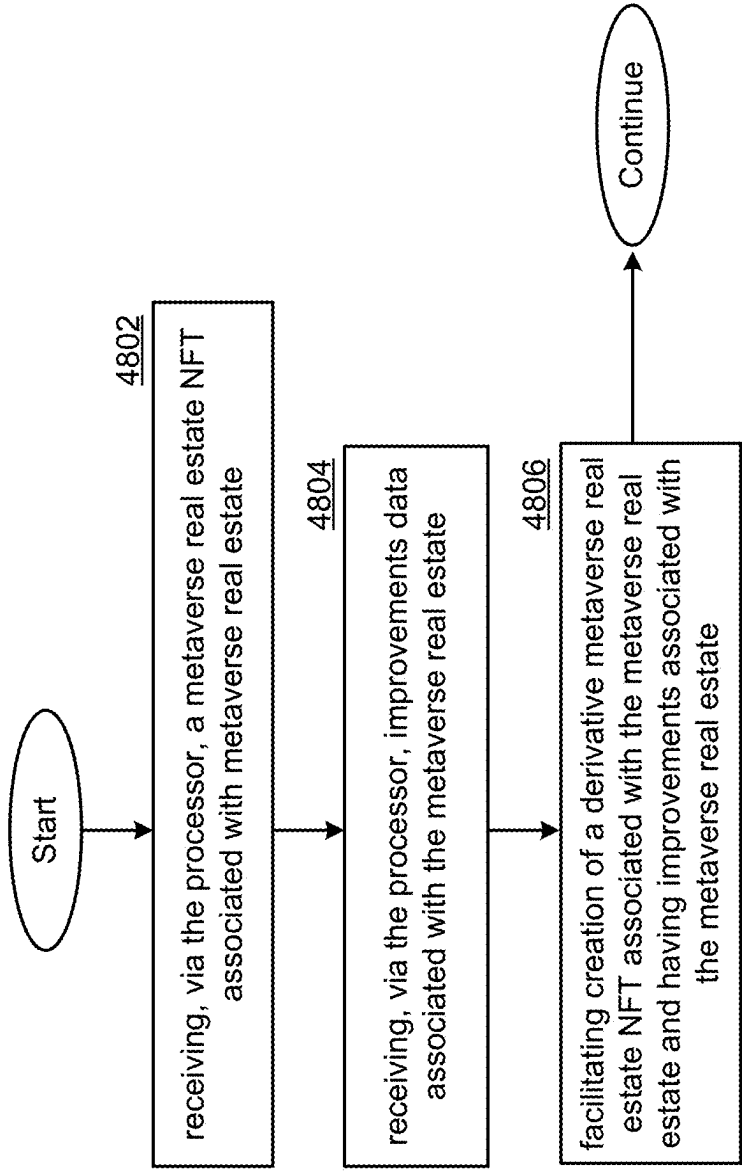


FIG. 9C



4800
FIG. 9E

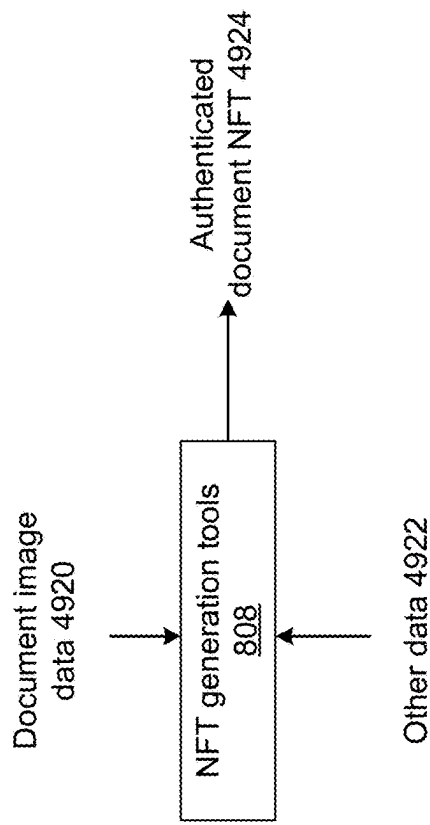
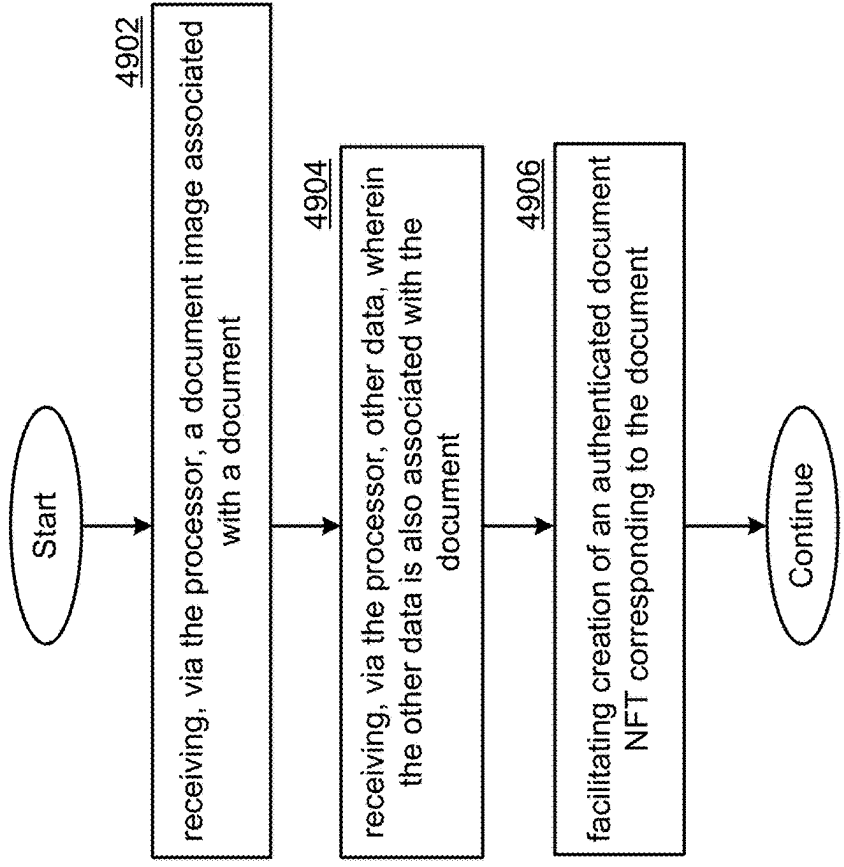


FIG. 10A



4900
FIG. 10B



FIG. 11B

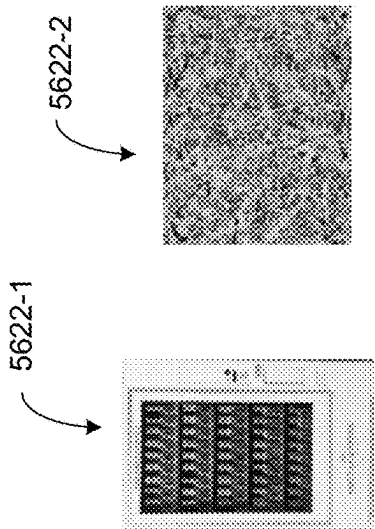


FIG. 11C

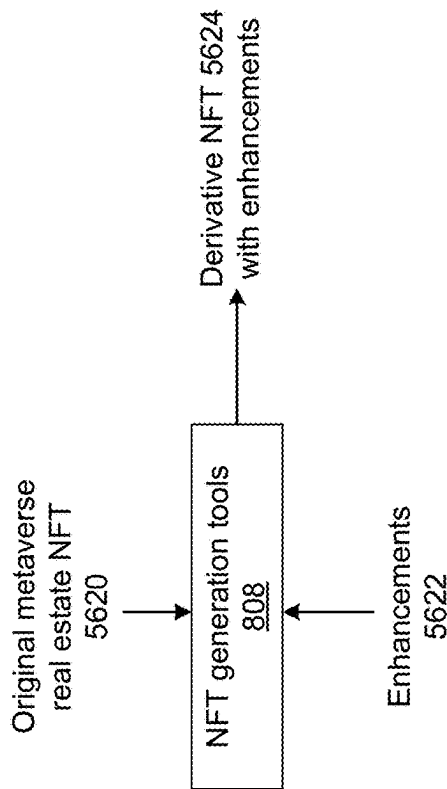


FIG. 11A

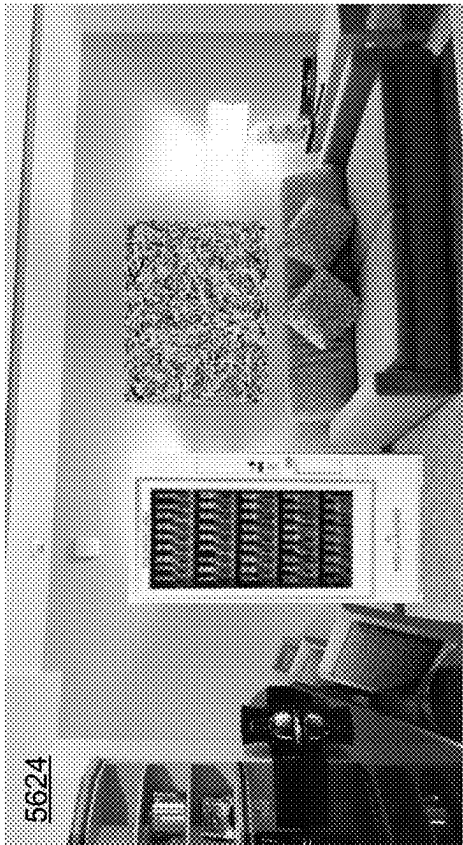
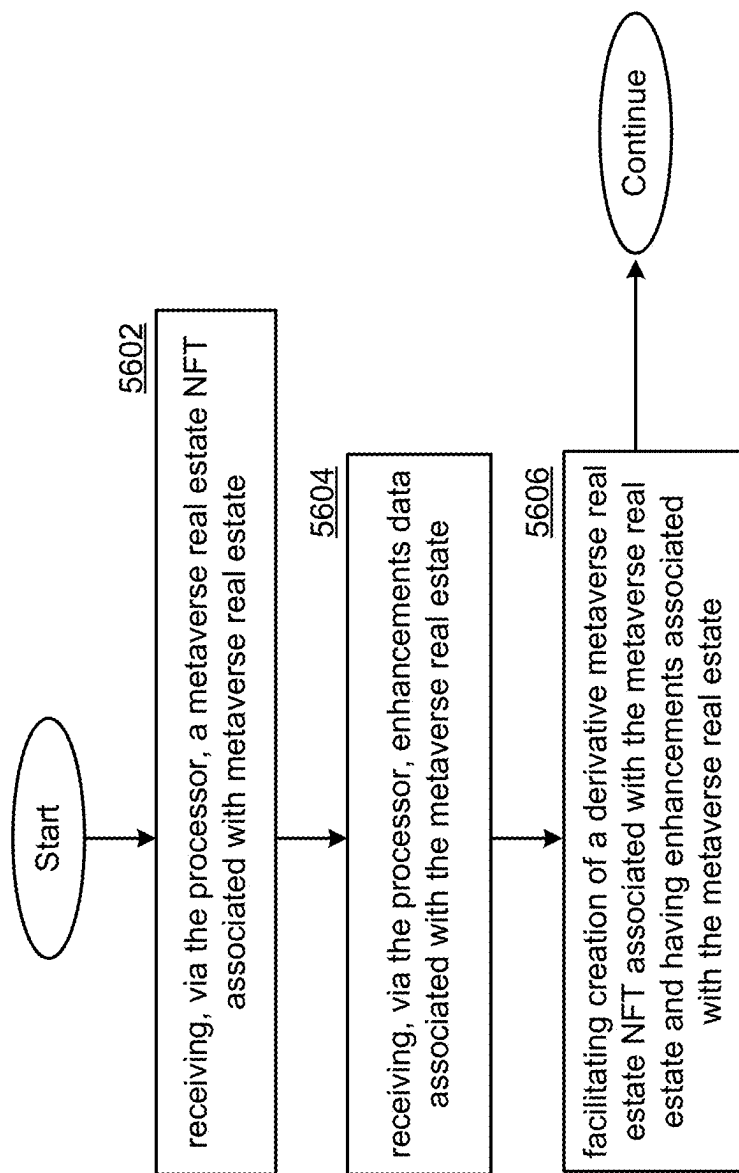


FIG. 11D



5600

FIG. 11E

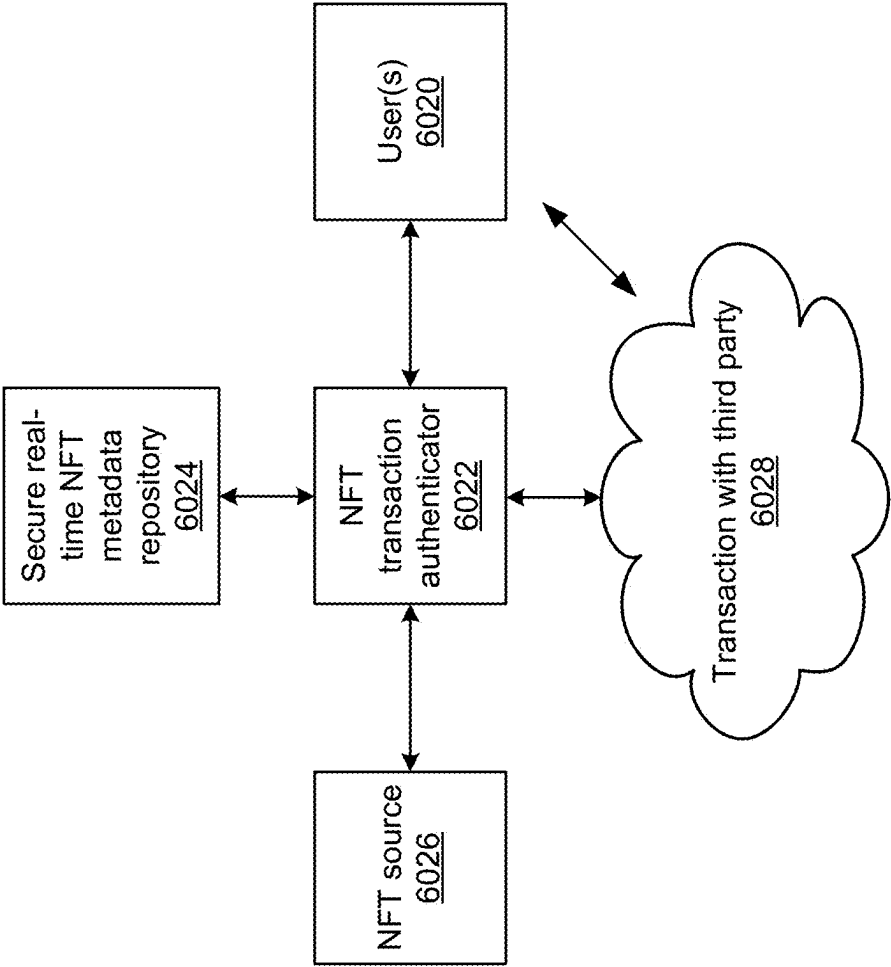
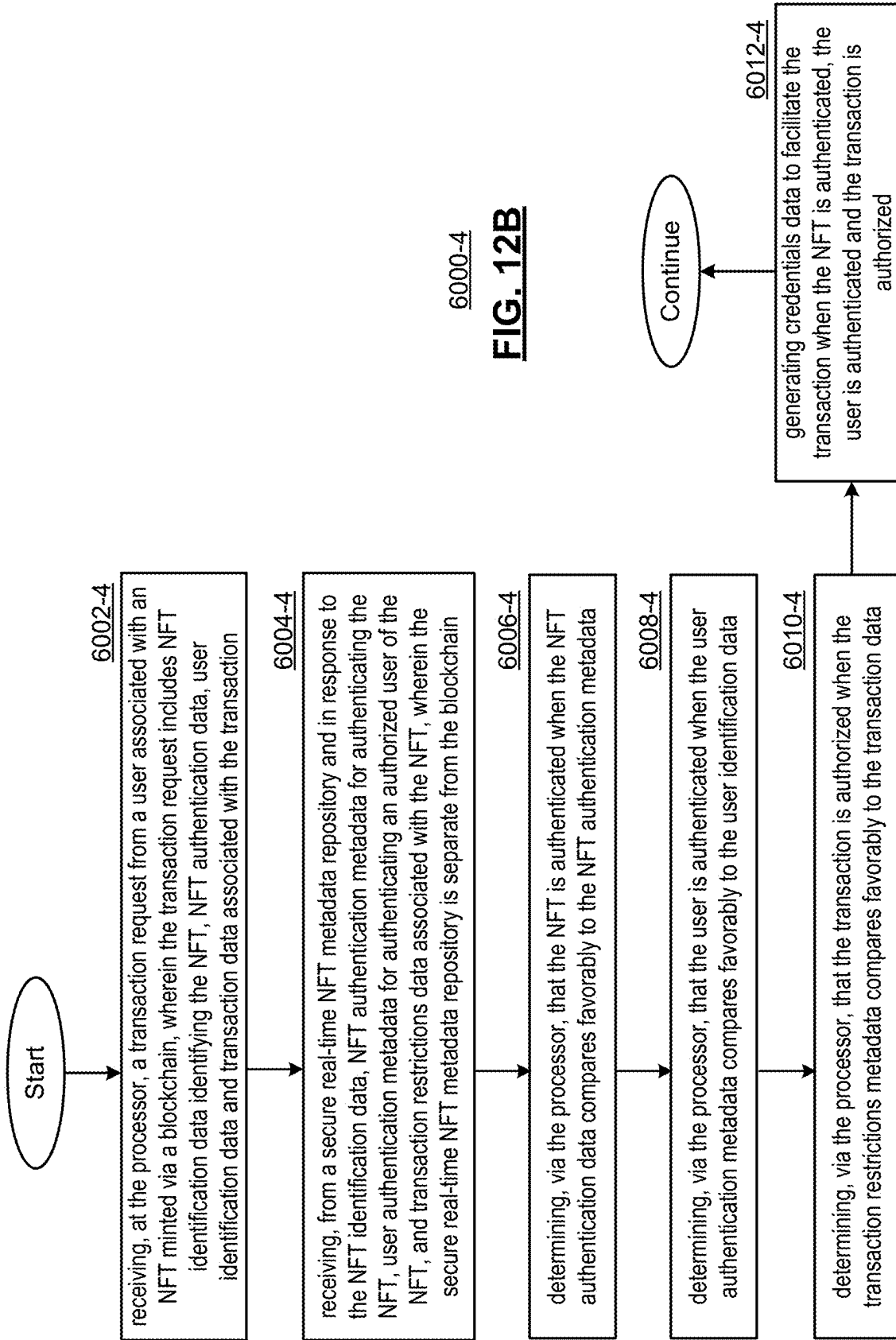


FIG. 12A



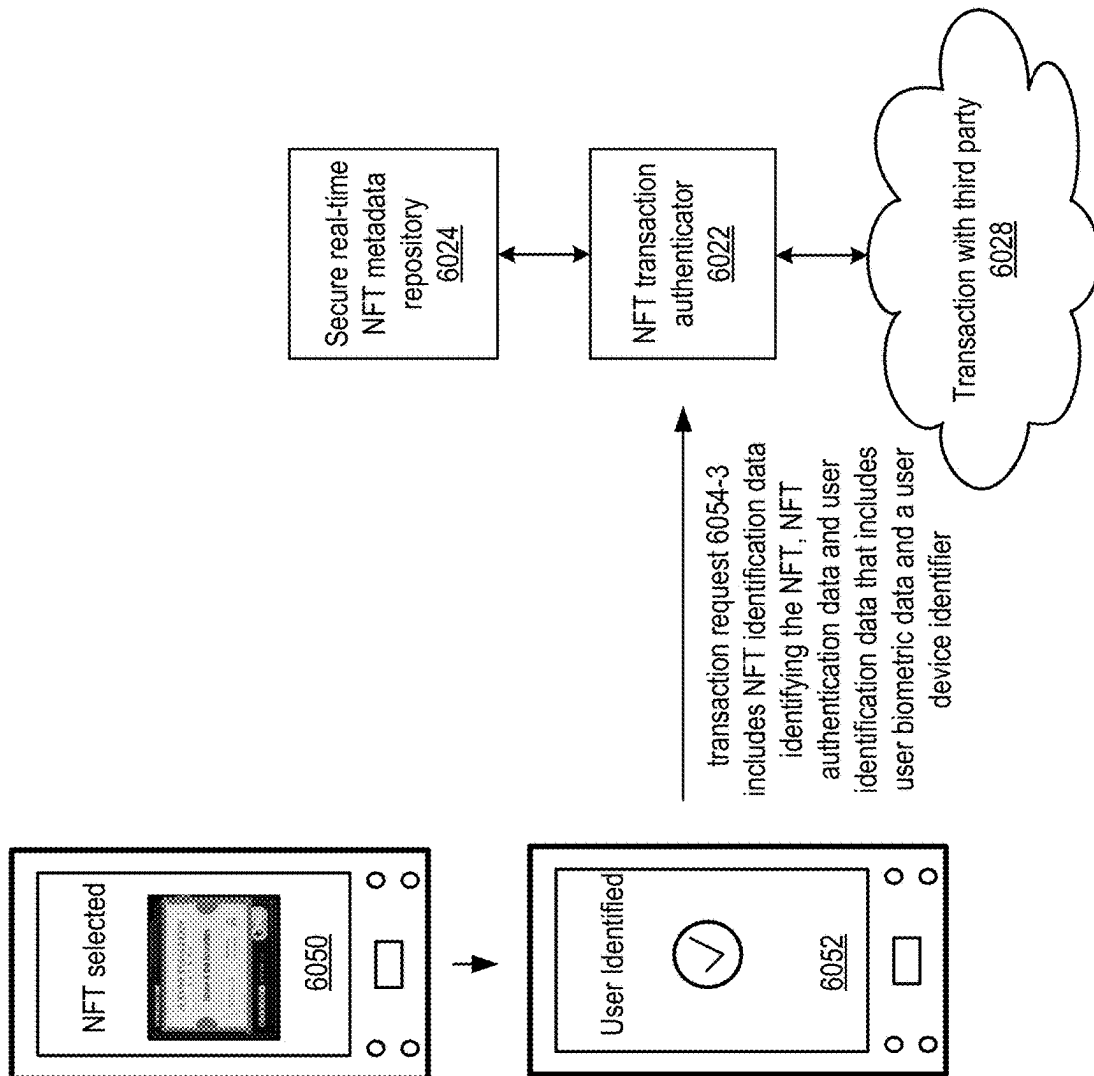


FIG. 12C

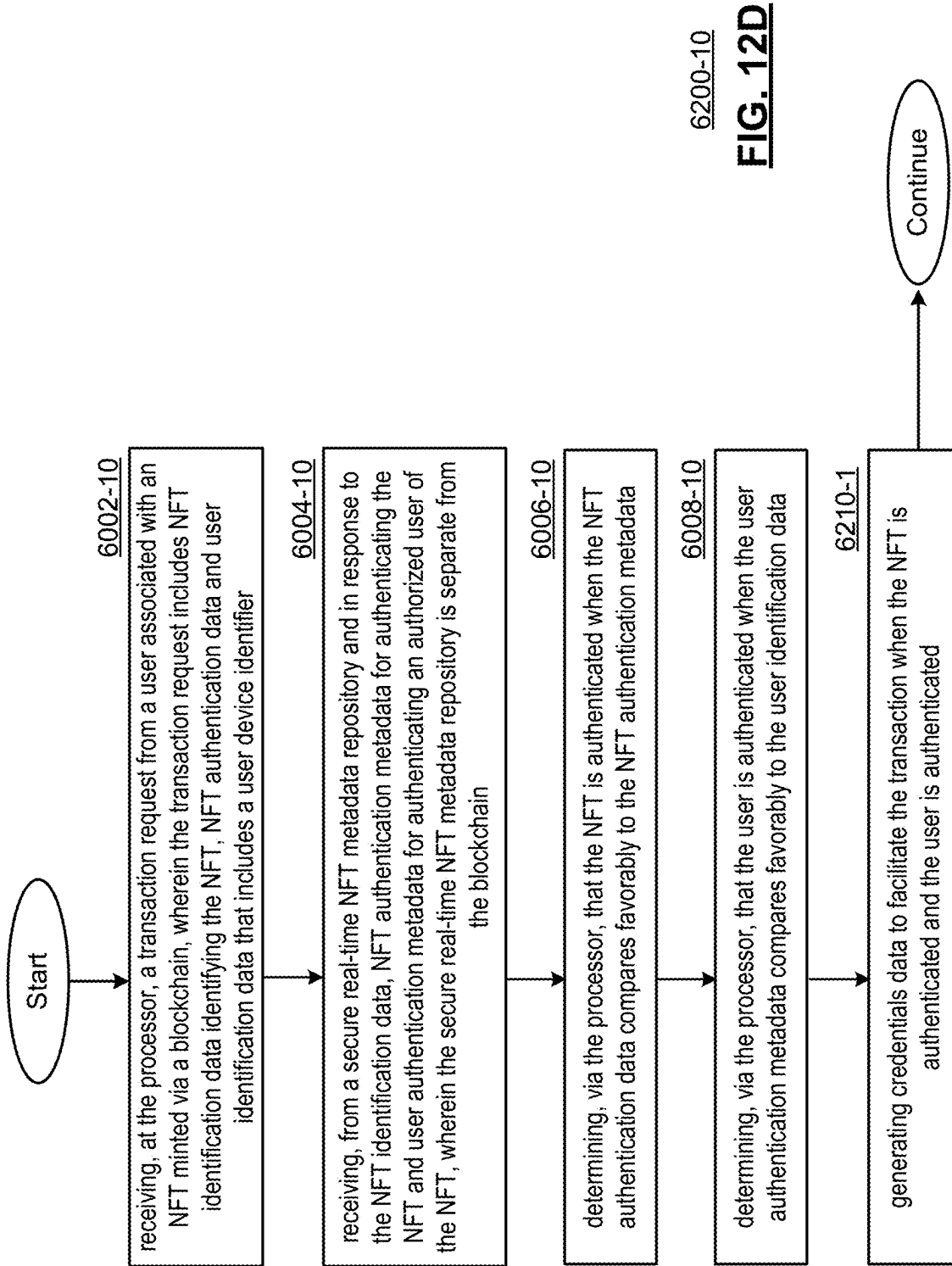




FIG. 13A

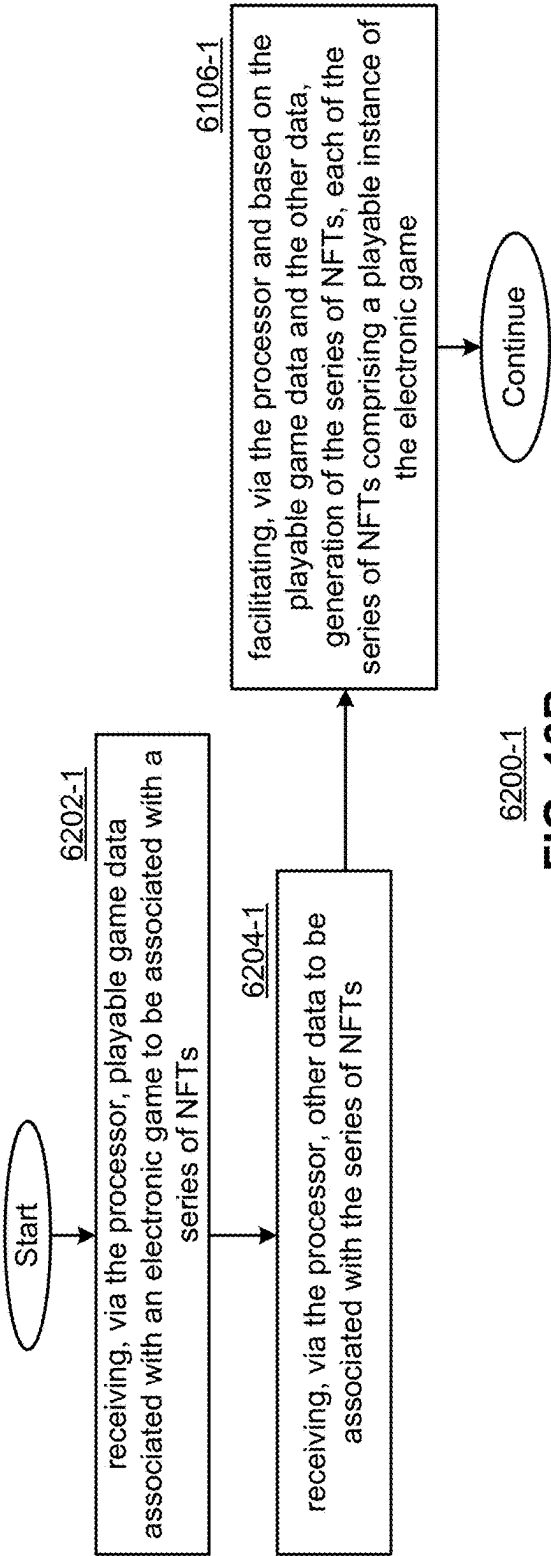


FIG. 13B

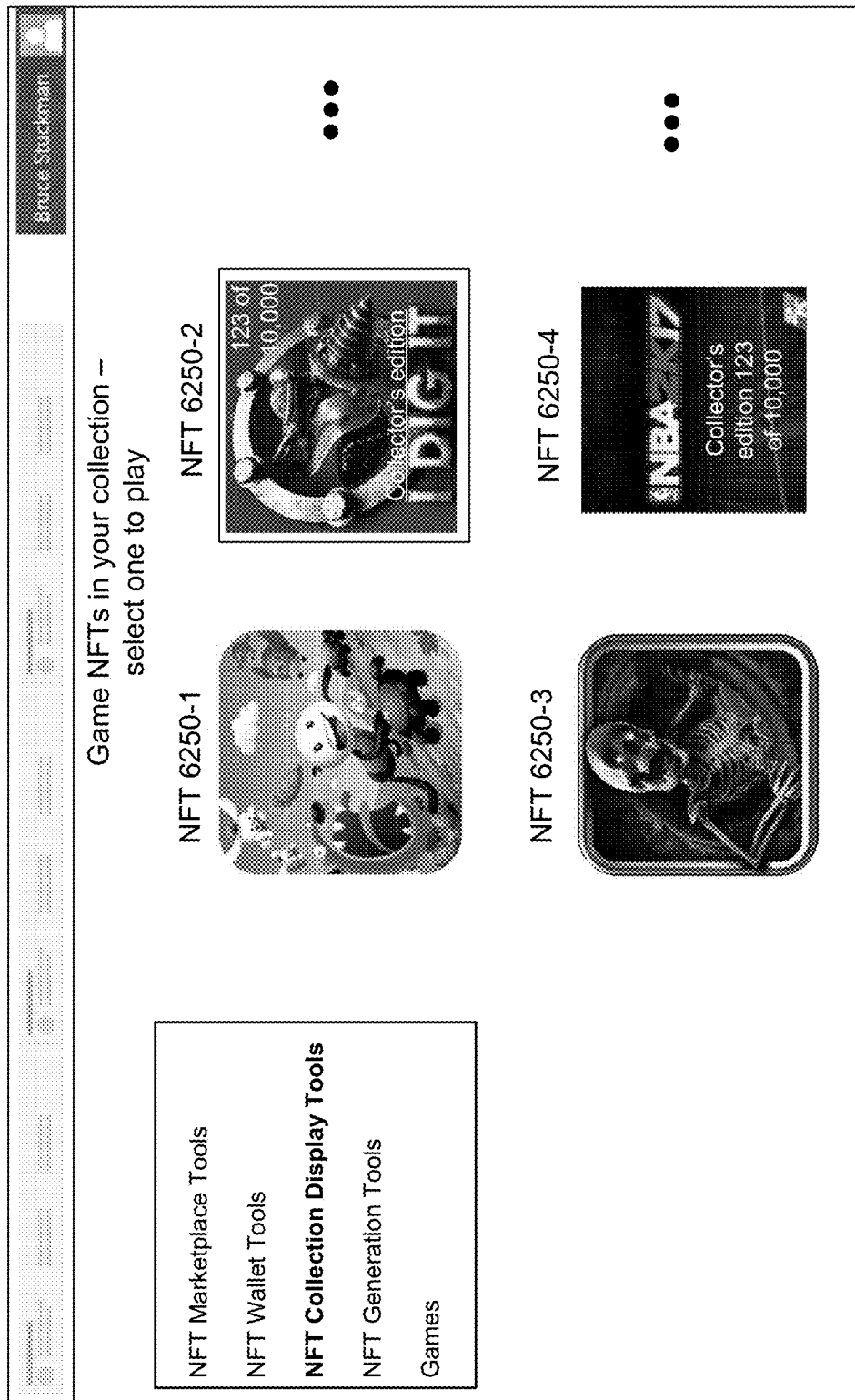


FIG. 13C

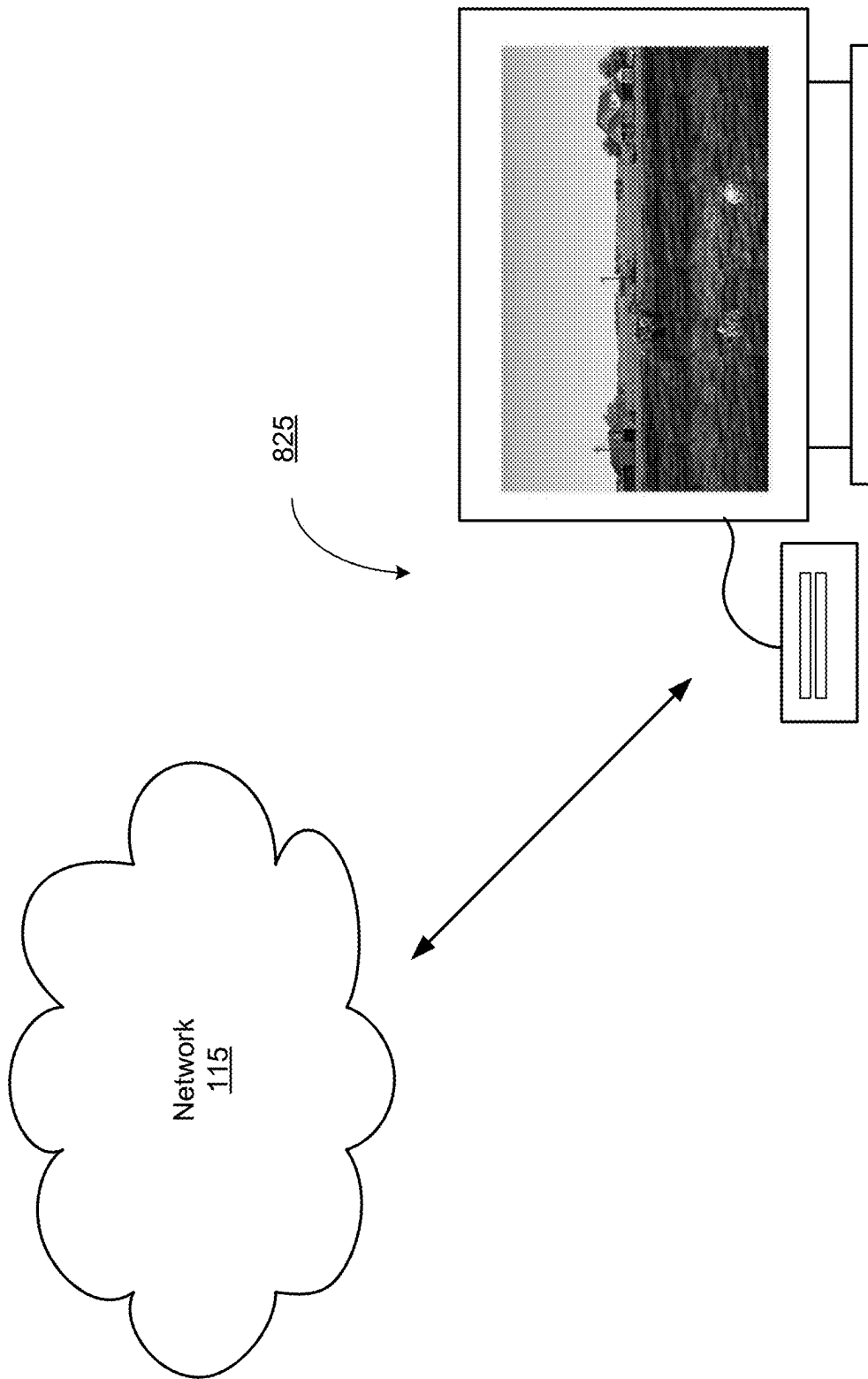
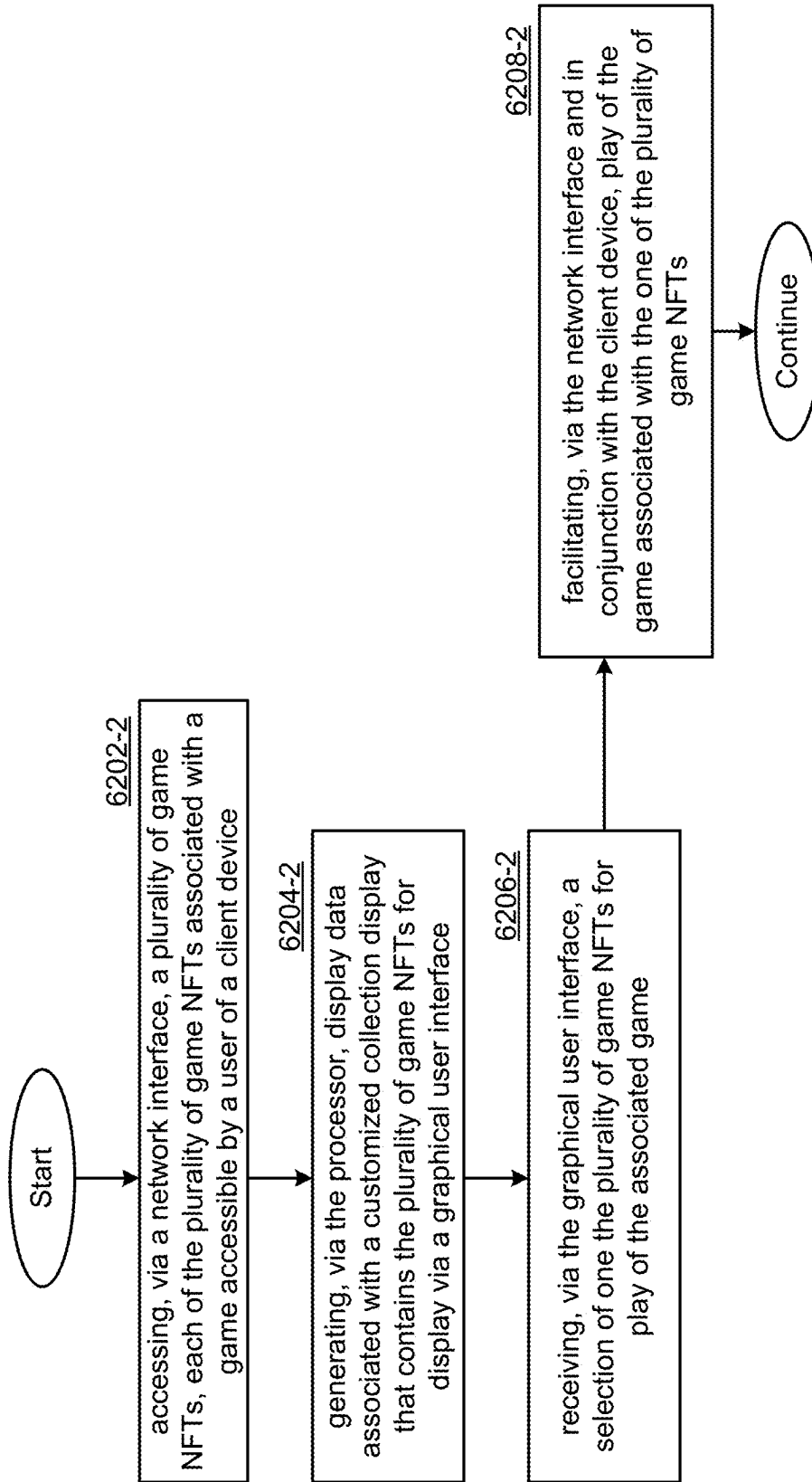


FIG. 13D

6200-2**FIG. 13E**

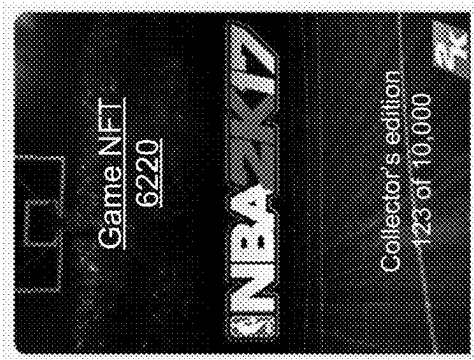


FIG. 13G

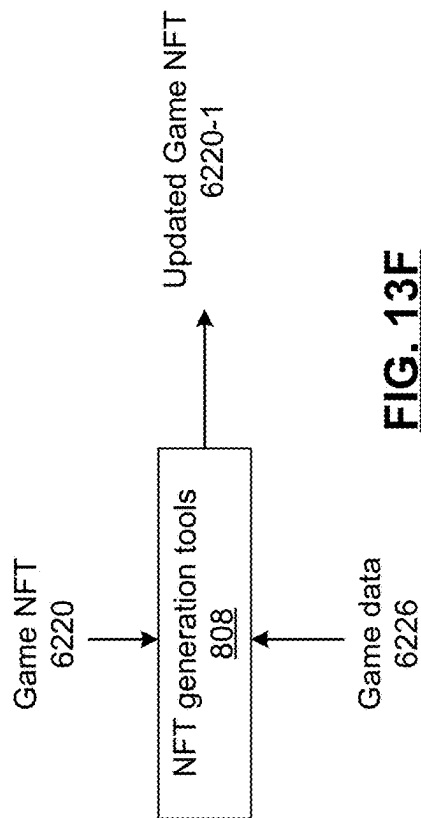
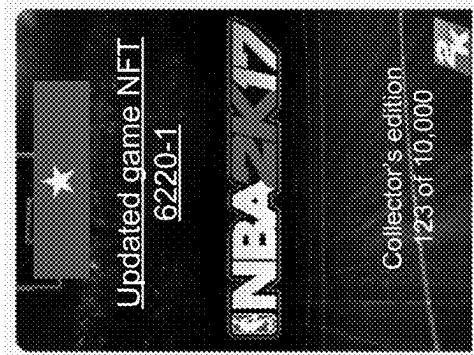


FIG. 13F

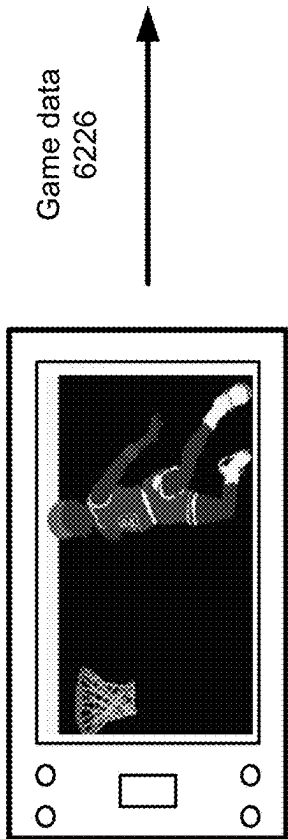
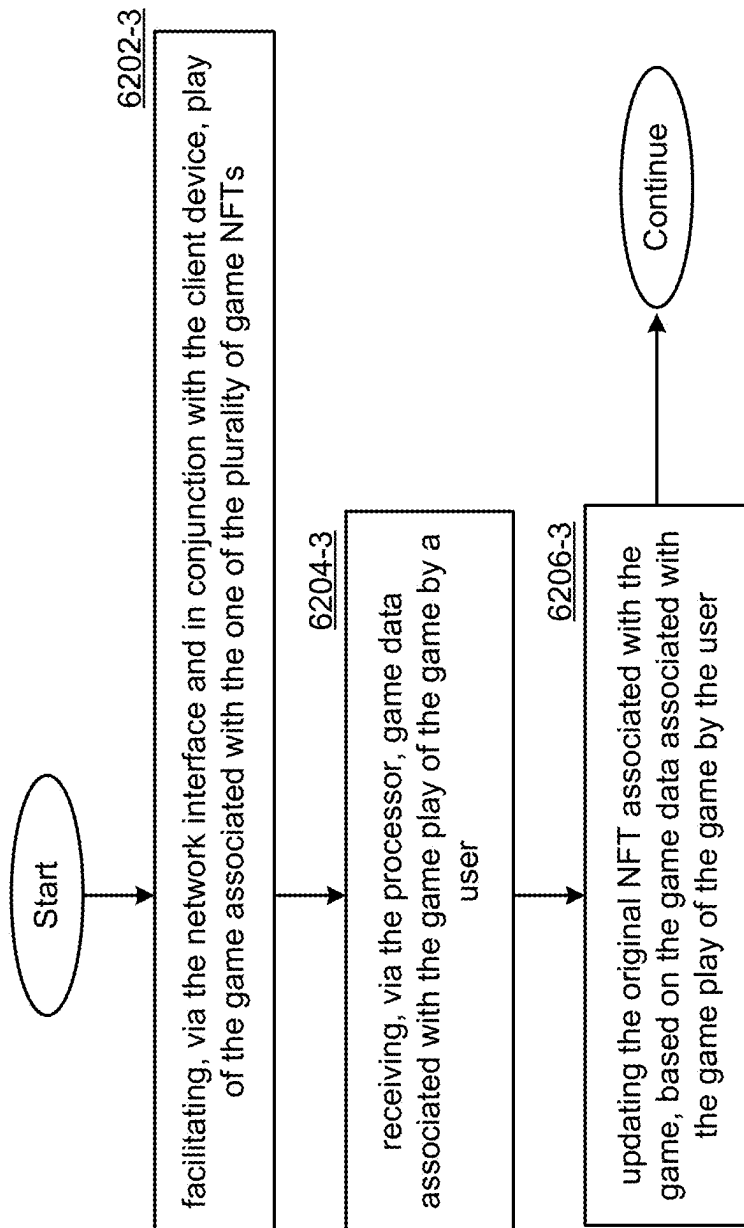


FIG. 13H

6200-3**FIG. 13I**

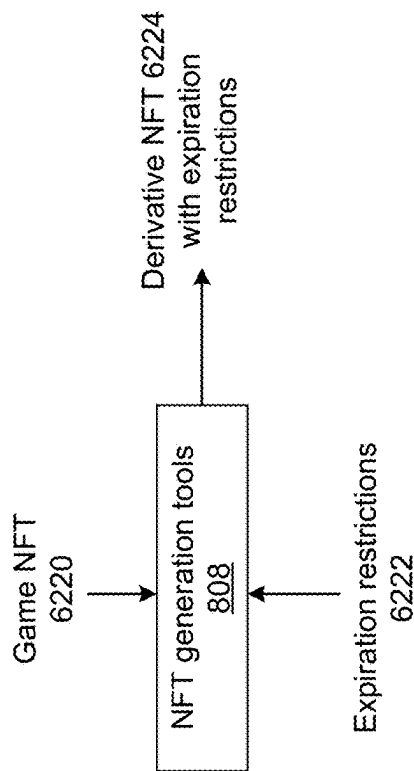


FIG. 13J



FIG. 13K

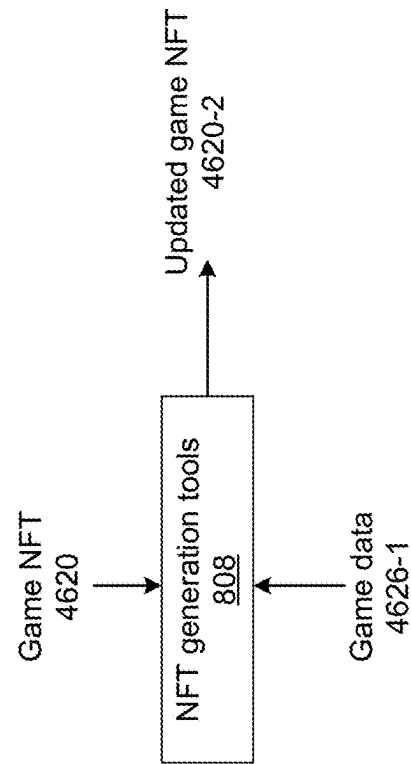


FIG. 13M

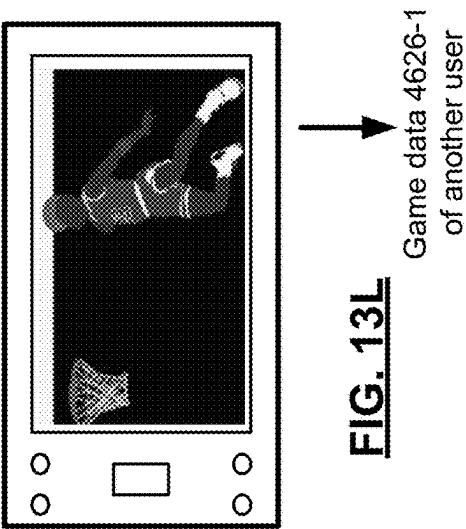
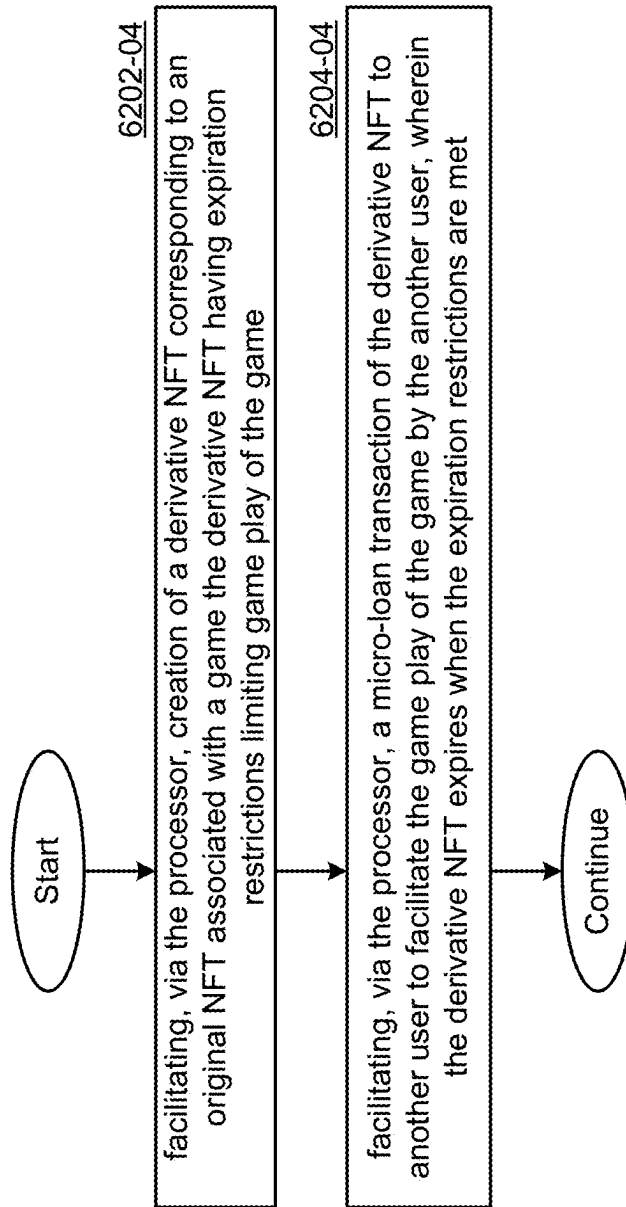
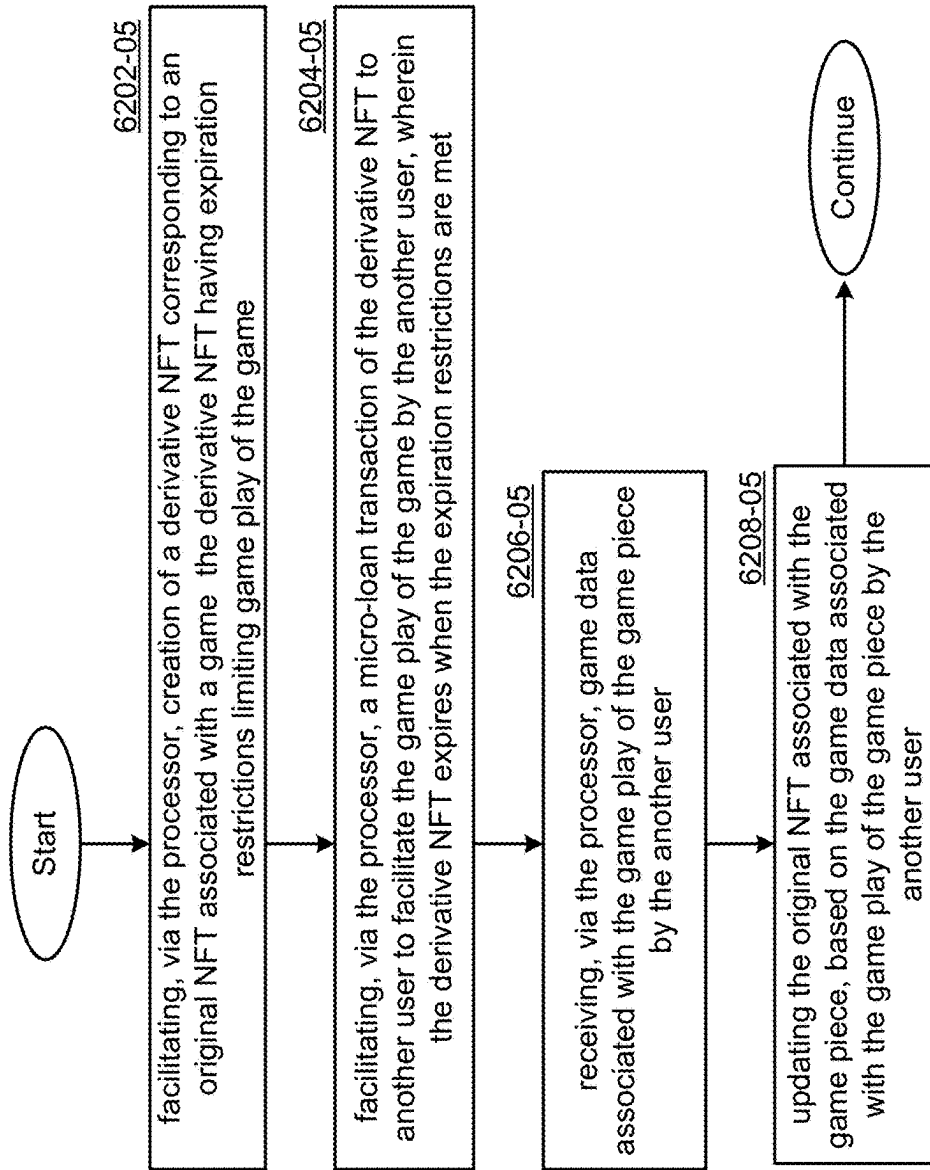


FIG. 13L



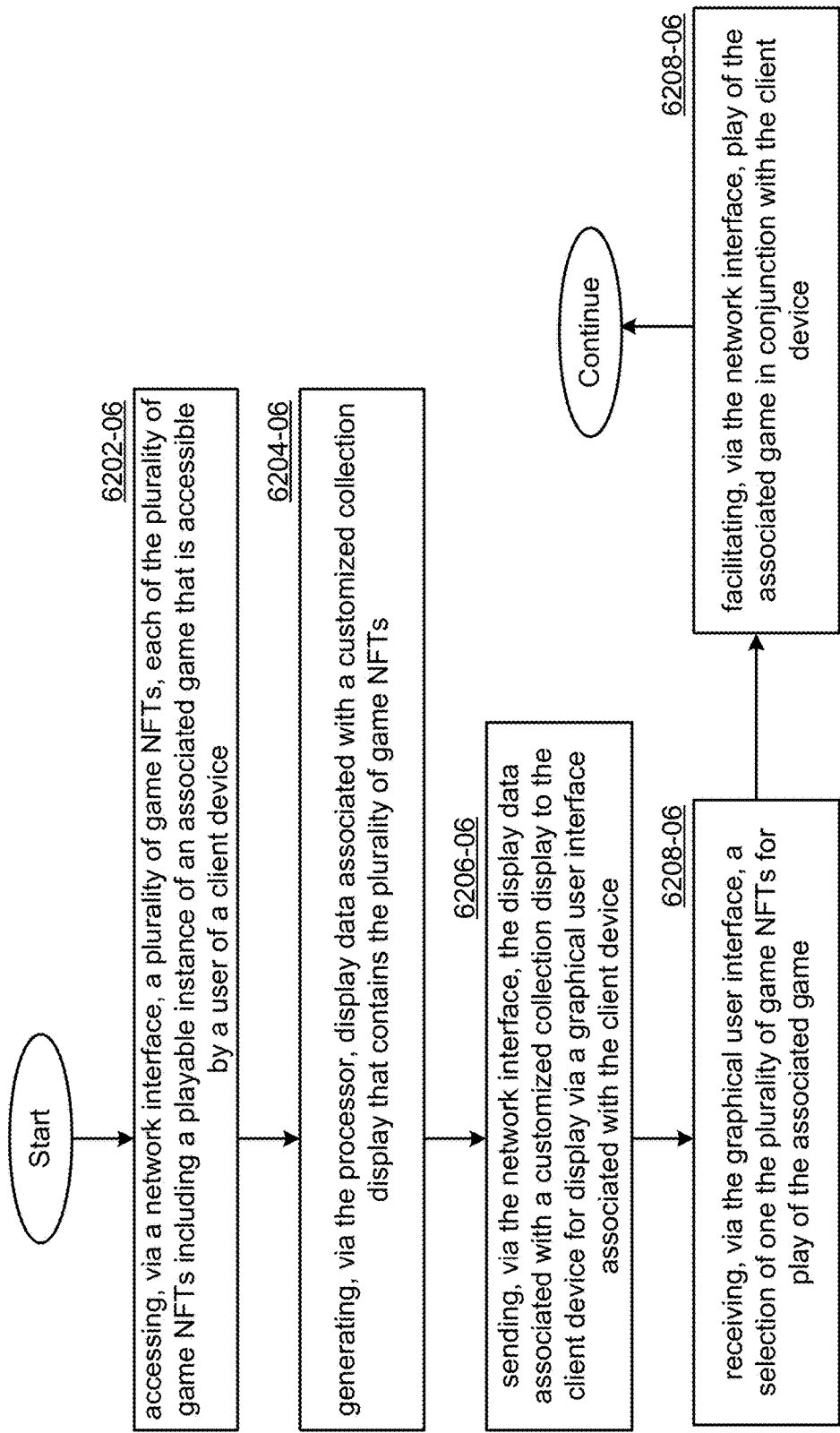
6200-04

FIG. 13N



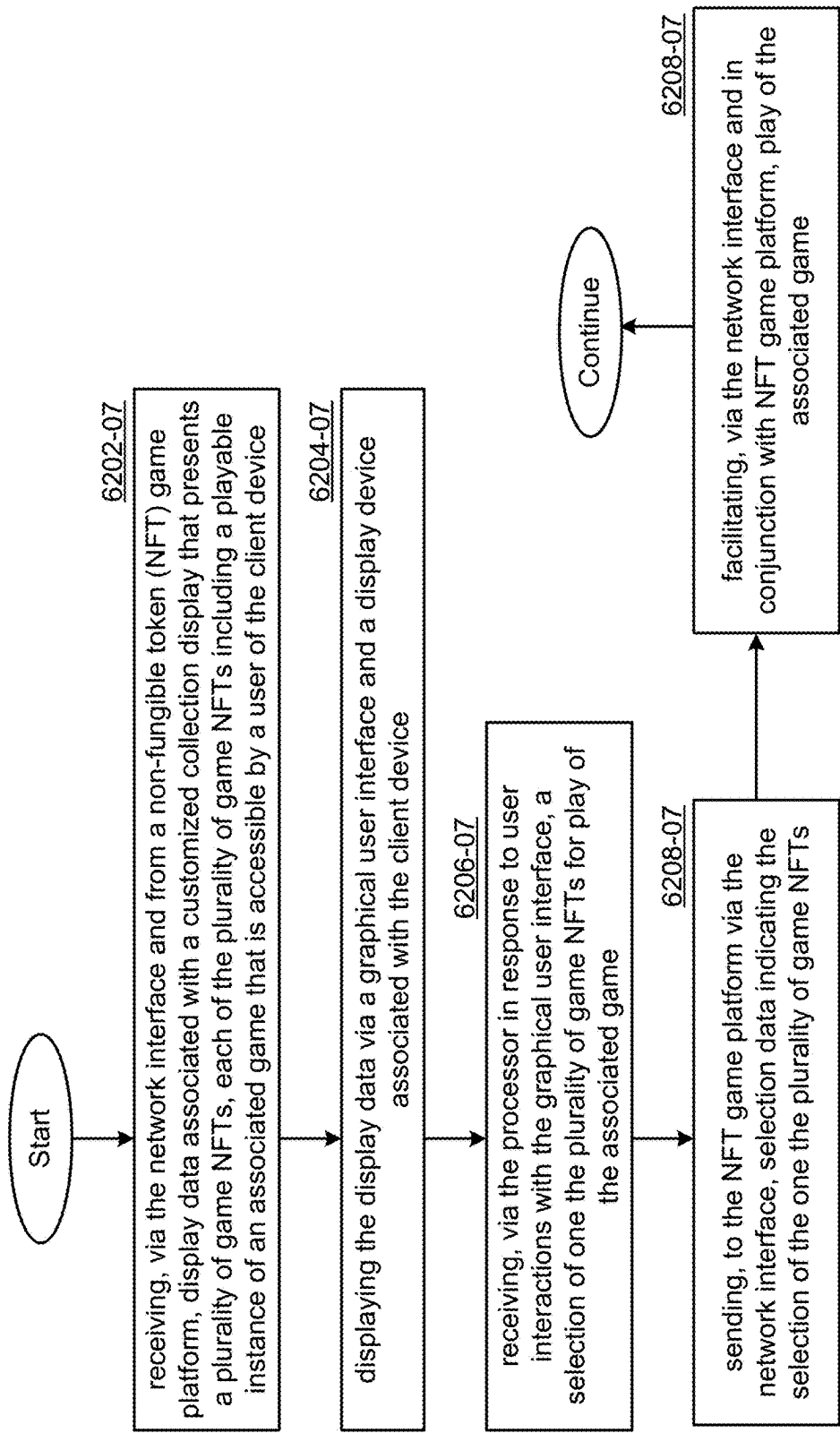
6200-05

FIG. 130



6200-06

FIG. 13P



6200-07

FIG. 13Q

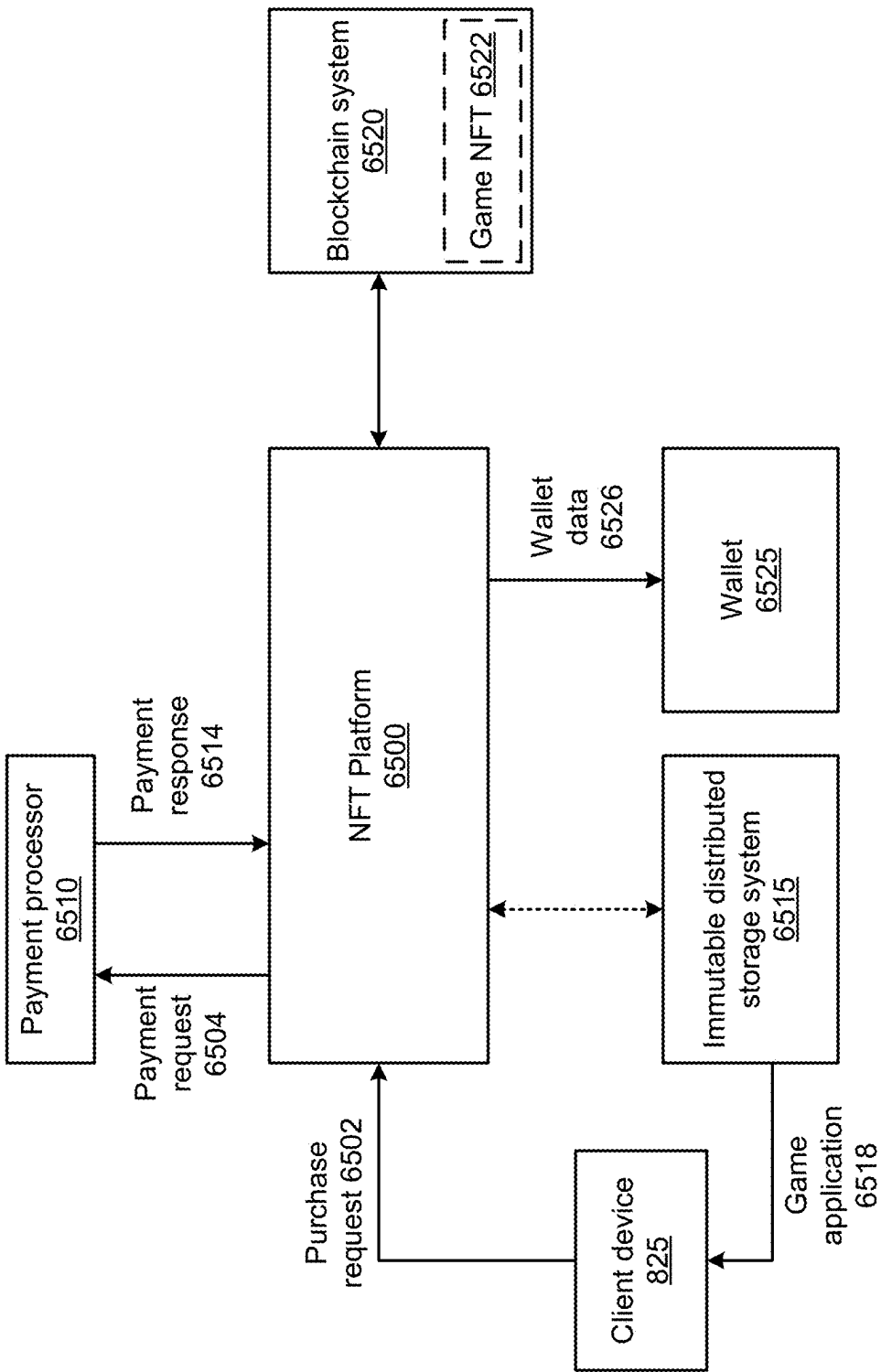
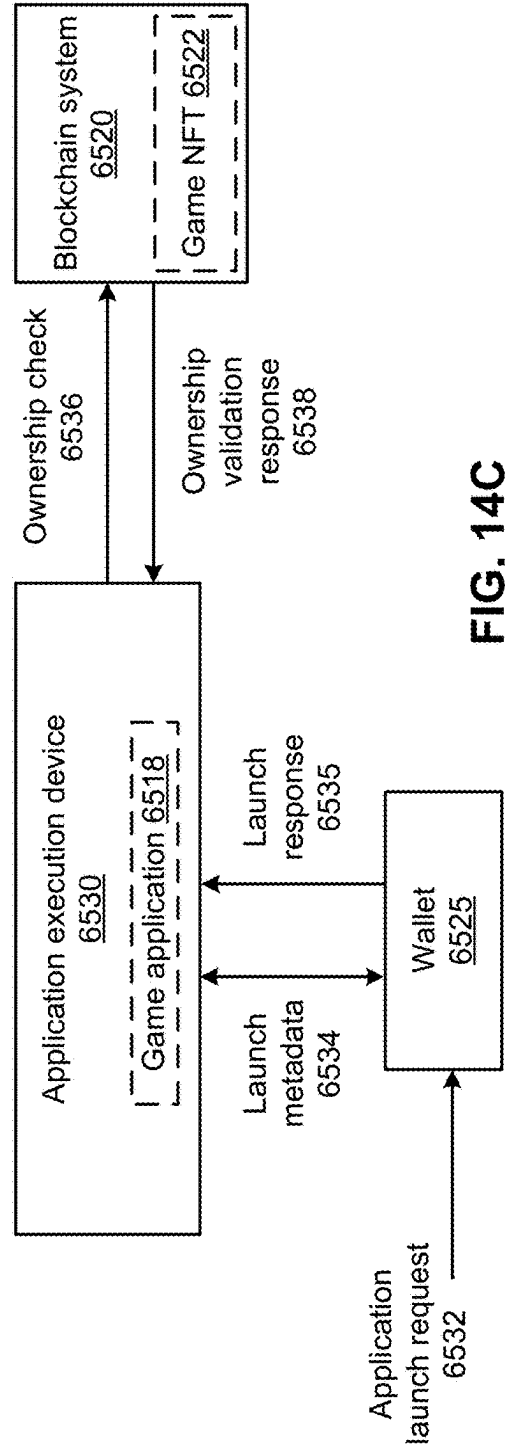
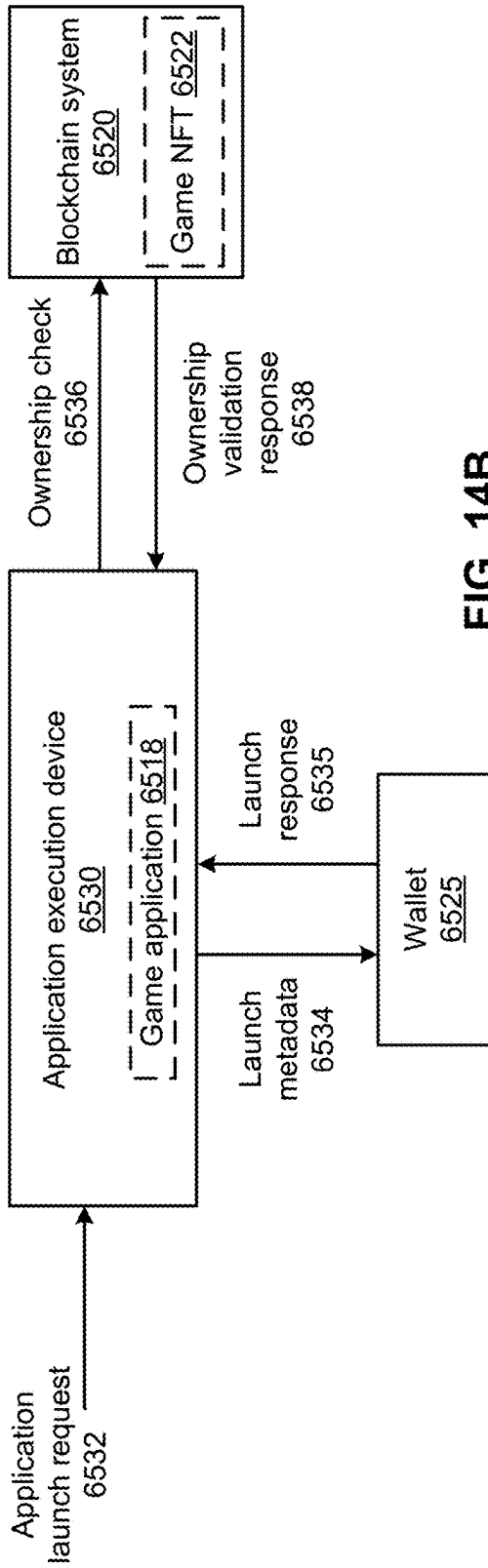


FIG. 14A



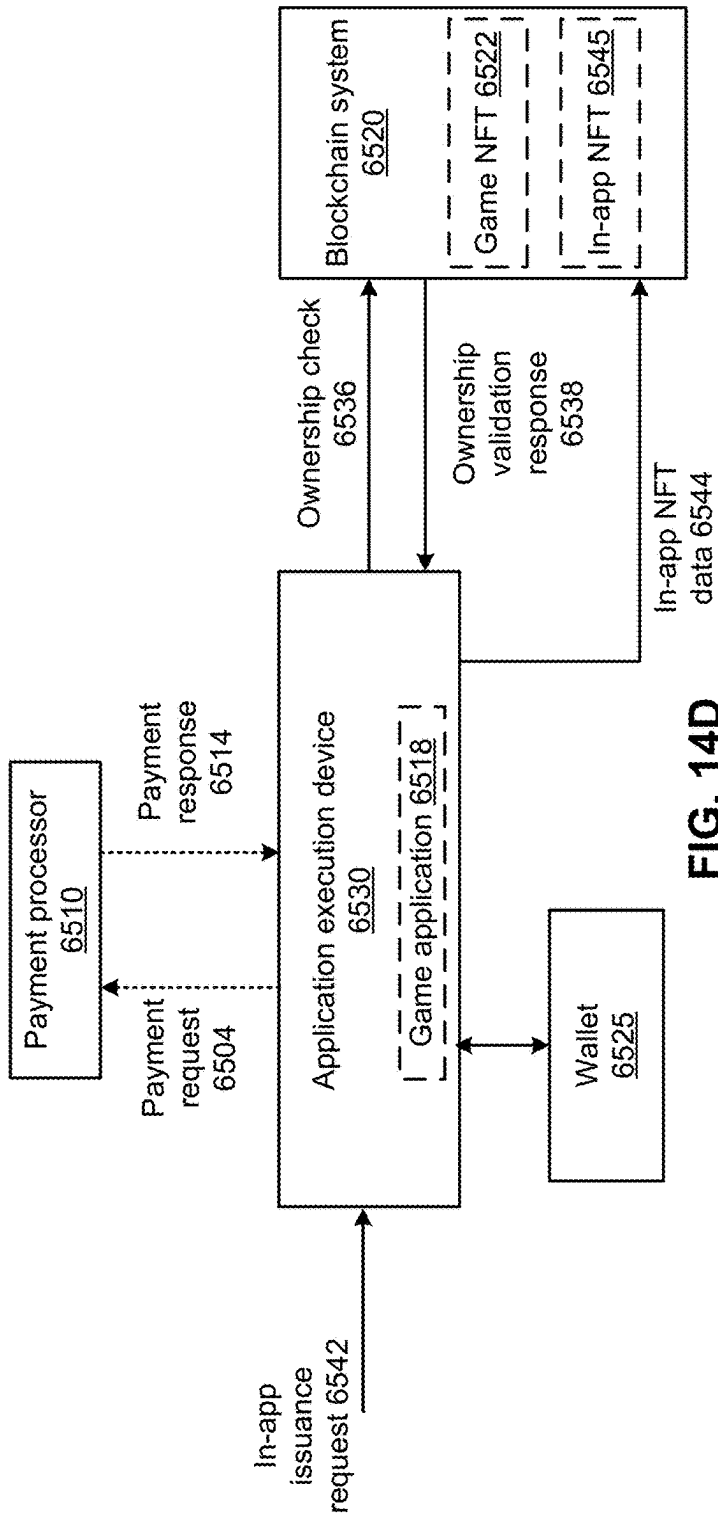


FIG. 14D

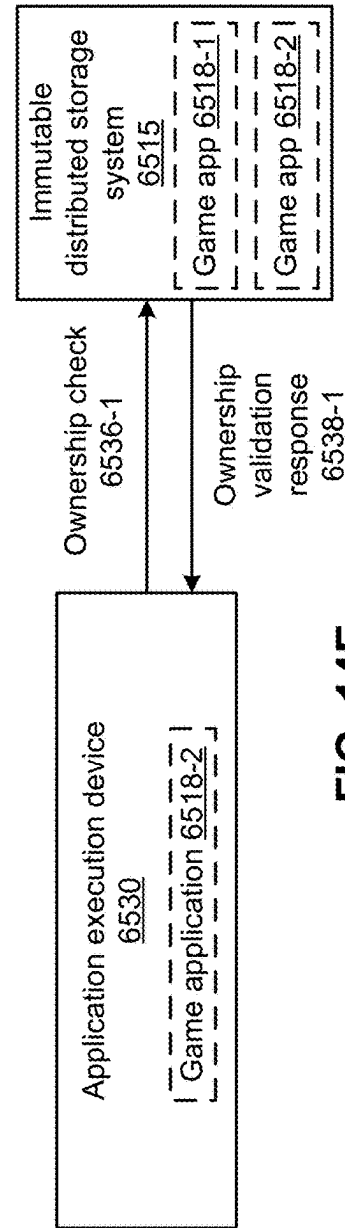


FIG. 14E

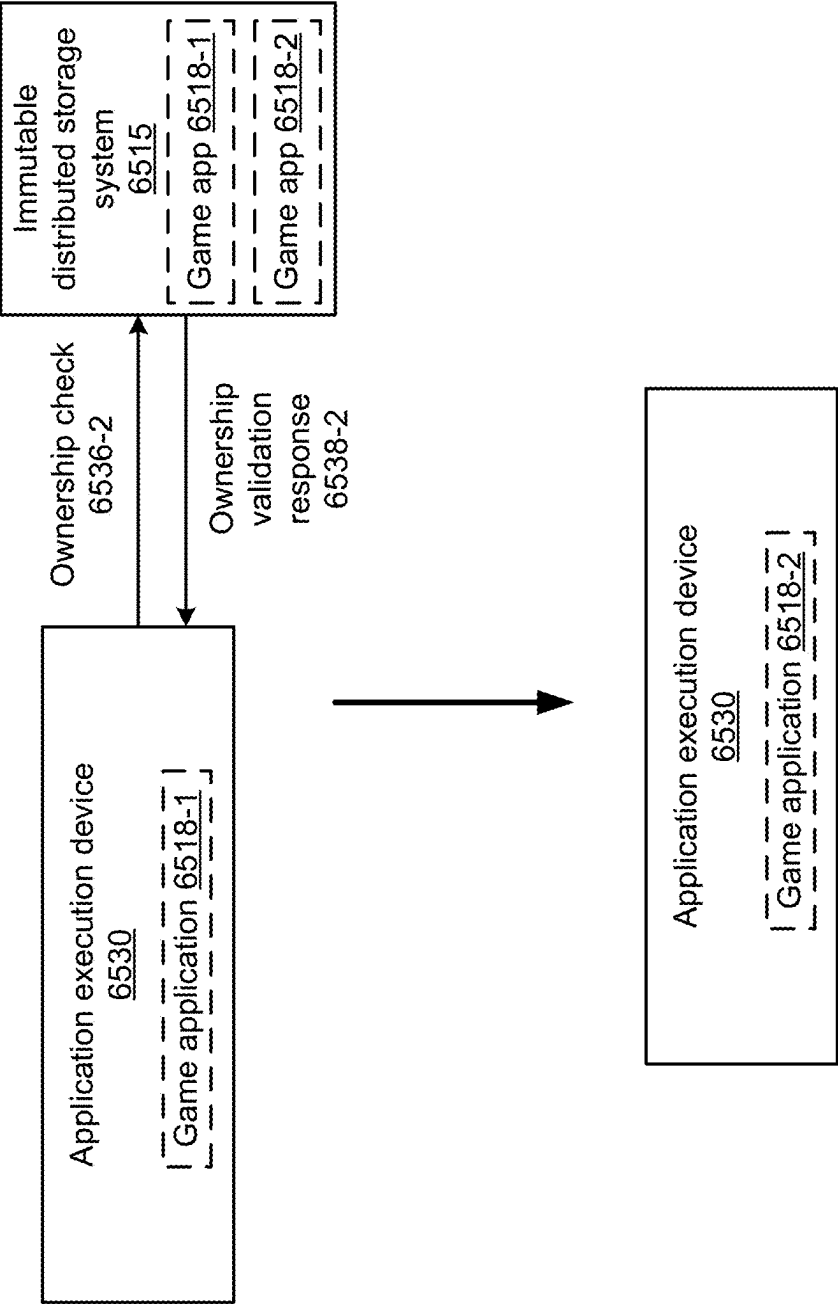
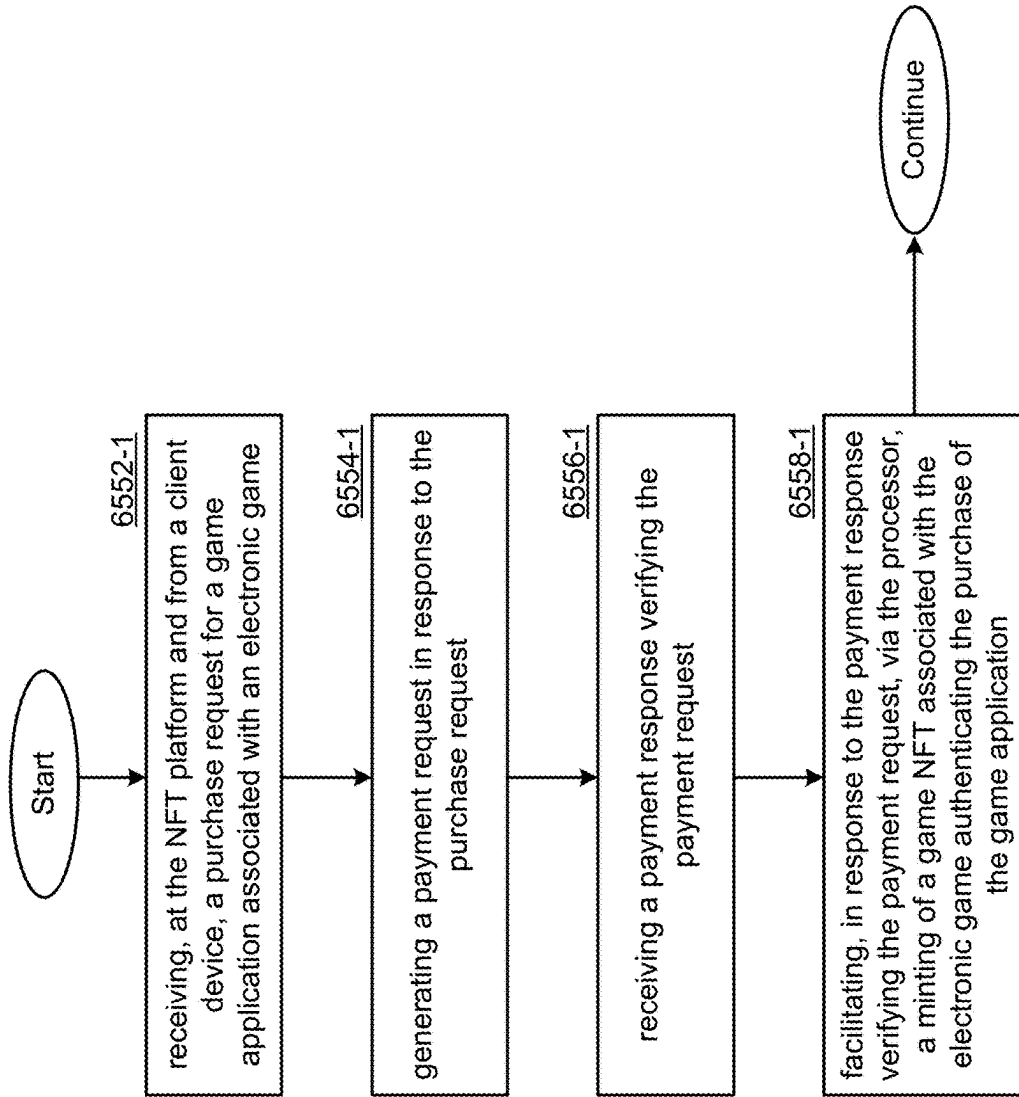
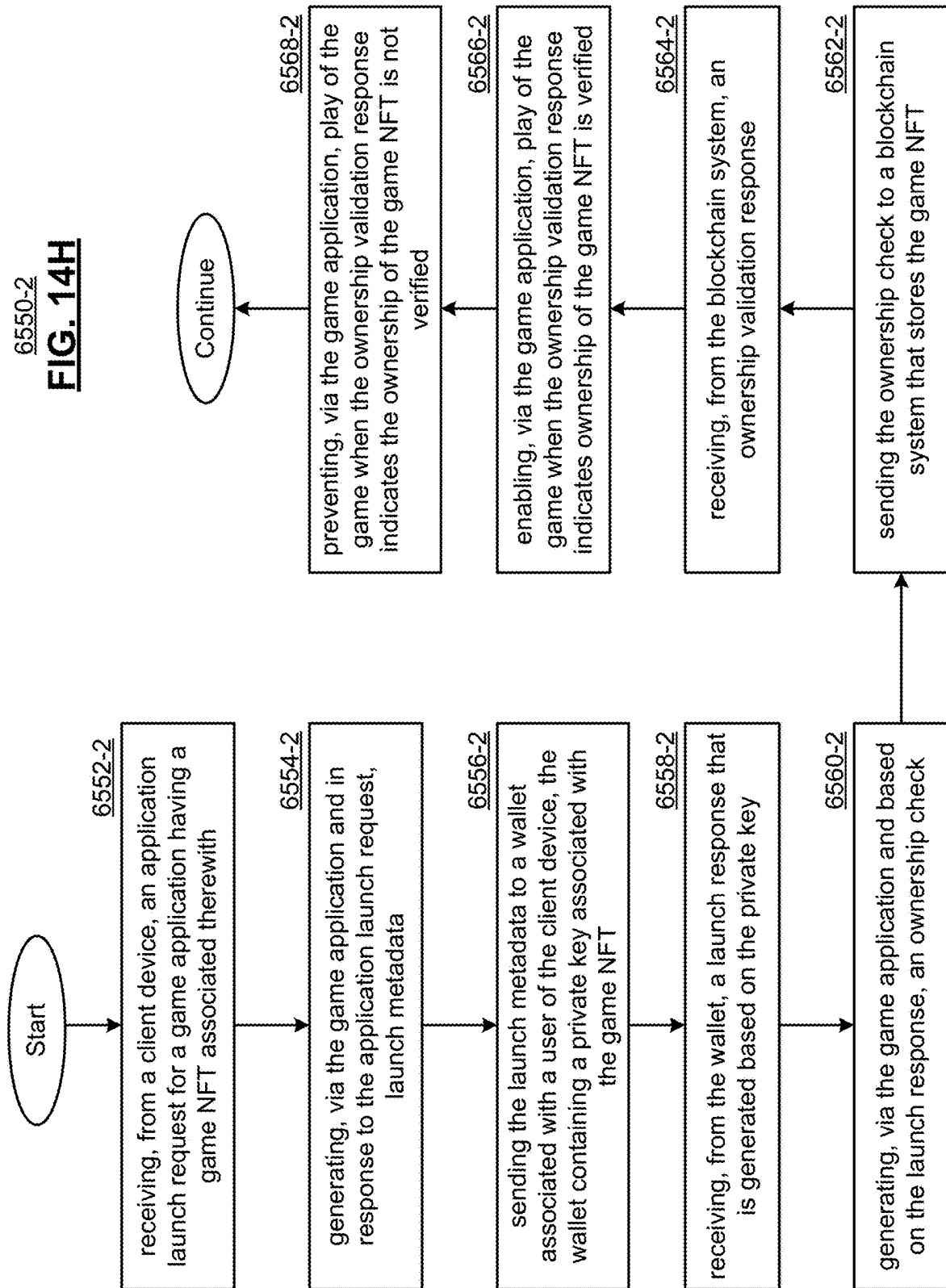
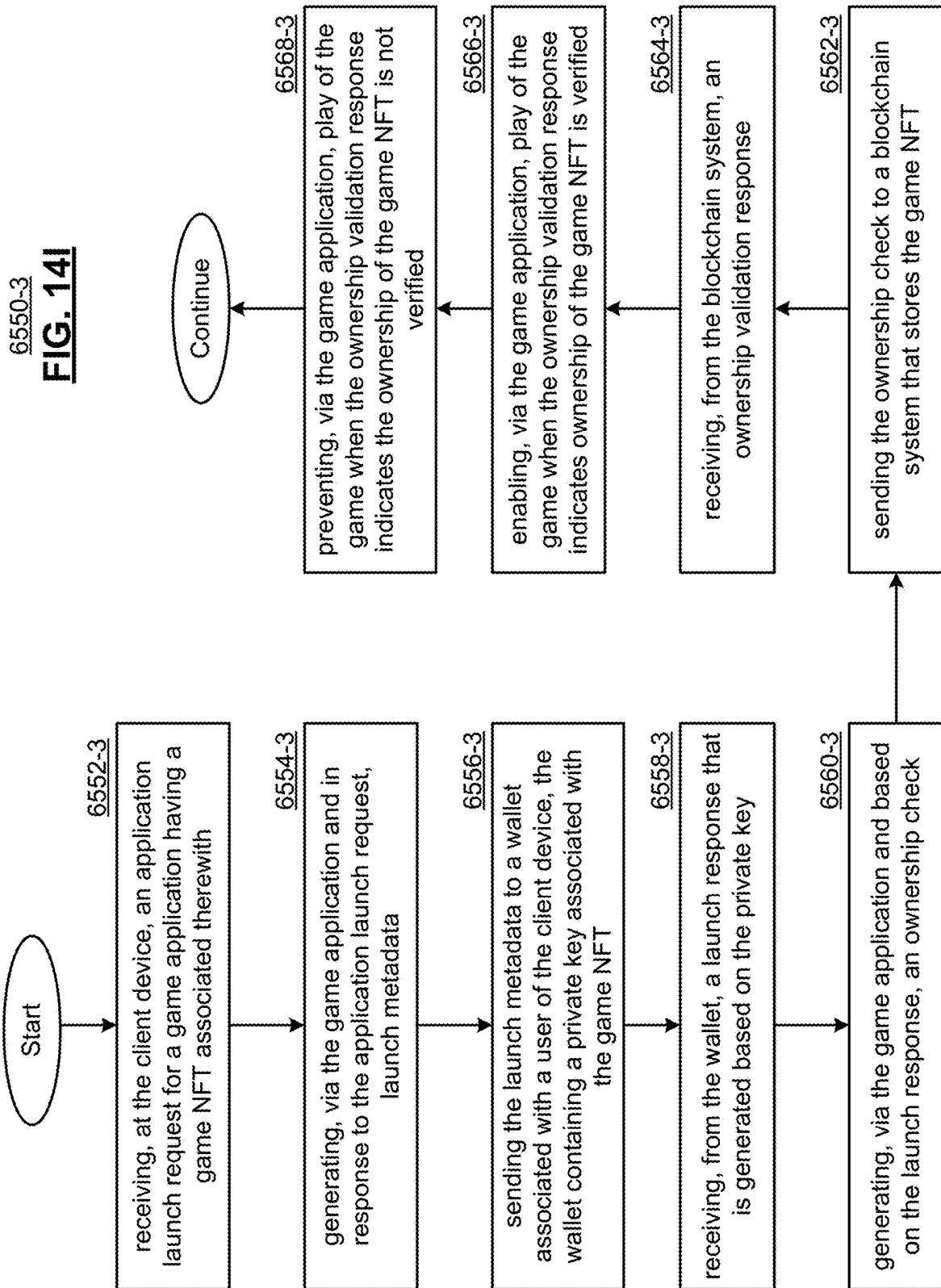


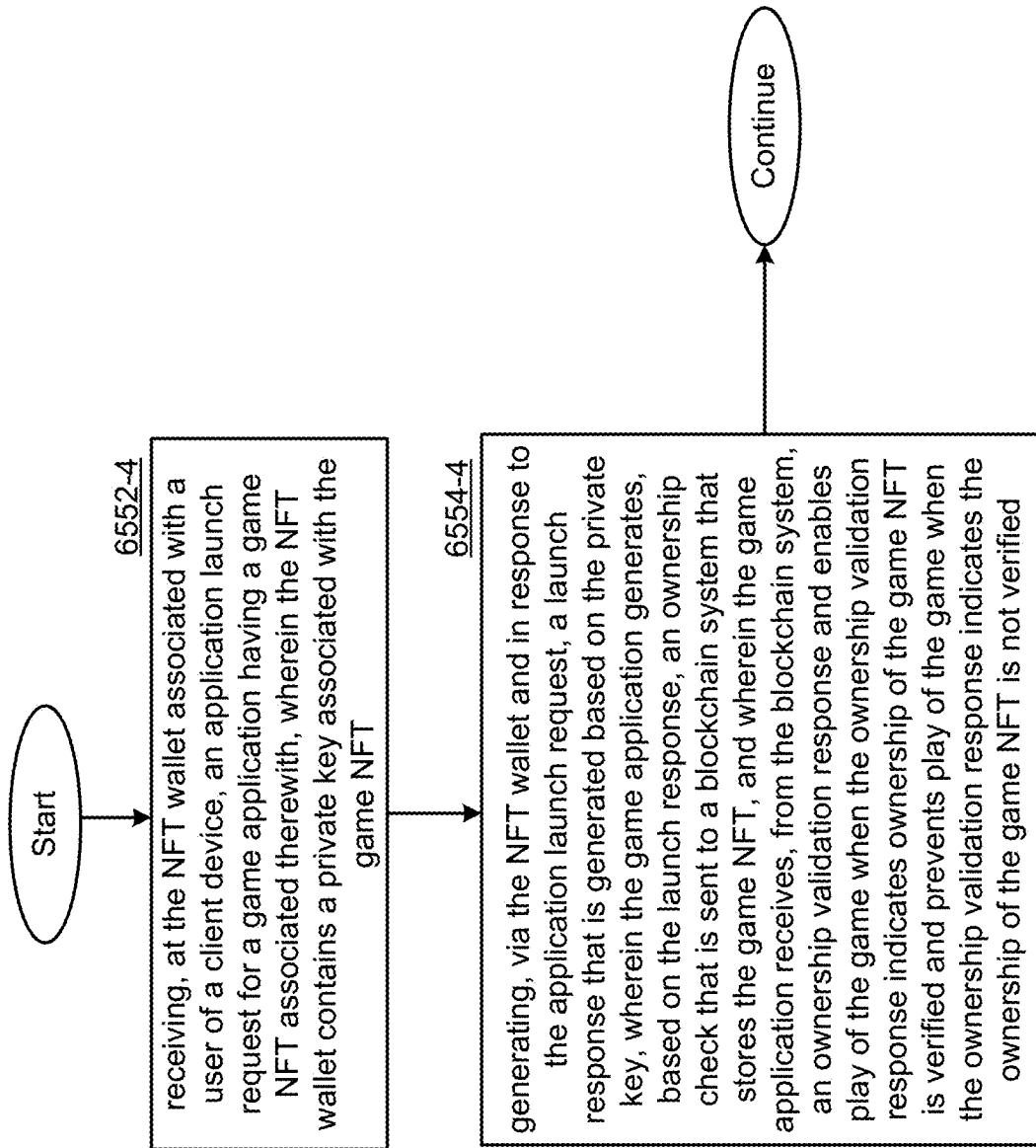
FIG. 14F

6550-1**FIG. 14G**

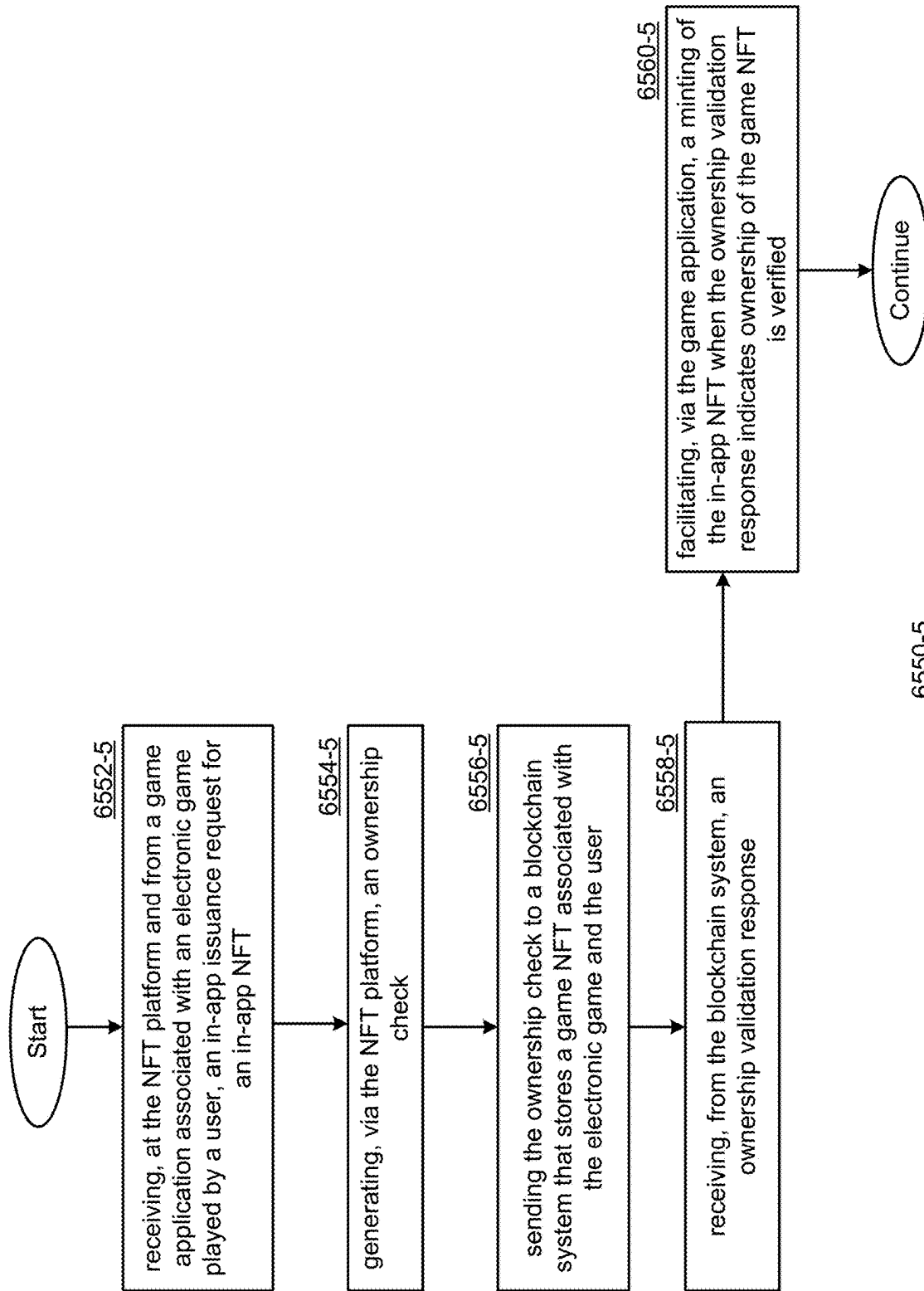
6550-2
FIG. 14H





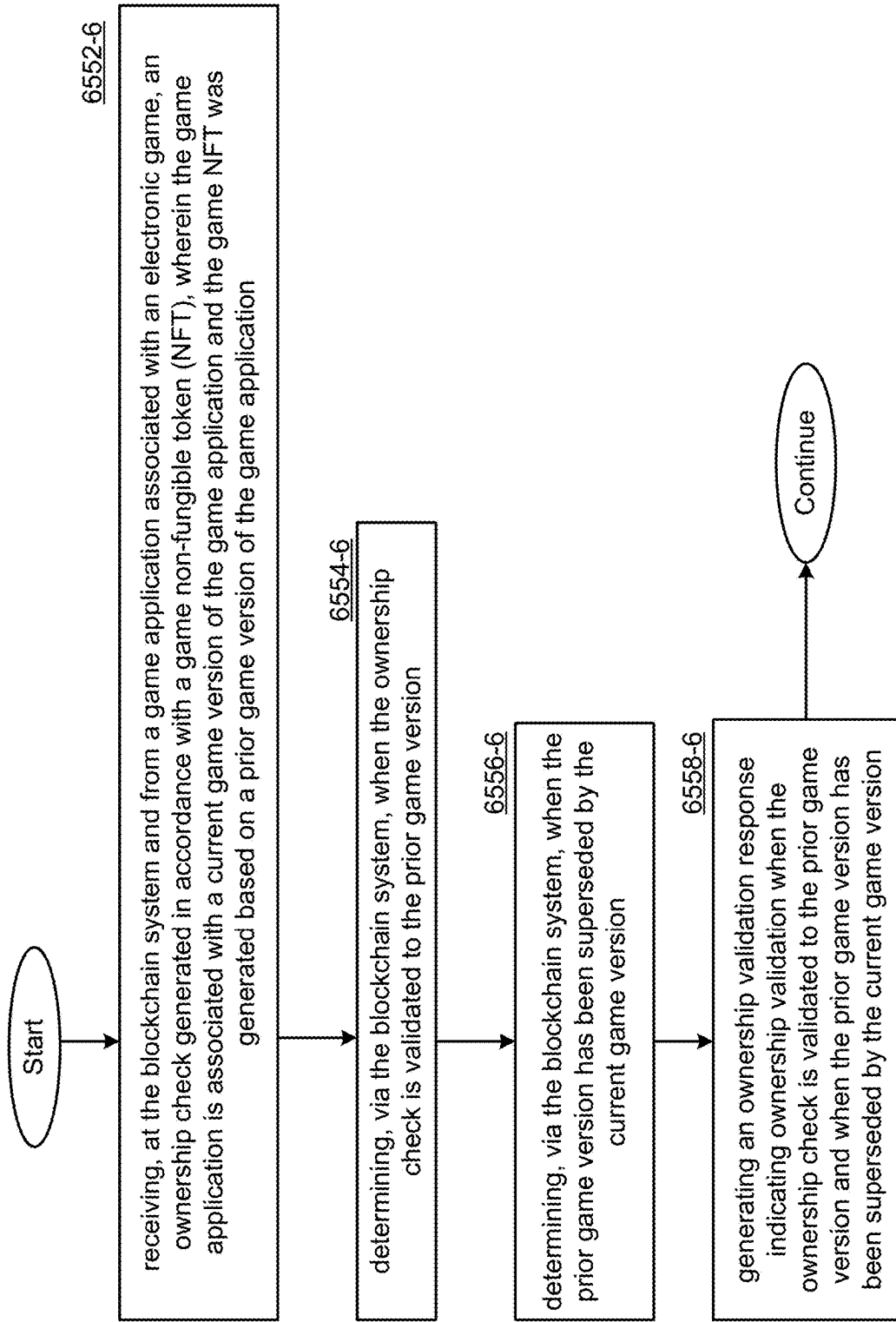


6550-4
FIG. 14J



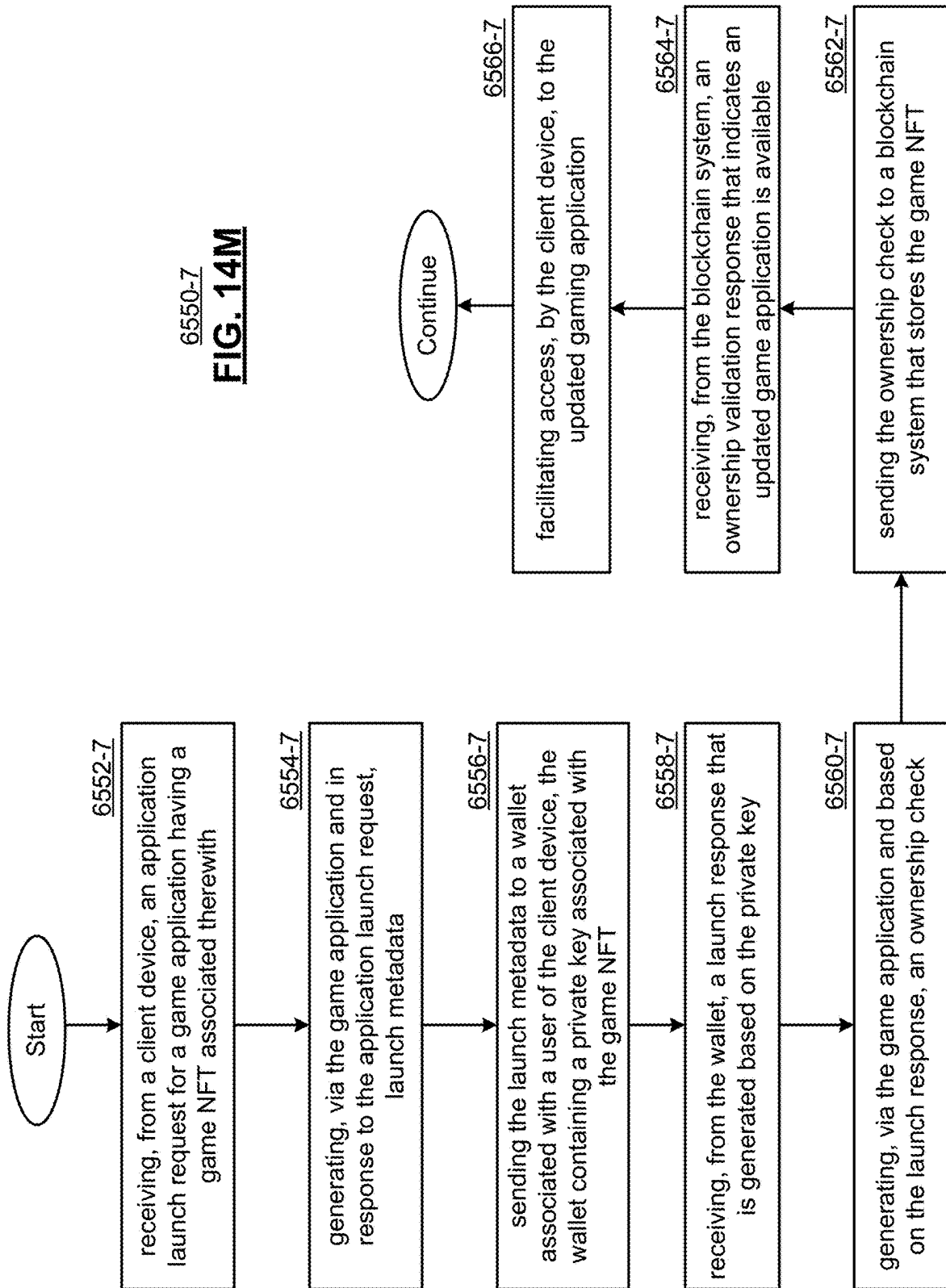
6550-5

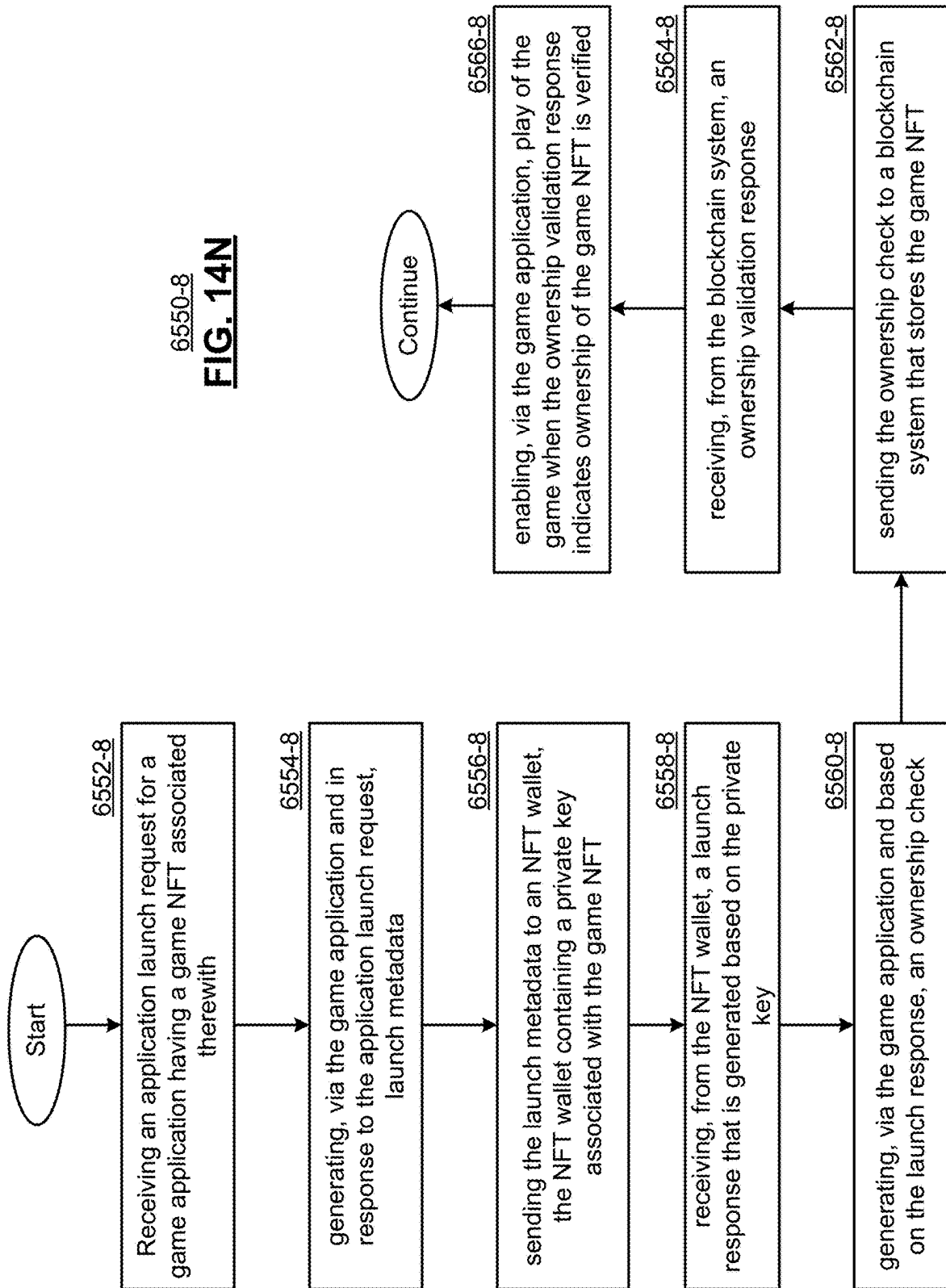
FIG. 14K



6550-6

FIG. 14L





METHOD FOR GENERATING NFTS IN RESPONSE TO IN-APP ISSUANCE REQUESTS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present U.S. Utility Patent Application claims priority pursuant to 35 U.S.C. § 120 as a continuation of U.S. Utility application Ser. No. 18/886,252, entitled “SYSTEM AND METHOD FOR VALIDATING GAME APPLICATION VERSIONS VIA A GAME NFT”, filed Sep. 16, 2024, which is a continuation of U.S. Utility application Ser. No. 17/661,362, entitled “SYSTEM FOR VALIDATING PLAY OF GAME APPLICATIONS VIA GAME NFTS AND METHODS FOR USE THEREWITH”, filed Apr. 29, 2022, issued as U.S. Pat. No. 12,121,820 on Oct. 22, 2024, which claims priority pursuant to 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/362,581, entitled “SYSTEM FOR VALIDATING PLAY OF GAME APPLICATIONS VIA GAME NFTS AND METHODS FOR USE THEREWITH”, filed Apr. 6, 2022, each of which are hereby incorporated herein by reference in their entirety and made part of the present U.S. Utility Patent Application for all purposes.

U.S. Utility application Ser. No. 17/661,362 also claims priority pursuant to 35 U.S.C. § 120 as a continuation-in-part of U.S. Utility application Ser. No. 17/656,084, entitled “FACILITATING PLAY OF GAME NETS VIA A CLIENT DEVICE”, filed Mar. 23, 2022, which claims priority pursuant to 35 U.S.C. § 119(e) to 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/262,536, entitled “NFT COLLECTION PLATFORM AND METHODS FOR USE THEREWITH”, filed Oct. 14, 2021; U.S. Provisional Application No. 63/262,762, entitled “NFT COLLECTION PLATFORM FOR PROCESSING NFTS AND DATA AND METHODS FOR USE THEREWITH”, filed Oct. 20, 2021; U.S. Provisional Application No. 63/262,858, entitled “NFT COLLECTION PLATFORM FOR PROCESSING NFTS AND DATA AND METHODS FOR USE THEREWITH”, filed Oct. 21, 2021; U.S. Provisional Application No. 63/263,844, entitled “NFT COLLECTION PLATFORM FOR RENDERING NFTS IN AN ENVIRONMENT AND METHODS FOR USE THEREWITH”, filed Nov. 10, 2021; U.S. Provisional Application No. 63/264,811, entitled “CLIENT DEVICE FOR NFTS AND METHODS FOR USE THEREWITH”, filed Dec. 2, 2021; U.S. Provisional Application No. 63/265,754, entitled “REAL ESTATE NFTS AND METHODS FOR USE THEREWITH”, filed Dec. 20, 2021; U.S. Provisional Application No. 63/266,478, entitled “AUTHENTICATED LICENSE NFTS AND METHODS FOR USE THEREWITH”, filed Jan. 6, 2022; U.S. Provisional Application No. 63/297,394, entitled “GENERATION OF ENHANCEMENT NFTS AND METHODS FOR USE THEREWITH”, filed Jan. 7, 2022; U.S. Provisional Application No. 63/302,757, entitled “GENERATION OF NFT GALLERIES AND METHODS FOR USE THEREWITH”, filed Jan. 25, 2022; U.S. Provisional Application No. 63/302,768, entitled “ENHANCEMENT OF REAL ESTATE NETS VIA NFT GALLERIES”, filed Jan. 25, 2022; U.S. Provisional Application No. 63/305,559, entitled “AUTHENTICATING NFT TRANSACTIONS VIA CONDITIONAL NFTS”, filed Feb. 1, 2022; U.S. Provisional Application No. 63/305,505, entitled “GENERATING A SECURE REAL-TIME NFT REPOSITORY FOR AUTHENTICATING NFT TRANSACTIONS”, filed Feb. 1, 2022; U.S. Provisional Application No. 63/306,405, entitled “GENERATING NFTS WITH FRACTIONAL

RIGHTS IN ASSETS”, filed Feb. 3, 2022; U.S. Provisional Application No. 63/306,412, entitled “GENERATING NFTS WITH FRACTIONAL RIGHTS IN A COLLECTABLE”, filed Feb. 3, 2022; U.S. Provisional Application No. 63/307,349, entitled “STREAMING OF VIDEO NFTS TO A CLIENT DEVICE”, filed Feb. 7, 2022; U.S. Provisional Application No. 63/308,546, entitled “NODE NFTS WITH TRANSACTION UPDATES”, filed Feb. 10, 2022; U.S. Provisional Application No. 63/309,382, entitled “GENERATING PLAYABLE GAME NFTS”, filed Feb. 11, 2022; and U.S. Provisional Application No. 63/310,376, entitled “GENERATING UPDATED TEAM NFTS BASED ON TEAM DATA”, filed Feb. 15, 2022, each of which are hereby incorporated herein by reference in their entirety and made part of the present U.S. Utility Patent Application for all purposes.

TECHNICAL FIELD

The present disclosure relates to processing systems and applications used in the collection, generation, display and use of non-fungible tokens (NFTs) or other digital tokens.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. 1A presents a block diagram representation of an example system.

FIG. 1B presents a block diagram representation of an example non-fungible token (NFT) collection platform.

FIG. 2 presents a block diagram representation of an example client device.

FIG. 3A presents a flowchart representation of an example method.

FIGS. 3B-3F and 3H present pictorial representations of example screen displays.

FIG. 3G presents a flow diagram representation of an example process.

FIG. 3I presents a flowchart representation of an example method.

FIG. 4A presents a flowchart representation of an example method.

FIG. 4B presents a pictorial representation of an example screen display.

FIG. 4C presents a flow diagram representation of an example process.

FIG. 5A presents a flowchart representation of an example method.

FIGS. 5B and 5C present pictorial representations of example screen displays.

FIG. 6 presents a block diagram/flow representation of an example of NFT generation.

FIG. 7 presents a block diagram representation of an example system.

FIG. 8 presents a block diagram representation of an example client device.

FIG. 9A presents a block diagram/flow representation of an example of NFT generation.

FIG. 9B presents a pictorial representation of an example NFT.

FIG. 9C presents a pictorial representation of an example improvement.

FIG. 9D presents a pictorial representation of an example NFT.

FIG. 9E presents a flowchart representation of an example method.

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FIG. 10A presents a block diagram/flow representation of an example of NFT generation.

FIG. 10B presents a flowchart representation of an example method.

FIG. 11A presents a block diagram/flow representation of an example of NFT generation.

FIG. 11B presents a pictorial representation of an example NFT.

FIG. 11C presents a pictorial representation of example enhancements.

FIG. 11D presents a pictorial representation of an example NFT.

FIG. 11E presents a flowchart representation of an example method.

FIG. 12A presents a block diagram of an example system. FIG. 12B presents a flowchart representation of an example method.

FIG. 12C presents a pictorial block diagram/flow representation of an example of transaction authentication.

FIG. 12D presents a flowchart representation of an example method.

FIG. 13A presents a pictorial representation of an example NFT.

FIG. 13B presents a flowchart representation of an example method.

FIG. 13C presents a pictorial representation of an example screen display.

FIG. 13D presents a pictorial/block diagram representation of an example system with screen display.

FIG. 13E presents a flowchart representation of an example method.

FIG. 13F presents a block diagram/flow representation of an example of NFT generation.

FIG. 13G presents a pictorial representation of an example NFT.

FIG. 13H presents a pictorial/flow representation of an example update to an NFT.

FIG. 13I presents a flowchart representation of an example method.

FIG. 13J presents a block diagram/flow representation of an example of NFT generation.

FIG. 13K presents a pictorial representation of an example derivative NFT.

FIG. 13L presents a pictorial representation of an example client device.

FIG. 13M presents a block diagram/flow representation of an example of NFT generation.

FIG. 13N presents a flowchart representation of an example method.

FIG. 13O presents a flowchart representation of an example method.

FIG. 13P presents a flowchart representation of an example method.

FIG. 13Q presents a flowchart representation of an example method.

FIG. 14A presents a block diagram representation of an example system.

FIG. 14B presents a block diagram representation of an example system.

FIG. 14C presents a block diagram representation of an example system.

FIG. 14D presents a block diagram representation of an example system.

FIG. 14E presents a block diagram representation of an example system.

FIG. 14F presents a block/flow diagram representation of an example system.

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FIG. 14G presents a flowchart representation of an example method.

FIG. 14H presents a flowchart representation of an example method.

FIG. 14I presents a flowchart representation of an example method.

FIG. 14J presents a flowchart representation of an example method.

FIG. 14K presents a flowchart representation of an example method.

FIG. 14L presents a flowchart representation of an example method.

FIG. 14M presents a flowchart representation of an example method.

FIG. 14N presents a flowchart representation of an example method.

DETAILED DESCRIPTION

FIG. 1A presents a block diagram representation of an example system in accordance with various examples. In particular, a system **850** is presented that includes an NFT collection platform **800** that communicates with client devices **825** via a network **115**. The network **115** can be the Internet or other wide area or local area network, either public or private. The client devices **825** can be computing devices such as laptops, smartphones, smart watches, tablets, desktops, or other computing devices associated with users, for example, buyers, sellers, collectors and/or users of NFTs.

NFT creation and ownership is growing worldwide. Existing platforms provide tools to create NFTs, marketplaces for buying and selling NFTs and wallets to securely hold them. Many NFT purchasers however, view NFTs similarly to cryptocurrency, purely as financial investments to be collected and later sold. Unlike cryptocurrencies which are fungible tokens, NFTs have a non-fungible (e.g. unique or limited) component that can correspond to photographs, graphics, art, video, games or other media or imagery that can be displayed, used in authentication and/or to provide other support for transactions and/or used in game play and for other utilitarian purposes, etc.

In the example shown, the NFT collection platform **800** includes a client device interface **802** for interacting with the client devices **825**, NFT marketplace tools **804**, NFT wallet tools **806**, NFT generation tools **808**, NFT collection display tools **810**, one or more games **812** that can be played using NFTs that have been collected including game NFTs that include and/or otherwise associated with, or correspond to, playable game content, and a database **814** for storing user and account information, preferences, display settings and other data associated with users of the NFT collection platform **800**. The NFT collection platform **800** can include or be implemented via one or more servers, a cloud computing system, an InterPlanetary File System (IPFS) or other decentralized or distributed computer system of individual computers and/or nodes and/or a centralized computing system.

For example, the client device interface **802** can operate in conjunction with each client device **825** and via network **115** to generate a graphical user interface. This graphical user interface is based on display data generated by the NFT collection platform **800** in a format for display on a display device associated with the client devices **825**. This graphical user interface generates input data that is received by the NFT collection platform **800** from the client devices **825** in response to user interaction with the graphical user interface.

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In various examples, the NFT collection platform **800** can serve the needs of a collector of NFTs by providing a system that offers new experiences that support various collector and/or use behaviors in the NFT world and metaverse including the collection, display, and use of NFTs and their corresponding content. The NFT collection display tools **810** can include a high-fidelity customizable page builder so each collector can personalize and display their NFTs in an environment. The NFT collection platform **800** can provide and support a collection social media site that, for example, is similar to MySpace, Facebook, or OnlyFans, but specifically directed to the arrangement, posting, sharing and/or display of collections of NFTs.

In the most basic mode of operation, the NFT collection platform **800** improves the technology of NFT systems by allowing the user to collect, arrange and display NFTs they have purchased so that the user, and other users, can enjoy and admire the NFTs they have collected. The NFT collection platform **800** not only allows users to curate, display and use their collections, to play games associated with their NFTs and/or also allows users to create a social/professional profile of their NFT property—enabling users to get creative with how their collections are presented, used and displayed. In addition, when a group of NFTs is collected and/or when a personalized/curated collection page, set and/or collection display is built, the NFT collection platform **800** improves the technology of NFT systems by allowing the user to mint a new “collection NFT” using blockchain-based, distributed computer network, and/or other crypto-based NFT creation techniques, for example and authenticated based on the user’s ownership of the underlying NFTs.

Furthermore, the NFT collection platform **800** allows users to buy, sell, loan, borrow and trade NFTs with other users, including their own collection NFTs created based on their own collections. In various examples, the NFT collection platform **800** may not include a NFT creator, marketplace or wallet itself. The NFT generation tools **808**, NFT wallet tools **806** and NFT marketplace tools **804** can reside on the NFT collection platform **800** and interface with one or more NFT wallet systems **820**, NFT marketplaces **822** and/or NFT creation systems **824** to enable transactions/interactions/abilities at a metaverse level, offering collectors new and unique experiences, pre and post purchase, across all different NFT’s and blockchains in one place. The NFT collection platform **800** improves the technology of NFT systems by allowing the user to perform the functions above in a fashion, that can be blockchain, wallet, and marketplace agnostic.

It should be noted that the system **850** can operate via blockchain-based technologies. In blockchain, a growing list of records, called “blocks”, are linked together using cryptography and spread over a decentralized computer system/network or other distributed network of participants. Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data (generally represented as a Merkle tree). The timestamp proves that the transaction data existed when the block was published in order to get into its hash. As blocks each contain information about the block previous to it, they form a chain, with each additional block reinforcing the ones before it. Therefore, blockchains are resistant to modification of their data because once recorded, the data in any given block cannot be altered retroactively without altering all subsequent blocks. The distributed nature of this process over different nodes, the contemporaneous nature of geographically distinct calculations, coupled with the extreme computational complexity

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of the required calculations means that these blockchain-based technologies cannot practically be performed by the human mind.

It should be noted, that while the NFT wallet system **820**, NFT marketplace **822** and NFT creation system **824** are shown as separate entities and as being external to the NFT collection platform **800**, other configurations are possible where two or more of these entities share a common platform and/or the functions and features of one or more of these entities can be incorporated within the NFT collection platform **800** itself.

It should also be noted that while client device interface **802**, NFT marketplace tools **804**, NFT wallet tools **806**, NFT generation tools **808**, NFT collection display tools **810**, games **812** and database **814** are shown as being internal to the NFT collection platform **800**, in other examples, any subset of the various elements of the NFT collection platform **800** can be implemented external to the NFT collection platform **800** and coupled to the other components via the network **115**. Furthermore, the NFT collection platform **800** can be implemented in a cloud computing configuration with any or all of the various elements of the NFT collection platform **800** implemented within the cloud.

The further operation of this system will be described in greater detail in conjunction with the figures that follow, including many optional functions and features and examples thereof.

FIG. 1B presents a block diagram representation of an NFT collection platform **800** in accordance with various examples. In particular, the NFT collection platform **800** includes a network interface **820** such as a 3G, 4G, 5G or other cellular wireless transceiver, a Bluetooth transceiver, a WiFi transceiver, UltraWideBand transceiver, WIMAX transceiver, ZigBee transceiver or other wireless interface, a Universal Serial Bus (USB) interface, an IEEE 1394 Firewire interface, an Ethernet interface or other wired interface and/or other network card or modem for communicating via the network **115**.

The NFT collection platform **800** also includes a processing module **830** and memory module **840** that stores an operating system (O/S) **844** such as an Apple, Unix, Linux or Microsoft operating system or other operating system, client device interface **802**, NFT marketplace tools **804**, NFT wallet tools **806**, NFT generation tools **808**, NFT collection display tools **810**, games **812** and database **814**. In particular, the O/S **844**, the client device interface **802**, NFT marketplace tools **804**, NFT wallet tools **806**, NFT generation tools **808**, NFT collection display tools **810**, and games **812** each include operational instructions that, when executed by the processing module **830**, cooperate to configure the processing module **830** into a special purpose device to perform the particular functions of the NFT collection platform **800** described herein.

The NFT collection platform **800** may include a user interface (I/F) **862** such as a display device, touch screen, key pad, touch pad, joy stick, thumb wheel, a mouse, one or more buttons, a speaker, a microphone, an accelerometer, gyroscope or other motion or position sensor, video camera or other interface devices that provide information to an administrator of the NFT collection platform **800** and that generate data in response to the administrator’s interaction with NFT collection platform **800**.

The processing module **830** can be implemented via a single processing device or a plurality of processing devices. Such processing devices can include a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, quantum computing device, field

programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on operational instructions that are stored in a memory, such as memory **840**. The memory module **840** can include a hard disc drive or other disc drive, read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that when the processing device implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. While a particular bus architecture is presented that includes a single bus **860**, other architectures are possible including additional data buses and/or direct connectivity between one or more elements. Further, the NFT collection platform **800** can include one or more additional elements that are not specifically shown.

FIG. 2 presents a block diagram representation of an example client device in accordance with various examples. In particular, a client device **825** is presented that includes a network interface **220** such as a 3G, 4G, 5G or other cellular wireless transceiver, a Bluetooth transceiver, a WiFi transceiver, UltraWideBand transceiver, WIMAX transceiver, ZigBee transceiver or other wireless interface, a Universal Serial Bus (USB) interface, an IEEE 1394 Firewire interface, an Ethernet interface or other wired interface and/or other network card or modem for communicating for communicating via network **115**.

The client device **825** also includes a processing module **230** and memory module **240** that stores an operating system (O/S) **244** such as an Apple, Unix, Linux or Microsoft operating system or other operating system, NFT data **246** associated with one or more NFTs owned by the user, and/or a collection application **248**. In particular, the O/S **244** and collection application **248** each include operational instructions that, when executed by the processing module **230**, cooperate to configure the processing module into a special purpose device to perform the particular functions of the client device **825** described herein.

The client device **825** also includes a user interface (I/F) **262** such as a display device, touch screen, key pad, touch pad, joy stick, thumb wheel, a mouse, one or more buttons, a speaker, a microphone, an accelerometer, gyroscope or other motion or position sensor, video camera or other interface devices that provide information to a user of the client device **825** and that generate data in response to the user's interaction with the client device **825**.

The processing module **230** can be implemented via a single processing device or a plurality of processing devices. Such processing devices can include a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, quantum computing device, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on operational instructions that are stored in a memory, such as memory **240**. The memory module **240** can include a hard disc drive or other disc drive, read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that when the processing device implements one or more of its functions via a state machine,

analog circuitry, digital circuitry, and/or logic circuitry, the memory storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. While a particular bus architecture is presented that includes a single bus **260**, other architectures are possible including additional data buses and/or direct connectivity between one or more elements. Further, the client device **825** can include one or more additional elements that are not specifically shown.

The client device **825** operates, via network interface **220**, network **115** and NFT collection platform **800**. In various examples, the client device **825** operates to display a graphical user interface generated based on display data from the NFT collection platform **800**, including corresponding screen displays. Furthermore, the graphical user interface can operate in response to interactions by a user to generate input data that is sent to the NFT collection platform **800** to control the operation of the NFT collection platform **800** and/or to provide other input.

It should be noted that while the client devices **825** and NFT collection platform **800** are shown as separate devices that communicate via the network **115**, it should be noted that any and all of the functionality attributed to the NFT collection platform **800**, including the NFT marketplace tools **804**, NFT wallet tools **806**, NFT generation tools **808**, NFT collection display tools **810**, games **812**, and database **814**, etc. can likewise be incorporated directly into the client device **825**. In this fashion, a client device **825** through the application of its operating system **244** and one or more applications can provide a graphical user interface to operate via network **115** but independently from any NFT collection platform to perform any of the functions and features described herein. In particular, the client device **825** can perform the functions of both the client device and the NFT collection platform **800** without requiring communications to be sent to the client device **825** from a NFT collection platform and communications sent to a NFT collection platform from the client device **825**.

FIG. 3A presents a flowchart representation of an example method in accordance with various examples. In particular, a method **300** for use in conjunction with any of the functions and features described herein for generating a collection NFT based on NFTs in a user's collection.

Step **302** includes importing, via a network interface, a plurality of NFTs associated with the user of the client device. Step **304** includes generating, via a processor and in response to user interactions with the graphical user interface, display data associated with a customized collection display that contains the plurality of NFTs. Step **306** includes sending, via the network interface, the display data associated with the customized collection display via the client device of the user. Step **308** includes facilitating creation of a collection NFT corresponding to the customized collection display that contains the plurality of NFTs.

FIGS. 3B-3F and 3H present pictorial representations of example screen displays. In particular, screen displays of a graphical user interface generated based on display data from the NFT collection platform **800** are shown. In the example below, three NFTs are used to design and create a new collection NFT based on a common theme, the 1999 Chicago Bulls team in the National Basketball association.

In FIG. 3B, the user "Bruce Stuckman" is logged into the NFT collection platform **800**. The user has interacted with NFT wallet tools, such as NFT wallet tools **804**, to import NFTs **100** and **102** from his wallet residing in NFT wallet system **820**. In this example, the NFTs **100** and **102** are

blockchain authenticated original photographs depicting two different images of the 1999 Chicago Bulls.

In FIG. 3C, the user has interacted with the NFT marketplace tools, such as NFT marketplace tools **804**, to select a new NFT **104** for purchase via an NFT marketplace such as NFT marketplace **822**. In FIG. 3D, the wallet tools are used again, this time to export the NFT **104** to the user's wallet. In FIG. 3E, the user has selected and used NFT collection display tools, such as NFT collection display tools **810**, to create a customized collection display **110** contain all three 1999 Chicago Bulls-related NFTs he now owns. In the example shown, the user has "dragged and dropped" NFTs he owns in a custom display window and has sized and arranged them into the particular collage that is shown. In other examples, the NFT collection display tools **810** can operate, based on metadata associated with the NFTs of a user indicating content, theme, color themes, subject matter, dates of creation, authorship, ownership, prior ownership, number of prior owners, size, resolution, and other NFT information and metadata, to automatically generate arrangements of custom collection display **110** that may be accepted by the user and/or that may be further arranged by the user to create the final customized collection display **110**.

In FIG. 3F, the user has selected NFT generation tools, such as NFT generation tools **808**, in order to facilitate, via NFT creation system **824** for example, the creation of a collection NFT from the customized collection display **110**.

FIG. 3G presents a flow diagram representation of an example process. In the example shown, the collection NFT **120** is created from the customized collection display **110**. As previously discussed, the NFT generation tools can operate by, for example, first authenticating the user's rights in the three NFTs and then creating, via NFT creation system **824**, a new NFT of the unique customized image with its own blockchain authentication. In various examples, the collection NFT **120** can contain metadata indicating, for example attributions to the sources of the original NFTs in the collection, the creator of the collection NFT, a date of creation, promotion data and coupons related to offers, privileges and/or discounts, title data with respect to title to tangible or intangible real or personal property, warrant data with respect to tangible or intangible real or personal property, transaction data regarding one or more transactions, and/or other metadata. This metadata can be protected via the blockchain and/or other crypto-based NFT creation technology that is employed to create and protect the collection NFT itself—with or without associated image data. In FIG. 3H, the wallet tools are used again, this time to export the collection NFT **120** to the user's wallet.

FIG. 3I presents a flowchart representation of an example method in accordance with various examples. In particular, a method **310** is presented for use in conjunction with any of the functions and features described herein for generating a collection NFT based on NFTs in a user's collection.

Step **312** includes importing, via a network interface, a plurality of NFTs associated with the user of the client device. Step **314** includes generating, via a processor and in response to metadata associated with the plurality of NFTs, display data associated with a customized collection display that contains the plurality of NFTs. Step **316** includes sending, via the network interface, the display data associated with the customized collection display via the client device of the user. Step **318** includes facilitating creation of a collection NFT corresponding to the customized collection display that contains the plurality of NFTs.

FIG. 4A presents a flowchart representation of an example method. In particular, a method **400** for use in conjunction

with any of the functions and features described herein in generating a collection NFT based on at least one NFT accessed via a temporary micro-loan.

Step **402** includes facilitating, via a processor and in response to user interactions with the graphical user interface, a temporary micro-loan of at least one NFT. Step **404** includes generating, via the processor and in response to user interactions with the graphical user interface, display data associated with a customized collection display that contains a plurality of NFTs including the at least one NFT. Step **406** includes sending, via the network interface, the display data associated with the customized collection display via the client device of the user. Step **408** includes facilitating creation of a collection NFT corresponding to the customized collection display that contains the plurality of NFTs including the at least one NFT.

FIG. 4B presents a pictorial representation of an example screen display. In particular, the user has used the marketplace tools to select NFTs **100-1**, **102-1**, and **104-1**. Instead of putting these NFTs up for purchase, the original owners have made them available for temporary micro-loan. This process allows the users can engage to, in exchange for a fee, "borrow" NFTs in order to create a collection NFT. After the Collection NFT is created, or upon the expiration of some predetermined time period (such as 15 minutes, 30 minutes, an hour, a day, etc.) the micro-loaned NFT(s) are returned, expired, deleted or destroyed. Transaction fees apply, a portion of which can be credited to the original NFT owner, the author, including an upfront cost, reward-based payment based on the use and/or performance of the micro-loaned NFT etc. In the alternative, a fixed fee could be charged to the user and credited to the original NFT owner. In various examples, the NFT collection platform **800** may be configured to operate with a single user and/or within a single wallet, and/or to otherwise prohibit the sale or borrowing of micro-loaned NFTs to avoid dilution of the value of the original itself. Furthermore, original and/or derivative NFTs can include restrictions on the total number of micro-loan transactions, the number of simultaneous/contemporaneous micro-loans, restrictions on types of microloan transactions such as normal use loans, staking loans, death match loans, loans less than a predetermined length of time, loans greater than a predetermined length of time, etc.

Consider the following example where an NFT is put up for loan. The proposed loan transaction can include restrictions including an expiration time and/or date, one or N time use in creating a collection NFT or in a game, tournament or challenge, etc. Once the loan is accepted by the borrower, an additional NFT, such as a derivative NFT of the original NFT being loaned, is created on a side chain, layer 1 or 2 blockchain (or "parachain") that can be different from (and/or independent from) the blockchain used to create the original NFT. This new NFT can be created and transferred to the wallet of the borrower via a smart contract that is based on the restrictions. A cryptocurrency market can be used to fund the transaction and/or to collateralize the micro-loan. The new NFT can then be set via the smart contract to automatically expire (e.g. be deleted from the wallet, destroyed or otherwise disabled), when the restrictions are met. In this fashion, if the purpose of the micro-loan is the creation of a collection NFT, the borrowed NFT can automatically expire once the collection NFT is created. It should be noted that the collection NFT can be created via the same blockchain platform (e.g. Ethereum) used to create the original NFT (e.g. not the parachain).

FIG. 4C presents a flow diagram representation of an example process where a collection NFT **120-1** is generated

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based on a customized collection display **110-1** created by the user based on the borrowed NFTs **100-1**, **102-1** and **104-1**. In various examples, the collection NFT **120-1** meta-data can also indicate the original sources of the micro-loans as well as the micro-loaned status of NFTs **100-1**, **102-1** and **104-1**. While not expressly shown, the collection NFTs based on one or more micro-loaned NFTs, can be created with a visual indication of the original vs. micro-loaned status of the NFTs as appropriate.

FIG. 5A presents a flowchart representation of an example method. In particular, a method **1300** for use in conjunction with any of the functions and features previously described facilitates the collection of endorsements associated with an NFT. Step **1302** includes importing, via a network interface, an NFT associated with the user of the client device. Step **1304** includes collecting, via a processor and in response to user interactions with the graphical user interface, endorsement data associated with the NFT. Step **1306** includes generating, via the processor and in response to user interactions with the graphical user interface, display data associated with a customized collection display that contains the NFT and the endorsement data. Step **1308** includes sending, via the network interface, the display data associated with the customized collection display via the client device of the user. Step **1310** includes facilitating creation of a collection NFT corresponding to the customized collection display.

FIGS. 5B and 5C present pictorial representations of example screen displays. In FIG. 5B, the user is interacting with the graphical user interface to use the NFT marketplace tools to seek an endorsement for a selected NFT. After an endorsement in the form of a signature has been received it can be appended to the customized collection display of the NFT. In FIG. 5C, the user interacts with the NFT generation tools to generate a collection NFT that includes both the NFT and the signature. In various examples, this collection NFT **120-6** metadata can also indicate an attribution associated the endorsement itself. Furthermore, the endorsement itself can be an original NFT, a derivative NFT, a micro-loaned NFT or other NFT.

FIG. 6 presents a block diagram/flow representation of an example of NFT generation. In the example shown, NFT generation tools **808** operate to convert display data **1502** and other data **1504** into an NFT **1506**. The display data **1502** can correspond to an original NFT, borrowed (e.g. temporary micro-loaned) NFT, a derivative NFT, a customized collection display **110**, an original image, and/or other derivatives or micro-loans thereof or other display or image data. In various examples, the NFT generation tools **808** can operate to verify the credentials of any NFTs whose images are contained in the display data **1502**, prior to creating the NFT **1506**. In this fashion, derivative NFTs can only be created when the source NFT or NFTs are verified-preventing the creation of unauthorized or counterfeit NFTs.

The other data **1504**, can be image data including signatures and other endorsement images, visual indications of derivative series, originality classification, attributions, or other image data, metadata of all kinds including metadata indicating one or more originality classifications, attributions, endorsement data, other derivative data indicating the series number and total number in a derivative series, restrictions on micro-loans or other derivatives, restrictions on derivatives with artistic effects, restrictions that derivatives must include attributions to the original source, restrictions on numbers of derivatives or micro-loans or the sizes of derivative series, restrictions on the creation of collection NFTs, the number of collection NFTs, the creation of collection NFTs including NFTs from other sources, from

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prohibited sources or with prohibited content, geographical restrictions, time restrictions (e.g., can be used to create derivatives or collection NFTs or can be temporarily micro-loaned for 1 month, one year, etc., other restrictions and/or other data associated with, or to be associated with, the display data **1502**.

This other data **1504** can be used to generate an NFT and/or combined with the display data **1502** to create a dataset that includes both the display data **1502** and the other data **1504**. This other data **1504** or combined dataset can be protected via the blockchain and/or other crypto-based NFT creation technology that is employed by the NFT generation tools **808** and via the NFT creation system **824** to create and protect the new NFT **1506** itself. It should be noted that the NFT **1506** can include a single derivative or a number of derivatives, including a limited series of derivatives. It should be noted further that some or all of the other data **1504**, including restriction data and/or attribution data, can be derived from one or more original NFTs whose images are associated with the display data **1502**. Furthermore, some or all of the other data **1504** can be generated in response to user interactions with a graphical user interface generated in conjunction with the NFT collection platform **800**.

FIG. 7 presents a block diagram representation of an example system. In particular, a system **2850** is presented that includes an NFT distribution platform **2800** that communicates with client devices **825** via a network **115**. The network **115** can be the Internet or other wide area or local area network, either public or private. The client devices **825** can be computing devices associated with users, for example, buyers, sellers, collectors, game players and/or other users of NFTs.

In the example shown, the NFT distribution platform **2800** includes a client device interface **2802** for interacting with the client devices **825**, NFTs **2804** to be distributed, and an operating system **2844**. One or more of the NFTs **2804** can have geographical restrictions as to distribution that are either part of the NFTs themselves or have restriction data that is stored separately.

The NFT distribution platform **2800** includes a network interface **2820** such as a 3G, 4G, 5G or another cellular wireless transceiver, a Bluetooth transceiver, a WiFi transceiver, UltraWideBand transceiver, WIMAX transceiver, ZigBee transceiver or other wireless interface, a Universal Serial Bus (USB) interface, an IEEE 1394 Firewire interface, an Ethernet interface or other wired interface and/or other network card or modem for communicating for communicating via the network **115**.

The NFT distribution platform **2800** also includes a processing module **2830** and memory module **2840** that stores an operating system (O/S) **2844** such as an Apple, Unix, Linux or Microsoft operating system or another operating system, the client device interface **2802**, and the NFTs **2804**. The O/S **2844** and the client device interface **802** each include operational instructions that, when executed by the processing module **830**, cooperate to configure the processing module **830** into a special purpose device to perform the particular functions of the NFT distribution platform **2800** described herein.

The NFT distribution platform **2800** may include a user interface (I/F) **2862** such as a display device, touch screen, key pad, touch pad, joy stick, thumb wheel, a mouse, one or more buttons, a speaker, a microphone, an accelerometer, gyroscope or other motion or position sensor, video camera or other interface devices that provide information to an administrator of the NFT distribution platform **2800** and that

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generate data in response to the administrator's interaction with NFT distribution platform **2800**.

The processing module **2830** can be implemented via a single processing device or a plurality of processing devices. Such processing devices can include a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, quantum computing device, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on operational instructions that are stored in a memory, such as memory **2840**. The memory module **2840** can include a hard disc drive or other disc drive, read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that when the processing device implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. While a particular bus architecture is presented that includes a single bus **2860**, other architectures are possible including additional data buses and/or direct connectivity between one or more elements. Further, the NFT distribution platform **2800** can include one or more additional elements that are not specifically shown.

For example, the client device interface **2802** can operate in conjunction with each client device **825** and via network **115** to generate a graphical user interface. This graphical user interface is based on display data generated by the NFT distribution platform **2800** in a format for display on a display device associated with the client devices **825**. This graphical user interface generates input data that is received by the NFT distribution platform **2800** from the client devices **825** in response to user interaction with the graphical user interface.

In various examples, the NFT distribution platform **2800** can operate to respond to input data from client devices in the form of read requests for NFTs and geolocation data such as GPS coordinates, connection to or proximity with a network element of network **115** or other location data indicating a location of the client device **825**. The NFT distribution platform **2800** sends the requested NFT(s) to the requesting client device—only when the geolocation data conforms with restriction data—for example, when the geolocation data indicates a position of the requesting client device within a limited area or proximity indicated by the restriction data.

FIG. **8** presents a block diagram representation of an example client device. In particular, a client device **825-1** is presented that functions similarly to client device **825**, and includes several elements of client device **825** that are referred to by common reference numerals. The client device **825-1** is capable of operating to client device **825** described herein.

In addition, the memory module **240** includes a wallet application (app) **250** that is capable of engaging in financial transactions including credit card transactions and traditional digital payments, is capable of holding crypto-currency and engaging in crypto-currency transactions and is further capable of storing one or more NFTs that are either original NFTs, derivative NFTs, borrowed (temporarily micro-loaned) NFTs, collection NFTs and/or combinations thereof. In various examples, the wallet app **250** is capable of operating in conjunction with the NFT collection platform

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800, the NFT distribution platform **2800**, the NFT wallet system **820**, the NFT marketplace **822**, and/or the NFT creation system **824** via network **115**.

As will be understood by one skilled in the art, unlike a normal wallet, which can physically hold cash, credit cards, etc., NFT wallets “store” NFTs by storing the NFT data necessary to access the NFT. So, even though a wallet can be said to store an NFT, technically the NFT content is stored on the blockchain, which can only be accessed via the NFT data in the wallet. This NFT data includes metadata, other off-chain data corresponding to the NFT and in particular, a private key. This private key can be considered an indicator of ownership of the NFT and is required to access the NFT via the blockchain. If the NFT data (including the private key) is lost, the NFT can no longer be accessed—and the NFT is itself “lost” for all intents and purposes, even though it remains immutably stored on the blockchain.

Also, the memory module **240** includes one or more game apps **252** that represent either stand alone games of the client device **825-1** or that operate in conjunction with the games **812** of the NFT collection platform **800** and/or interface with the NFT distribution platform **2800**. This allows, for example, a user of client device **825-1** to engage in (e.g. play) games associated with NFT content, and engage in other activities that involve the acquisition, collection, display, distribution, and/or use of one or more NFTs that are either original NFTs, derivative NFTs, borrowed (temporarily micro-loaned) NFTs, collection NFTs and/or combinations thereof.

Furthermore, the network interface **220** includes one more geolocations elements **222** such as a GPS receiver, a ultra-wideband (UWB) transceiver, a Bluetooth transceiver and/or other component(s) that facilitate the generation of geolocation data and/or facilitate other location-based services. Consider the case where the client device **825** is a smartphone or tablet and the wallet app **250** is an Apple or Android wallet or mobile wallet card that is in a Apple or Android wallet. Once the wallet app **250** is activated, NFTs can be easily added to the wallet. In addition, the wallet app **250** can access the location services of the device, and for example, generate push notifications regarding NFTs that are available near the current location.

In various examples, the geolocation data generating in such a fashion can facilitate the generation of geolocation data discussed in conjunction with the operation of NFT distribution platform **2800**. In particular, the NFT distribution platform **2800** can automatically detect the presence of the user at a venue based on geolocation data received from the user's client device **825-1** and automatically prompted the user to click to send a read request. In this fashion, the user can be geo-authorized, before the request. In other examples, the NFT distribution platform **2800** can distribute NFTs to client devices **825** based on payments, authentication and/or other criteria that does not rely on geolocation data.

Furthermore, while the client device **825-1** and NFT collection platform **800** (or NFT distribution platform) are shown as separate devices that communicate via the network **115**, it should be noted that any and all of the functionality attributed to the NFT collection platform **800** (or NFT distribution platform), including the NFT marketplace tools **804**, NFT wallet tools **806**, NFT generation tools **808**, NFT collection display tools **810**, games **812**, and database **814**, etc. can likewise be incorporate directly into the client device **825**. In this fashion, a client device **825** through the application of its operating system **244** and one or more applications can provide a graphical user interface to operate

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via network **115** but independently from any NFT collection platform to perform any of the functions and features described herein. In particular, the client device **825** can perform the functions of both the client device and the NFT collection platform **800** without requiring communications to be sent to the client device **825** from a NFT collection platform (or NFT distribution platform) and communications sent to a NFT collection platform (or NFT distribution platform) from the client device **825**.

In addition, NFT generation tools **808** can be used to protect, encrypt and/or authenticate any digital information that could be stored in the wallet app **250**, including for example rewards cards, coupons, movie tickets, event tickets, boarding passes, public transit cards, student ID cards, credit cards, debit cards, prepaid cards, and loyalty cards. In addition, the functionality of the wallet app **250** can be further expanded to protect other information such as vehicle titles, warranty cards, driver's licenses and other IDs, vaccination records, prescriptions, and/or other medical records, social security cards, financial records, authentication tokens, insurance cards, passwords, user IDs and/or other images and information of a personal and/or sensitive nature. Any of these types of digital information can be protected via an NFT or other blockchain transaction in conjunction, with or without associated image or display data, and with or without metadata and/or "other data" as that term has been used herein in association with the NFT generation tools **808**.

FIG. **9A** presents a block diagram/flow representation of an example of NFT generation. In the example shown, an original metaverse real estate NFT **4820** corresponds to a portion/plot of real estate that is used in a game or metaverse application. An example is shown in FIG. **9B**. The original metaverse real estate NFT **4820** can be purchased by the user, created or leveled-up via game play or otherwise acquired by the user of a metaverse application or game.

The NFT generation tools **808** are used to generate a derivative NFT **4824** or other metaverse real estate NFT that is based on the improvements **4822** shown in FIG. **9C**. In the example shown the improvements **4822** correspond to a building or other structure, however, other real estate improvements can likewise be implemented. In various example, the derivative NFT **4824** with improvements shown in FIG. **9D** can be created on a parachain or other sidechain that is different from the blockchain on which the original metaverse real estate NFT **4820** resides or the same blockchain on which the original metaverse real estate NFT **4820** resides.

FIG. **9E** presents a flowchart representation of an example method. In particular, a method **4800** is presented for use in conjunction with any of the functions and features described herein. Step **4802** includes receiving, via the processor, a metaverse real estate NFT associated with metaverse real estate. Step **4804** includes receiving, via the processor, improvements data associated with the metaverse real estate. Step **4806** includes facilitating creation of a derivative NFT associated with the metaverse real estate and having improvements associated with metaverse real estate.

FIG. **10A** presents a block diagram/flow representation of an example of NFT generation. In the example shown, document image data **4920** and other data corresponding to a document are used to generate an authenticated document NFT **4924** via the NFT generation tools **808**. These authenticated document NFTs **4924** can be stored in an NFT wallet associated with the mobile phone or other client device associated with the user and can be used, for example, in place of coupons, cards, legal documents, medical docu-

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ments, financial documents, IDs, credit cards, licenses and/or other important documents associated with a user that normally exist in non-digital, e.g. paper or plastic form. The authenticated document NFT **4924** can be used to prevent fraud and/or promote privacy in transactions via secure user and/or document authentication. In various examples, the authenticated document NFT **4924** can be presented and analyzed via secured blockchain or other crypto transactions at the time of a transaction in order to authenticate the identity of the user and/or to verify the accuracy and authentic nature of the other data **4922** and/or to facilitate the security of the transaction.

FIG. **10B** presents a flowchart representation of an example method. In particular, a method **4900** is presented for use in conjunction with any of the functions and features described herein. Step **4902** includes receiving, via the processor, a document image associated with a document. Step **4904** includes receiving, via the processor, other data, wherein the other data is also associated with the document. Step **4906** includes facilitating creation of an authenticated document NFT corresponding to the document.

FIG. **11A** presents a block diagram/flow representation of an example of NFT generation. The NFT generation tools **808** are used to generate a derivative NFT **5624** based on an original metaverse real estate NFT **5620** and based on the enhancement data **5622**.

In the example shown in FIG. **11B**, an original metaverse real estate NFT **5620** corresponds to real estate having an interior room that is used in a game or metaverse application. The original metaverse real estate NFT **5620** can be purchased by the user, created or leveled-up via game play or otherwise acquired by the user of a metaverse application or game.

The NFT generation tools **808** are used to generate a derivative NFT **5624** or other metaverse real estate NFT that is based on the enhancements data **5622** shown in FIG. **11C**. In the example shown the enhancements **5622-1** and **5622-2** correspond to a Moet & Chandon vending machine and a Jackson Pollock painting that are acquired either as image data or as individual NFTs. While particular enhancements are shown, other real estate enhancements can likewise be implemented including statues and other art, rugs, lamps, furniture and other furnishings and accessories, outdoor objects, appliances, knick-knacks, machinery and other virtual objects for decorating or finishing an office, home, factory, venue or other real estate.

In the example shown in FIG. **11D**, the user has placed the enhancements **5622-1** and **5622-2** as desired before creating the derivative NFT **5624**. The derivative NFT **5624** with improvements shown can be created on a parachain or other sidechain that is different from the blockchain on which the original metaverse real estate NFT **5620** resides or the same blockchain on which the original metaverse real estate NFT **5620** resides.

FIG. **11E** presents a flowchart representation of an example method. In particular, a method **5600** is presented for use in conjunction with any of the functions and features described herein. Step **5602** includes receiving, via the processor, a metaverse real estate NFT associated with metaverse real estate. Step **5604** includes receiving, via the processor, enhancement data associated with the metaverse real estate. Step **5606** includes facilitating creation of a derivative NFT associated with the metaverse real estate and having enhancements associated with metaverse real estate.

FIG. **12A** presents a block diagram of an example system. In particular, a system is shown that can be implemented similarly to, or in conjunction with, NFT collection platform

800. The system includes an NFT transaction authenticator 6022 and a secure real-time NFT metadata repository 6024. In various examples, the NFT transaction authenticator 6022 and the secure real-time NFT metadata repository 6024 can be implemented via one or more modules that include a network interface, processing circuitry and memory. The secure real-time NFT metadata repository 6024 stores NFT metadata received in conjunction with NFTs created via metadata source 6026, such as one or more NFT creation systems 824. The NFTs are associated with one or more users 6020.

In operation, the NFT transaction authenticator 6022 responds to transaction requests from a user associated with an NFT to authenticate the NFT and the user and to otherwise determine the validity of the transaction that is requested. If the user and the NFT are both authenticated, and the requested transaction is otherwise permissible (e.g., not restricted by conditions on use or other transaction restrictions), then the NFT transaction authenticator 6022 responds by issuing credentials to facilitate the transaction with a third party 6028. As will be discussed herein, the maintenance and use of the secure real-time NFT metadata repository 6024 allows authentication of NFT related transactions in real-time-avoiding possible delays in performing, for example, complex blockchain transactions via an NFT source 6026 where the NFT was minted and/or otherwise maintained.

Consider the following example where an NFT is created via NFT source 6026. In addition to other NFT data, the NFT has metadata that uniquely identifies the NFT, a hash or other NFT authentication metadata that can be used to authenticate the NFT and/or transaction restriction metadata indicating possible restrictions on transactions/use conditions involving the NFT. Furthermore, when the NFT is created and/or acquired by a user, user-specific user authentication metadata is acquired or created and stored on the blockchain with the NFT with the other metadata. This user authentication metadata can include one or more passwords, answers to security questions, identifiers of recognized devices such as a device identifier of a personal cellphone, laptop, tablet, computer or other known and/or trusted device, one or more trusted networks of the user, other multifactor authentication data such as personal information, known answers to security questions, biometric data related to fingerprints, retinal scans, facial features or other biometrics of the user and/or other user authentication data that can be used to determine if a user is the owner of the NFT or otherwise an authorized user and in particular, whether or not the user is (or is not) who they claim to be.

The metadata associated with the NFT is indexed by NFT identifier and stored on the secure real-time NFT metadata repository 6024 for use by the NFT transaction authenticator 6022 in authenticating NFT-related transactions. This metadata is available from the repository on a real-time basis (e.g., is available with an acceptable amount of latency associated with a corresponding transaction). The metadata in the secure real-time NFT metadata repository 6024 is also synced periodically with the NFT via the NFT source 6026 to reflect any changes in the NFT itself. While some metadata, such as an NFT identifier, NFT authentication metadata and/or transaction restriction metadata may be made accessible to the user who holds the NFT, in various examples, the user authentication metadata in particular, can be encrypted in such a fashion that is decryptable by the secure real-time NFT metadata repository 6024—but not by the user. In various examples, the secure real-time NFT metadata repository 6024 lacks a general network connection and is

connected to the NFT transaction authenticator 6022 via a dedicated and/or otherwise secured connection or is otherwise walled-off from other network connections of the NFT transaction authenticator 6022. This helps prevent unauthorized tampering with the sensitive data stored therein.

When a user 6020 proposes an NFT-related transaction, the NFT transaction authenticator 6022 collects from the user as part of the transaction request (a) an identifier of the NFT, and NFT authentication data corresponding to the NFT (b) user authentication data user, and (c) information on the proposed transaction. The NFT transaction authenticator 6022 determines whether or not the identifier corresponds to a valid NFT. If so, it retrieves the metadata associated with the NFT from secure real-time NFT metadata repository 6024. The NFT transaction authenticator 6022 authenticates the NFT by comparing the NFT authentication data to the NFT authentication metadata to determine if they match. The NFT transaction authenticator 6022 can also authenticate the user 6020 to the NFT by comparing the user authentication data to the user authentication metadata to determine if they match. If authentication succeeds, the NFT transaction authenticator 6022 facilitates the transaction with the third party 6028 by authorizing completion of the transaction, e.g. by issuing a credential to the third party 6028. The credential can include any message, object, or data structure that vouches for the identity of the user, the authenticity of the NFT and/or the validity of the transaction, through some method of security, trust and/or authentication.

In this fashion, the NFT transaction authenticator 6022 can authenticate transactions such as access to a flight via a driver's license or passport NFT, sale of a vehicle, real estate via a title NFT, a credit, debit or gift card transaction via a credit, debit or gift card NFT, the sale of a stock or bond via a stock or bond certificate NFT, warranty transactions via a warranty card NFT, access to events via venue ticket NFTs and/or vaccination card NFTs, coupon redemption via a coupon NFT, access to a vehicle, dwelling or office via a key NFT, etc. Furthermore, the NFT transaction authenticator 6022 can authenticate transactions such as sales and/or micro-loans of NFTs itself.

It should be noted that some NFTs are conditional, e.g. that have restrictions on their use and/or the transactions that are permitted. Depending on the type of transaction, the NFT transaction authenticator 6022 can also operate to compare transaction data received from the user to transaction restrictions metadata to determine if a transaction is permitted—before it is authorized. In this fashion, a credit, debit or gift card transaction can be halted if an expiration date or transaction limit has been exceeded. A warranty transaction can be halted if the warranty has expired. A key access for a pool, gym or office can be halted based on date, day of week or time of day restrictions, etc.

Furthermore, in sales transactions and/or micro-loans that have been authorized, the NFT transaction authenticator 6022 can also operate to note the pending sale or micro-loan in the secure real-time NFT metadata repository 6024 or otherwise place a hold on the NFT to prevent another sales or micro-loan until the repository is updated in a future sync with the NFT source 6026. Furthermore, in some cases the NFT corresponds to an expendable asset such as a venue ticket, gift card, coupon, etc. The NFT transaction authenticator 6022 can also operate to determine that the NFT is expended, based on the transaction restrictions metadata and the transaction data. In response to such a determination, NFT transaction authenticator 6022 can then update the secure real-time NFT metadata repository to indicate the

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NFT is expended. In any of these cases above, the secure real-time NFT metadata repository may facilitate updating of the NFT source **6026** to reflect an authorized transaction via notifications and/or that an NFT has been expended.

FIG. **12B** presents a flowchart representation of an example method. In particular, a method **6000-4** is presented for use in conjunction with any of the functions and features described herein. Step **6002-4** includes receiving, at the processor, a transaction request from a user associated with an NFT minted via a blockchain, wherein the transaction request includes NFT identification data identifying the NFT, NFT authentication data, user identification data and transaction data associated with the transaction. Step **6004-4** includes receiving, from a secure real-time NFT metadata repository and in response to the NFT identification data, NFT authentication metadata for authenticating the NFT, user authentication metadata for authenticating an authorized user of the NFT, and transaction restrictions data associated with the NFT, wherein the secure real-time NFT metadata repository is separate from the blockchain.

Step **6006-4** includes determining, via the processor, that the NFT is authenticated when the NFT authentication data compares favorably to the NFT authentication metadata. Step **6008-4** includes determining, via the processor, that the user is authenticated when the user authentication metadata compares favorably to the user identification data. Step **6010-4** includes determining, via the processor, that the transaction is authorized when the transaction restrictions metadata compares favorably to the transaction data. Step **6012-4** includes generating credentials data to facilitate the transaction when the NFT is authenticated, the user is authenticated and the transaction is authorized.

FIG. **12C** presents a pictorial block diagram/flow representation of a further example of transaction authentication. In this case, the transaction request **6054-1** includes NFT identification data identifying the NFT, NFT authentication data, and also user identification data that includes both a user device identifier and user biometric data that was collected via the client device **825** as part of its own user authentication process.

Again, the NFT transaction authenticator **6022** can operate in conjunction with the secure real-time NFT metadata repository **6024** for example, to authenticate the transaction with the third party **6028**. In this additional case, the user authentication mechanisms of the client device are used in the transaction of the authentication process. Once the user is identified by a trusted device, the device identifier of the trusted device can be in combination with the user biometrics collected by that device to provide further security for the proposed transaction.

FIG. **12D** presents a flowchart representation of an example method. In particular, a method **6000-10** is presented for use in conjunction with any of the functions and features described herein. Step **6002-10** includes receiving, at the processor, a transaction request from a user associated with an NFT minted via a blockchain, wherein the transaction request includes NFT identification data identifying the NFT, NFT authentication data and user identification data that includes a user device identifier. Step **6004-10** includes receiving, from a secure real-time NFT metadata repository and in response to the NFT identification data, NFT authentication metadata for authenticating the NFT and user authentication metadata for authenticating an authorized user of the NFT, wherein the secure real-time NFT metadata repository is separate from the blockchain.

Step **6006-10** includes determining, via the processor, that the NFT is authenticated when the NFT authentication data

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compares favorably to the NFT authentication metadata. Step **6008-10** includes determining, via the processor, that the user is authenticated when the user authentication metadata compares favorably to the user identification data. Step **6010-10** includes generating credentials data to facilitate the transaction when the NFT is authenticated and the user is authenticated.

In FIG. **13A**, an NFT includes (e.g. that has associated content stored on-chain or off-chain) corresponding to a playable version (e.g. a game application, game binary, game data, etc.) of an electronic game (or more simply, a “game”), such as an online game, cloud-based game, a game application for a mobile device, a video game playable on a game console or personal computer and this is rendered on a streaming platform, in the cloud, or locally at the client device itself. The electronic game can be a metaverse game or other metaverse experience, a single player game, a multiplayer game, an action/adventure game, a real-time strategy game, a shooter game, a multiplayer online battle arena, a role-playing game, simulation or sports game, a puzzle, party game and/or other game.

In various examples, the NFT includes (e.g. that has associated content stored on-chain or off-chain) corresponding to a game application having playable game data such as graphics files, game status information, executable code or other instructions that operate in conjunction with a platform and/or a client device to execute the play of the game itself. Furthermore, the NFT may include authentication data with respect to a user that “owns” the game NFT and this particular copy of the game and entitles the user or other holder of the NFT to play the game and other metadata and other data as discussed herein. In the example shown, the game NFT includes metadata that indicates that this copy corresponds to a special collector’s edition of the game and, in particular, one of a limited series release of NFTs corresponding to one of the first 10,000 copies of the game and/or game NFT.

While described above as a “game NFT”, such an NFT can have any of the metadata, other data, and/or operate similarly to any of the other types of NFTs corresponding to videos, images, documents, game pieces, game characters, metaverse features, and/or other objects, etc. that are described herein.

FIG. **13B** presents a flowchart representation of an example method. In particular, a method **6200-1** is presented for use in conjunction with any of the functions and features described herein. Step **6202-1** includes receiving, via the processor, playable game data associated with an electronic game to be associated with a series of NFTs. Step **6204-1** includes receiving, via the processor, other data to be associated with the series of NFTs. Step **6206-1** includes facilitating, via the processor and based on the playable game data and the other data, generation of the series of NFTs, each of the series of NFTs comprising a playable instance of the electronic game.

FIG. **13C** presents a pictorial representation of an example screen display. In the example shown, a user of an NFT game platform has accessed, via a client device associated with the user such as client device **825**, a plurality of game NFTs associated with the user. The NFT game platform can be implemented via NFT collection platform **800**, NFT distribution platform **2800**, and/or other NFT platform that can include or operate in conjunction with NFT transaction authenticator **6022**, NFT source **6026**, secure real-time NFT metadata repository **6024**, NFT wallet system **820**, NFT marketplace **822**, NFT creation system **824**, and/or other tools, devices and systems described herein.

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In this case, the user's collection of game NFTs includes NFTs **4150-1**, **4150-2**, **4150-3**, and **4150-4**, etc. In various embodiments the collection display can be arranged by the user or automatically sorted into categories such as genre's or other groupings based on metadata associated with each of the NFTs.

In the screen display shown, the platform has generated display data associated with a customized collection display that contains the plurality of game NFTs for display via a graphical user interface associated with the client device. Furthermore, the user has interacted with the graphical user interface and indicated a selection of one the plurality of game NFTs for play—in this case, the game "I Dig It". The selection is received by the platform which responds by interacting, via the network **115**, with a client device **825** to support play of the selected game by the user—as shown in FIG. **13D**. This support can include streaming game video to, and receiving game input from, the client device **625** via the network **115**.

The NFT collection platform **800** and/or associated client device improves the technology of NFT systems and existing gaming platforms by allowing the user to generate, store, collect, display and use game NFTs that include playable content. The use of NFTs in association with playable game content allows users to "own" playable versions of games that can be streamed to a user's client device, to own and trade collectable limited game edition NFTs, to loan authenticated versions of games for the play of others, and/or to authenticate the user's ownership in the game and the game NFT.

FIG. **13E** presents a flowchart representation of an example method. In particular, a method **6200-2** is presented for use in conjunction with any of the functions and features described herein. Step **6202-2** includes accessing, via a network interface, a plurality of game NFTs, each of the plurality of game NFTs associated with a game accessible by a user of a client device. Step **6204-2** includes generating, via the processor, display data associated with a customized collection display that contains the plurality of game NFTs for display via a graphical user interface.

Step **6206-2** receiving, via the graphical user interface, a selection of one the plurality of game NFTs for play of the associated game. Step **6208-2** includes facilitating, via the network interface and in conjunction with the client device, play of the game associated with the one of the plurality of game NFTs.

FIG. **13F** presents a block diagram/flow representation of an example of NFT generation. In particular, game data **6226** has been collected in association with the user's (player's) play of a game corresponding to game NFT **6220**. At some time, such as the end of a game, when a player reaches certain goals or achievements in the game, upon expiration of a predetermined time period, upon accumulation of a predetermined amount of game play (e.g. a number of games played, an amount of play time, etc.) or other criteria, the game data **6226** is used to generate an update to the game NFT **6220** as the updated game NFT **6220-1**. In this fashion, the player's game play and/or performance and achievements can be used to update the game NFT so that a player can pick up where he/she left off, have a mechanism to display some of the performance/achievements in conjunction with the game NFT itself.

As used herein, the term "updated" NFT can be used to represent changes that are made to an original NFT—i.e. the NFT that is the subject of the update. In addition, an "updated" NFT can be a new NFT, such as a derivative NFT, an all new NFT or other NFT that is minted based on the

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content, images, metadata and/or other data or based merely on the existence of the original NFT. This new NFT can be minted on the original blockchain used to mint the original NFT or via a parachain/side chain that is different from the original blockchain.

In FIG. **13G**, an example of Game NFT **6220** corresponding to an NBA game is presented. In FIG. **13H**, game data **6226** is generated based on the player's play of the game and used to generate an updated game NFT **6220-1**. In the example shown, updated game NFT **6220-1** visually reflects that the player has reached the "one-star" level.

FIG. **13I** presents a flowchart representation of an example method. In particular, a method **6200-3** is presented for use in conjunction with any of the functions and features described herein. Step **6202-3** includes facilitating, via the network interface and in conjunction with the client device, play of the game associated with the one of the plurality of game NFTs. Step **6204-3** includes receiving, via the processor, game data associated with the game play of the game by a user. Step **6206-3** includes updating the original NFT associated with the game piece, based on the game data associated with the game play of the game by the user.

FIG. **13J** presents a block diagram/flow representation of an example of NFT generation. In the example shown, NFT generation tools **808** are used to generate a derivative NFT **6224** based on the expiration restrictions **6222** that is shown in FIG. **13K**. The expiration restrictions can correspond to, for example, an expiration time and/or date, one or N time use in a game, tournament or challenge, etc. This derivative NFT **4624** can be created on a parachain or other sidechain that is different from the blockchain on which the original game NFT **6220** resides. Furthermore, this derivative NFT **6224** can be micro-loaned or otherwise leased or loaned to another user for game play—subject to the expiration restrictions. In this fashion, for example, other users can play the game in exchange for a price (payment, bounty, experience, etc.).

In FIG. **13L**, the derivative NFT **6224** has been temporarily loaned to another user who plays the game. At the end of the loan (e.g. when the expiration restrictions are met) the loaned game is automatically deleted, destroyed, disabled or otherwise "burned" and game data **6226** indicating the end of the escrow, stats and experience or other game performance metrics earned by the other user via the game are retrieved and can be stored by the user on the blockchain or other decentralized computer network with the original NFT **6220** to generate updated NFT **6220'**, as shown in FIG. **13M**.

FIG. **13N** presents a flowchart representation of an example method. In particular, a method **6200-4** is presented for use in conjunction with any of the functions and features described herein. Step **6202-4** includes facilitating, via the processor, creation of a derivative NFT corresponding to an original NFT associated with a game the derivative NFT having expiration restrictions limiting game play of the game. Step **6204-4** includes facilitating, via the processor, a micro-loan transaction of the derivative NFT to another user to facilitate the game play of the game by the other user, wherein the derivative NFT expires when the expiration restrictions are met.

FIG. **13O** presents a flowchart representation of an example method. In particular, a method **6200-5** is presented for use in conjunction with any of the functions and features described herein. Step **6202-5** includes facilitating, via the processor, creation of a derivative NFT corresponding to an original NFT associated with a game the derivative NFT having expiration restrictions limiting game play of the game. Step **6204-5** includes facilitating, via the processor, a

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micro-loan transaction of the derivative NFT to another user to facilitate the game play of the game by the another user, wherein the derivative NFT expires when the expiration restrictions are met. Step **6206-5** includes receiving, via the processor, game data associated with the game play of the game piece by the another user. Step **6208-5** includes updating the original NFT associated with the game piece, based on the game data associated with the game play of the game piece by the other user.

FIG. 13P presents a flowchart representation of an example method. In particular, a method **6200-6** is presented for use in conjunction with any of the functions and features described herein. Step **6202-6** includes accessing, via a network interface, a plurality of game NFTs, each of the plurality of game NFTs including a playable instance of an associated game that is accessible by a user of a client device. Step **6204-06** includes generating, via the processor, display data associated with a customized collection display that contains the plurality of game NFTs. Step **6206-06** includes sending, via the network interface, the display data associated with a customized collection display to the client device for display via a graphical user interface associated with the client device.

Step **6208-06** includes receiving, via the graphical user interface, a selection of one the plurality of game NFTs for play of the associated game. Step **62010-06** includes facilitating, via the network interface, play of the associated game in conjunction with the client device.

In various examples, facilitating the play of the associated game includes streaming, via the network interface, game video associated with the one of the plurality of game NFTs to the client device and receiving from the client device, via the network interface, game data associated with the game play of the game by the user. The customized collection display that presents the plurality of game NFTs can be configured in response to user interactions with the graphical user interface, configured based on metadata associated with the plurality of game NFTs and/or example, be configured based on genres associated with the plurality of game NFTs or other categories or customized arrangements, either automatically or by selection from the user.

In various examples, the method can further include updating the game NFT based on the game play of the associated game by the user, for example, by minting an updated game NFT that may be minted on via parachain that can be different from the blockchain that minted the game NFT. The method can further include: facilitating creation of a derivative NFT corresponding to a game NFT of the plurality of game NFTs, the derivative NFT having expiration restrictions limiting game play of the associated game; and facilitating a micro-loan transaction of the derivative NFT to another user to facilitate the game play of the associated game by the another user, wherein the derivative NFT expires when the expiration restrictions are met. These facilitating steps can be accomplished via NFT marketplace tools **804** and NFT generation tools **808** in conjunction with, for example, one or more NFT wallet systems **820**, one or more NFT marketplaces **822** and one or more NFT creation systems **824**.

FIG. 13Q presents a flowchart representation of an example method. In particular, a method **6200-7** is presented for use in conjunction with any of the functions and features described herein. Step **6202-7** includes receiving, via the network interface and from a non-fungible token (NFT) game platform, display data associated with a customized collection display that presents a plurality of game NFTs, each of the plurality of game NFTs including a playable

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instance of an associated game that is accessible by a user of the client device. Step **6204-07** includes displaying the display data via a graphical user interface and a display device associated with the client device.

Step **6206-07** includes receiving, via the processor in response to user interactions with the graphical user interface, a selection of one the plurality of game NFTs for play of the associated game. Step **6208-7** includes sending, to the NFT game platform via the network interface, selection data indicating the selection of the one the plurality of game NFTs. Step **6210-07** includes facilitating, via the network interface and in conjunction with NFT game platform, play of the associated game.

The NFT game platform can be implemented via NFT collection platform **800**, NFT distribution platform **2800**, and/or other NFT platform that can include or operate in conjunction with NFT transaction authenticator **6022**, NFT source **6026**, secure real-time NFT metadata repository **6024**, NFT wallet system **820**, NFT marketplace **822**, NFT creation system **824**, and/or other tools, devices and systems described herein.

In various examples, facilitating the play of the associated game includes receiving, via the network interface, game video associated with the one of the plurality of game NFTs that is streamed from the NFT game platform and/or sending to the NFT game platform, via the network interface, game data associated with the game play of the game by the user that is generated by the client device.

FIG. 14A presents a block diagram representation of an example system. The system includes a NFT platform **6500**, such as NFT collection platform **800**, NFT distribution platform **2800** or other NFT platform that operates to facilitate the creation of new NFTs such as game NFTs or other NFTs associated with games or other media and/or the play of games associated with one or more game NFTs that have been purchased or acquired by a player, collector or other user. The system further includes an immutable distributed storage system **6515**, a client device **825**, a payment processor **6510** or other marketplace or marketplace tool, a blockchain system **6520**, such as NFT creation system **824** or other decentralized computer system and a wallet **6526**, such as NFT wallet system **820** or other wallet.

In operation, the immutable distributed storage system **6515**, such as an interplanetary file system (IPFS) or other decentralized computer system/network stores a game application **6518** (e.g. playable game data) such as application binary and data corresponding to an electronic game, such as a game **812** or other electronic game. This form of storage promotes an incorruptibility and perpetual storage of the game. When a client device generates a purchase request **6502** for a game application associated with an electronic game that is received by the NFT platform **6500**. The NFT platform **6500** responds by generating a payment request **6504** and receives a payment response **6514** indicating whether or not the payment request is verified. In various examples, in response to the payment response verifying the payment request, the platform system **6500** facilitates the minting of a game NFT **6522** associated with the electronic game. In addition, or in alternative, one or more game NFTs can be minted prior to sale. In this fashion, for example, a limited and/or numbered series of game NFTs can be minted and offered for sale to collectors. Furthermore, pre-minting a number of NFTs for sale can help expedite the sales because the minting of the game NFTs does not need to occur in real-time.

In various examples, minting of the game NFT **6522** includes generation of a private key associated with the

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game NFT that is stored with NFT metadata and/or other wallet data **6526** in the wallet **6526**. In other examples, an existing private key can be retrieved and used for the same purpose. The game NFT **6522** serves to authenticate the purchase of the game application **6518** by the user of the client device **825** and can be used to enable play of the electronic game via the client device **825**, for example, by verification of the NFT authorization via the application binary of the game application **6518**, before the game is fully launched for game play via the client device **825**. This improves the technology of game systems by providing a secure and incorruptible methodology for authenticating ownership of a game and/or authenticating the user to the game as the actual game owner.

While the immutable distributed storage system **6515** and the blockchain system **6520** are shown as separate elements, these two systems can be combined, for example, when the game NFT **6522** and the game application **6518** are stored via the same blockchain. Furthermore, while the game application **6518** is shown as being sent to client device **825**, in other examples, the immutable distributed storage system **6515** and the NFT platform **6500** can be in direct or network communication and the game application **6518** can be executed by the NFT platform **6500** for game play via the client device **825**. Other configurations, combinations and connections are likewise possible.

FIG. 14B presents a block diagram representation of an example system. In the example shown, an application execution device **6530** stores the game application **6518** for execution having a game NFT **6522** associated therewith. Similar devices shown in FIG. 65A are represented by similar reference numerals. The application execution device can be a client device **825**, an NFT platform **800** or other NFT platform or other device for executing the game application **6518** to facilitate play of the game.

In operation, the application execution device **6530** operates in response to application launch request **6532** for the game application **6518**. This application launch request **6532** can be received (internally) from a user interface associated with the application execution device **6530** (e.g. a client device or NFT platform) and/or received from an external device such as a client device, when for example, the application execution device **6530** corresponds to an NFT platform in communication with a client device. Once execution of the gaming application is initiated in response to the application launch request **6532**, launch metadata **6534** is generated, for example, to determine if the launch request can be validated. The launch metadata **6534** is sent to the wallet **6525** containing a private key associated with the game NFT **6522** associated with a user, such as a user of the application execution device and/or the application execution device **6530**. The application execution device **6530** receives from the wallet **6526**, a launch response **6535** that is generated based on the private key. The game application generates, based on the launch response **6536**, an ownership check **6536** that is sent to the blockchain system **6520** that stores the game NFT **6522**. An ownership validation response **6538** is received from the blockchain system **6520** that enables play of the game via the game application **6518** when the ownership validation response **6538** indicates ownership of the game NFT is verified and prevents play of the game via the game application **6518** when the ownership validation response **6538** indicates ownership of the game NFT is not verified. When not verified, the game application **6518** can, for example, shut itself down. In this fashion, the game application **6518** can self-validate on launch to verify

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ownership using game NFT data in a user's wallet which holds the private key to the game NFT **6522**.

In various examples, the launch metadata **6534** can include a current date and time, expiration information (if any) and a version number of the game application **6518**, a counter that indicates a number of times the application has been executed, and/or other data corresponding to the game NFT **6522**, the game application **6518** and/or the launch. The launch response **6535** can include a public key associated with the game NFT **6522**, a signed hash that is based on the private key, the launch metadata **6534**, and/or other data/metadata to be used to validate the NFT data from the wallet **6525** to the game NFT **6522**. In this fashion, the private key is not shared directly with the application binary—and instead is merely used by the wallet **6525** in a secure fashion.

The ownership check **6536** can include information from the launch response **6535**, additional owner/user information, such as user authentication data, associated with the user and/or owner of the game NFT **6522** and or other and/or other data/metadata to be used to validate the NFT data from the wallet **6525** to the game NFT **6522**. In various examples, the blockchain system **6520** can include an NFT transaction authenticator **6022**, a secure real-time NFT metadata repository **6024** and an NFT source **6026** that actually stores the game NFT **6522** on a blockchain or other decentralized computer system/network and user authentication/NFT validation can take place, for example as described in conjunction with FIGS. 12A-12D.

FIG. 14C presents a block diagram representation of an example system. This system operates similarly to the system of FIG. 65B, however, the application launch request **6532** is received directly from the wallet **6525**.

FIG. 14D presents a block diagram representation of an example system. These examples address a scenario where NFT's can be issued in-application (in-app NFTs). These in-app NFTs can correspond to collected or collectable items, achievements, purchased items, and or other NFTs that can be purchased or collected in-app. For example, first launch for a new user can issue an unsealing/open box NFT that is the digital equivalent of opening the box of a physical good and collecting a prize. These in-app NFT's can be tied to the user/owner of the game NFT **6522** and/or tied to a particular instance of the game application **6518**. This allows some NFTs to be transferred when ownership of the game application is transferred via sale or micro-loan of the game NFT **6518**. In other examples, the in-app NFTs can be independently sold and disassociated with the game NFT **6522**.

In various examples, the application execution device **6530** receives an in-app issuance request **6542** from a gaming application **6518** that was generated either automatically or in response to user interaction with the game. The application execution device **6530** responds by facilitating, based on in-app NFT data **6544**, the minting of an in-game NFT **6545** associated with the electronic game based on an authentication of the ownership of the game application **6518** via ownership of the associated game NFT **6522** by the user/owner that is playing the game.

In various examples, the system operates by: generating, via the application execution device **6530**, an ownership check **6536**; sending the ownership check **6536** to a blockchain system **6520** that stores a game NFT **6522** associated with the electronic game and the user; receiving, from the blockchain system **6520**, an ownership validation response **6538**; and facilitating, via the game application **6518**, a

minting of the in-app NFT **6545** when the ownership validation response **6538** indicates ownership of the game NFT **6518** is verified.

In this example shown, an ownership check **6536** is initiated not only on launch of the game application **6518**, but also when an in-app issuance request **6542** is initiated to generate a in-app NFT. The in-app NFT **6546** is only minted when a favorable ownership validation response **6538** is received indicating validated ownership and/or user authentication. Furthermore, a favorable payment response **6514** may be required when the in-app issuance request **6542** has an associated a cost. Otherwise, if payment is denied or ownership validation fails, the requested in-app NFT **6542** is not minted.

FIG. **14E** presents a block diagram representation of an example system. In various examples, game application **6518** starts with a root version that can be designed as **6518-1**. As bugs are fixed and new updates are released, new versions of the game application (**6518-2**, **8518-3** . . .) can be created

Consider the following example, where the immutable storage system **6515** includes an IPFS file system. The game application **6518-1** can be uploaded and an IPFS hash of the content (a "CID," or content address) is generated that is used to make an IPFS URL (ipfs://<cid>). This IPFS URL can be included in the NFT data of the game NFT to refer to off-chain data (e.g., the metadata field in game NFT, the field in your metadata corresponding to the content—in this case the game application) as a pointer to the content itself. When there is a request to retrieve the game application **6518-1**, the NFT data can be accessed from the decentralized IPFS network from any peer that has the content. CIDs reference immutable content stored via a blockchain so that the game application that is accessed is the content referenced in the game NFT. The game application **6518-1** can be fetched directly, via a public IPFS gateway, or by using an IPFS Desktop application or via an IPFS command line.

When a new version of the game application, designated as **6518-2** is released, a new blockchain can be created with each block in the chain superseding the corresponding blocks from the previous chain. A game NFT for the prior version of the game application **6518-1** can be used to check the blockchain to validate the newer version of the game application **6518-2**. For example, when a blockchain for a new version of the game application is created, its metadata can include links/references to one (or all) prior versions that are stored in the immutable storage system **6515**. In this fashion, an ownership check associated with a newer version can determine that a prior version has been superseded when the newer version includes references to that prior version. While a direct ownership check based on older NFT may initially fail because the corresponding older version has been superseded, the immutable storage system **6515** can then attempt ownership checks associated with one or more prior versions until either ownership is validated for one of the prior versions or there are no remaining prior versions to check.

Considering the example shown, the immutable storage system **6515** stores versions **6518-1** and **6518-2** of the game application via a blockchain or other decentralized computer network. When the game application **6518-2** is launched by an application execution device **6530**, an ownership check **6536-1** is generated in accordance with a game NFT generated based on a prior game version of the game application **6518-1**. The immutable distributed storage system **6515** operates by determining when the ownership check **6536-1** is validated to the prior game version **6518-1** and further-

more determining when the prior game version **6518-1** has been superseded by the current game version **6518-2**. The ownership validation response **6538-1** is generated to indicate ownership validation only when both the ownership check **6536-1** is validated to the prior game version **6518-1** and when the prior game version **6518-2** has been superseded by the current game version. These two steps can be performed via the one process described above or via separate processes.

FIG. **14F** presents a block/flow diagram representation of an example system. In various examples the immutable distributed storage system **6515** can also determine when updates of a game application are available and facilitate the download of such updates.

Consider the following example. When the game application **6518-1** is launched by an application execution device **6530**, an ownership check **6536-2** is generated in accordance with a game NFT generated based on the (original) game version of the game application **6518-1**. The immutable distributed storage system **6515** operates by determining when the ownership check **6536-2** is validated to the game version **6518-1** and furthermore determining when the game version **6518-1** has been superseded by the current game version **6518-2**. In this case, the ownership check can be validated on the first instance since the launch metadata/response data corresponds to the version of the game application that is being launched. The immutable distributed storage system **6515** can, for example, also search the metadata, a version list associated with the game application or other data structure, to determine and identify another version of the game that supersedes the version being launched. If both conditions are met, the ownership validation response **6538-2** is generated to indicate that the updated game application **6518-2** is available. Furthermore, the immutable distributed storage system **6515** can facilitate access, by the application execution device **6530**, to the updated gaming application **6518-2**. This facilitation can be performed by coordinating with the application execution device **6530** to download the game application **6518-2** as shown. The updated gaming application **6518-2** can then be launched and validated as previously described in conjunction with FIG. **14E**.

FIG. **14G** presents a flowchart representation of an example method. In particular, a method **6550-1** is presented for use in conjunction with any of the functions and features described herein. Step **6552-1** includes receiving, at the NFT platform and from a client device, a purchase request for a game application associated with an electronic game. Step **6554-1** includes generating a payment request in response to the purchase request. Step **6556-1** includes receiving a payment response verifying the payment request. Step **6558-1** includes facilitating, in response to the payment response verifying the payment request, via the processor, a minting of a game NFT associated with the electronic game authenticating the purchase of the game application.

In various examples, the application data associated with the game application is stored via an immutable distributed storage system. The game NFT can operate to enable play of the electronic game via the client device. Minting of the game NFT can include generation or retrieval of a private key to be associated with the game NFT and the method can further include facilitating storage, via a wallet associated with a user of the client device, of the private key associated with the game NFT.

FIG. **14H** presents a flowchart representation of an example method. In particular, a method **6550-2** is presented for use in conjunction with any of the functions and features

described herein. Step **6552-2** includes receiving, from a client device, an application launch request for a game application having a game NFT associated therewith. Step **6554-2** includes generating, via the game application and in response to the application launch request, launch metadata. Step **6556-2** includes sending the launch metadata to a wallet associated with a user of the client device, the wallet containing a private key associated with the game NFT.

Step **6558-2** includes receiving, from the wallet, a launch response that is generated based on the private key. Step **6560-2** includes generating, via the game application and based on the launch response, an ownership check. Step **6562-2** includes sending the ownership check to a blockchain system that stores the game NFT. Step **6564-2** includes receiving, from the blockchain system, an ownership validation response. Step **6566-2** includes enabling, via the game application, play of the game when the ownership validation response indicates ownership of the game NFT is verified. Step **6568-2** includes preventing, via the game application, play of the game when the ownership validation response indicates the ownership of the game NFT is not verified.

In various examples, the application data associated with the game application is stored via an immutable distributed storage system. The game NFT can enable play of the electronic game via the client device. The game NFT can include generation of a private key associated with the game NFT and the method can further include facilitating storage, via a wallet associated with a user of the client device, of the private key associated with the game NFT.

FIG. **14I** presents a flowchart representation of an example method. In particular, a method **6550-3** is presented for use in conjunction with any of the functions and features described herein. Step **6552-3** includes receiving, at the client device, an application launch request for a game application having a game NFT associated therewith. Step **6554-3** includes generating, via the game application and in response to the application launch request, launch metadata. Step **6556-3** includes sending the launch metadata to a wallet associated with a user of the client device, the wallet containing a private key associated with the game NFT. Step **6558-3** includes receiving, from the wallet, a launch response that is generated based on the private key.

Step **6560-3** includes generating, via the game application and based on the launch response, an ownership check. Step **6562-3** includes sending the ownership check to a blockchain system that stores the game NFT. Step **6564-3** includes receiving, from the blockchain system, an ownership validation response. Step **6566-3** includes enabling, via the game application, play of the game when the ownership validation response indicates ownership of the game NFT is verified. Step **6568-3** includes preventing, via the game application, play of the game when the ownership validation response indicates the ownership of the game NFT is not verified.

FIG. **14J** presents a flowchart representation of an example method. In particular, a method **6550-4** is presented for use in conjunction with any of the functions and features described herein. Step **6552-4** includes receiving, at the NFT wallet associated with a user of a client device, an application launch request for a game application having a game NFT associated therewith, wherein the NFT wallet contains a private key associated with the game NFT. Step **6554-4** includes generating, via the NFT wallet and in response to the application launch request, a launch response that is generated based on the private key, wherein the game application generates, based on the launch response, an

ownership check that is sent to a blockchain system that stores the game NFT, and wherein the game application receives, from the blockchain system, an ownership validation response and enables play of the game when the ownership validation response indicates ownership of the game NFT is verified and prevents play of the game when the ownership validation response indicates the ownership of the game NFT is not verified.

FIG. **14K** presents a flowchart representation of an example method. In particular, a method **6550-5** is presented for use in conjunction with any of the functions and features described herein. Step **6552-5** includes receiving, at the NFT platform and from a game application associated with an electronic game played by a user, an in-app issuance request for an in-app NFT. Step **6554-5** includes generating, via the NFT platform, an ownership check. Step **6556-5** includes sending the ownership check to a blockchain system that stores a game NFT associated with the electronic game and the user. Step **6558-5** includes receiving, from the blockchain system, an ownership validation response. Step **6560-5** includes facilitating, via the game application, a minting of the in-app NFT when the ownership validation response indicates ownership of the game NFT is verified.

In various examples, the method further includes: generating a payment request in response to the purchase request; and receiving a payment response verifying the payment request; wherein the ownership check is generated in response to the payment response. The in-app NFT can have in-app NFT metadata associated therewith, wherein the in-app NFT metadata includes one or more of: an achievement data associated with the electronic game and the user, an application identifier associated with the electronic game, a user identifier associated with the user or an instance identifier indicating an instance of the electronic game associated with the user.

FIG. **14L** presents a flowchart representation of an example method. In particular, a method **6550-6** is presented for use in conjunction with any of the functions and features described herein. Step **6552-6** includes receiving, at the blockchain system and from a game application associated with an electronic game, an ownership check generated in accordance with a game non-fungible token (NFT), wherein the game application is associated with a current game version of the game application and the game NFT was generated based on a prior game version of the game application. Step **6554-6** includes determining, via the blockchain system, when the ownership check is validated to the prior game version. Step **6556-6** includes determining, via the blockchain system, when the prior game version has been superseded by the current game version. Step **6558-6** includes generating an ownership validation response indicating ownership validation when the ownership check is validated to the prior game version and when the prior game version has been superseded by the current game version.

FIG. **14M** presents a flowchart representation of an example method. In particular, a method **6550-7** is presented for use in conjunction with any of the functions and features described herein. Step **6552-7** includes receiving, from a client device, an application launch request for a game application having a game NFT associated therewith. Step **6554-7** includes generating, via the game application and in response to the application launch request, launch metadata. Step **6556-7** includes sending the launch metadata to a wallet associated with a user of the client device, the wallet containing a private key associated with the game NFT. Step **6558-7** includes receiving, from the wallet, a launch response that is generated based on the private key. Step

6560-7 includes generating, via the game application and based on the launch response, an ownership check.

Step **6562-7** includes sending the ownership check to a blockchain system that stores the game NFT. Step **6564-7** includes receiving, from the blockchain system, an ownership validation response that indicates an updated game application is available. Step **6566-7** includes facilitating access, by the client device, to the updated gaming application.

FIG. 14N presents a flowchart representation of an example method. In particular, a method **6550-8** is presented for use in conjunction with any of the functions and features described herein. Step **6552-8** includes receiving an application launch request for a game application having a game non-fungible token (NFT) associated therewith. Step **6554-8** includes generating, via the game application and in response to the application launch request, launch metadata. Step **6556-8** includes sending the launch metadata to an NFT wallet, the NFT wallet containing a private key associated with the game NFT. Step **6568-8** includes receiving, from the NFT wallet, a launch response that is generated based on the private key.

Step **6560-8** includes generating, via the game application and based on the launch response, an ownership check. Step **6562-8** includes sending the ownership check to a blockchain system that stores the game NFT. Step **6564-8** includes receiving, from the blockchain system, an ownership validation response. Step **6566-8** includes enabling, via the game application, play of the game when the ownership validation response indicates ownership of the game NFT is verified.

In various examples, the method includes preventing, via the game application, play of the game when the ownership validation response indicates the ownership of the game NFT is not verified. The application data associated with the game application can be stored via an immutable distributed storage system and be downloaded to a client device or NFT platform for execution of the game application. The application launch request can be generated by a client device, and wherein the game application is executed via the client device or via an NFT platform in communication with the client device.

In various examples, the ownership validation response indicates when an updated game application is available and the method further includes: facilitating access to the updated gaming application when the updated game application is available. The application launch request can be generated by a client device, wherein the gaming application is executed by the client device, and wherein facilitating access to the updated gaming application includes downloading the updated gaming application to the client device for execution via the client device. The application launch request can be generated by a client device, wherein the gaming application is executed by an NFT platform in communication with the client device, and wherein facilitating access to the updated gaming application includes downloading the updated gaming application to the NFT platform for execution via the NFT platform.

While the forgoing systems and methods of FIGS. 14A-14N have focused on game NFTs, these systems and methods could likewise be applied to NFTs associated with other applications, other media and/or other types of NFTs. Furthermore, the applications and techniques associated with NFTs previously described in this application, and in copending U.S. application Ser. No. 17/656,084 entitled, FACILITATING PLAY OF GAME NETS VIA A CLIENT DEVICE, can likewise be applied to game NFTs as well.

As used herein, the terms “game piece” and “game character” can be used interchangeably to refer to a game character, an ancillary article or other article or object that is used in a game.

As used herein “blockchain” and “blockchain node” refer to traditional blockchain technology. However, other decentralized computer network technologies that, for example, maintain a secure and decentralized record of transactions and/or otherwise protect the security of digital information can likewise be employed. In various examples, a blockchain can operate to collect information together in groups, such as blocks, that hold sets of information. These blocks have certain storage capacities and, when filled, are closed and linked to the previously filled blocks, forming a chain of data. New information that follows that freshly added block is compiled into newly formed blocks that, once filled, can also be added to the chain.

As used herein the term “tool” corresponds to a utility, application and/or other software routine that performs one or more specific functions in conjunction with a computer.

It is noted that terminologies as may be used herein such as bit stream, stream, signal sequence, etc. (or their equivalents) have been used interchangeably to describe digital information whose content corresponds to any of a number of desired types (e.g., data, video, speech, text, graphics, audio, etc. any of which may generally be referred to as “data”).

As may be used herein, the terms “substantially” and “approximately” provides an industry-accepted tolerance for its corresponding term and/or relativity between items. For some industries, an industry-accepted tolerance is less than one percent and, for other industries, the industry-accepted tolerance is 10 percent or more. Other examples of industry-accepted tolerance range from less than one percent to fifty percent. Industry-accepted tolerances correspond to, but are not limited to, component values, integrated circuit process variations, temperature variations, rise and fall times, thermal noise, dimensions, signaling errors, dropped packets, temperatures, pressures, material compositions, and/or performance metrics. Within an industry, tolerance variances of accepted tolerances may be more or less than a percentage level (e.g., dimension tolerance of less than $\pm 1\%$). Some relativity between items may range from a difference of less than a percentage level to a few percent. Other relativity between items may range from a difference of a few percent to magnitude of differences.

As may also be used herein, the term(s) “configured to”, “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as “coupled to”.

As may even further be used herein, the term “configured to”, “operable to”, “coupled to”, or “operably coupled to” indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term “associated

with”, includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

As may be used herein, the term “compares favorably”, indicates that a comparison between two or more items, signals, etc., indicates an advantageous relationship that would be evident to one skilled in the art in light of the present disclosure, and based, for example, on the nature of the signals/items that are being compared. As may be used herein, the term “compares unfavorably”, indicates that a comparison between two or more items, signals, etc., fails to provide such an advantageous relationship and/or that provides a disadvantageous relationship. Such an item/signal can correspond to one or more numeric values, one or more measurements, one or more counts and/or proportions, one or more types of data, and/or other information with attributes that can be compared to a threshold, to each other and/or to attributes of other information to determine whether a favorable or unfavorable comparison exists. Examples of such an advantageous relationship can include: one item/signal being greater than (or greater than or equal to) a threshold value, one item/signal being less than (or less than or equal to) a threshold value, one item/signal being greater than (or greater than or equal to) another item/signal, one item/signal being less than (or less than or equal to) another item/signal, one item/signal matching another item/signal, one item/signal substantially matching another item/signal within a predefined or industry accepted tolerance such as 1%, 5%, 10% or some other margin, etc. Furthermore, one skilled in the art will recognize that such a comparison between two items/signals can be performed in different ways. For example, when the advantageous relationship is that signal 1 has a greater magnitude than signal 2, a favorable comparison may be achieved when the magnitude of signal 1 is greater than that of signal 2 or when the magnitude of signal 2 is less than that of signal 1. Similarly, one skilled in the art will recognize that the comparison of the inverse or opposite of items/signals and/or other forms of mathematical or logical equivalence can likewise be used in an equivalent fashion. For example, the comparison to determine if a signal $X > 5$ is equivalent to determining if $-X < -5$, and the comparison to determine if signal A matches signal B can likewise be performed by determining $-A$ matches $-B$ or $\text{not}(A)$ matches $\text{not}(B)$. As may be discussed herein, the determination that a particular relationship is present (either favorable or unfavorable) can be utilized to automatically trigger a particular action. Unless expressly stated to the contrary, the absence of that particular condition may be assumed to imply that the particular action will not automatically be triggered. In other examples, the determination that a particular relationship is present (either favorable or unfavorable) can be utilized as a basis or consideration to determine whether to perform one or more actions. Note that such a basis or consideration can be considered alone or in combination with one or more other bases or considerations to determine whether to perform the one or more actions. In one example where multiple bases or considerations are used to determine whether to perform one or more actions, the respective bases or considerations are given equal weight in such determination. In another example where multiple bases or considerations are used to determine whether to perform one or more actions, the respective bases or considerations are given unequal weight in such determination.

As may be used herein, one or more claims may include, in a specific form of this generic form, the phrase “at least one of a, b, and c” or of this generic form “at least one of a, b, or c”, with more or less elements than “a”, “b”, and “c”.

In either phrasing, the phrases are to be interpreted identically. In particular, “at least one of a, b, and c” is equivalent to “at least one of a, b, or c” and shall mean a, b, and/or c. As an example, it means: “a” only, “b” only, “c” only, “a” and “b”, “a” and “c”, “b” and “c”, and/or “a”, “b”, and “c”.

As may also be used herein, the terms “processing module”, “processing circuit”, “processor”, “processing circuitry”, and/or “processing unit” may be a single processing device or a plurality of processing devices. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. The processing module, module, processing circuit, processing circuitry, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, processing circuitry, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, processing circuitry, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network). Further note that if the processing module, module, processing circuit, processing circuitry and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, processing circuitry and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

One or more examples have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims. Further, the boundaries of these functional building blocks have been arbitrarily defined for convenience of description. Alternate boundaries could be defined as long as the certain significant functions are appropriately performed. Similarly, flow diagram blocks may also have been arbitrarily defined herein to illustrate certain significant functionality.

To the extent used, the flow diagram block boundaries and sequence could have been defined otherwise and still perform the certain significant functionality. Such alternate

definitions of both functional building blocks and flow diagram blocks and sequences are thus within the scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules and components herein, can be implemented as illustrated or by discrete components, application specific integrated circuits, processors executing appropriate software and the like or any combination thereof.

In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with one or more other routines. In addition, a flow diagram may include an “end” and/or “continue” indication. The “end” and/or “continue” indications reflect that the steps presented can end as described and shown or optionally be incorporated in or otherwise used in conjunction with one or more other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

The one or more examples are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical example of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the examples discussed herein. Further, from figure to figure, the examples may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.

Unless specifically stated to the contrary, signals to, from, and/or between elements in a figure of any of the figures presented herein may be analog or digital, continuous time or discrete time, and single-ended or differential. For instance, if a signal path is shown as a single-ended path, it also represents a differential signal path. Similarly, if a signal path is shown as a differential path, it also represents a single-ended signal path. While one or more particular architectures are described herein, other architectures can likewise be implemented that use one or more data buses not expressly shown, direct connectivity between elements, and/or indirect coupling between other elements as recognized by one of average skill in the art.

The term “module” is used in the description of one or more of the examples. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more sub-modules, each of which may be one or more modules.

As may further be used herein, a computer readable memory includes one or more memory elements. A memory element may be a separate memory device, multiple memory devices, or a set of memory locations within a memory device. Such a memory device may be a read-only memory, random access memory, volatile memory, non-

volatile memory, static memory, dynamic memory, flash memory, cache memory, a quantum register or other quantum memory and/or any other device that stores data in a non-transitory manner. Furthermore, the memory device may be in a form of a solid-state memory, a hard drive memory or other disk storage, cloud memory, thumb drive, server memory, computing device memory, and/or other non-transitory medium for storing data. The storage of data includes temporary storage (i.e., data is lost when power is removed from the memory element) and/or persistent storage (i.e., data is retained when power is removed from the memory element). As used herein, a transitory medium shall mean one or more of: (a) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for temporary storage or persistent storage; (b) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for temporary storage or persistent storage; (c) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for processing the data by the other computing device; and (d) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for processing the data by the other element of the computing device. As may be used herein, a non-transitory computer readable memory is substantially equivalent to a computer readable memory. A non-transitory computer readable memory can also be referred to as a non-transitory computer readable storage medium.

One or more functions associated with the methods and/or processes described herein can be implemented via a processing module that operates via the non-human “artificial” intelligence (AI) of a machine. Examples of such AI include machines that operate via anomaly detection techniques, decision trees, association rules, expert systems and other knowledge-based systems, computer vision models, artificial neural networks, convolutional neural networks, support vector machines (SVMs), Bayesian networks, genetic algorithms, feature learning, sparse dictionary learning, preference learning, deep learning and other machine learning techniques that are trained using training data via unsupervised, semi-supervised, supervised and/or reinforcement learning, and/or other AI. The human mind is not equipped to perform such AI techniques, not only due to the complexity of these techniques, but also due to the fact that artificial intelligence, by its very definition—requires “artificial” intelligence—i.e. machine/non-human intelligence.

One or more functions associated with the methods and/or processes described herein involve NFTs that are generated (“minted”) and secured via blockchain or other decentralized computer network technology. The distributed nature of these technologies over different nodes, the contemporaneous nature of geographically distinct calculations, coupled with the extreme computational complexity of the required calculations means that these decentralized computer network technologies cannot practically be performed by the human mind.

One or more functions associated with the methods and/or processes described herein can be implemented as a large-scale system that is operable to receive, transmit and/or process data on a large-scale. As used herein, a large-scale refers to a large number of data, such as one or more kilobytes, megabytes, gigabytes, terabytes or more of data that are received, transmitted and/or processed. Such receiv-

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ing, transmitting and/or processing of data cannot practically be performed by the human mind on a large-scale within a reasonable period of time, such as within a second, a millisecond, microsecond, a real-time basis or other high speed required by the machines that generate the data, receive the data, convey the data, store the data and/or use the data.

One or more functions associated with the methods and/or processes described herein can require data to be manipulated in different ways within overlapping time spans. The human mind is not equipped to perform such different data manipulations independently, contemporaneously, in parallel, and/or on a coordinated basis within a reasonable period of time, such as within a second, a millisecond, microsecond, a real-time basis or other high speed required by the machines that generate the data, receive the data, convey the data, store the data and/or use the data.

One or more functions associated with the methods and/or processes described herein can be implemented in a system that is operable to electronically receive digital data via a wired or wireless communication network and/or to electronically transmit digital data via a wired or wireless communication network. Such receiving and transmitting cannot practically be performed by the human mind because the human mind is not equipped to electronically transmit or receive digital data, let alone to transmit and receive digital data via a wired or wireless communication network.

One or more functions associated with the methods and/or processes described herein can be implemented in a system that is operable to electronically store digital data in a memory device. Such storage cannot practically be performed by the human mind because the human mind is not equipped to electronically store digital data.

One or more functions associated with the methods and/or processes described herein may operate to cause an action by a processing module directly in response to a triggering event—without any intervening human interaction between the triggering event and the action. Any such actions may be identified as being performed “automatically”, “automatically based on” and/or “automatically in response to” such a triggering event. Furthermore, any such actions identified in such a fashion specifically preclude the operation of human activity with respect to these actions—even if the triggering event itself may be causally connected to a human activity of some kind.

While particular combinations of various functions and features of the one or more examples have been expressly described herein, other combinations of these features and functions are likewise possible. The present disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

What is claimed is:

1. A method for use with a non-fungible token (NFT) platform that includes a processor and a memory, the method comprising:
 - receiving, at the NFT platform and from a game application associated with an electronic game played by a user, an in-app issuance request for an in-app NFT;
 - generating, via the NFT platform, an ownership check;
 - sending the ownership check to a blockchain system that stores a game NFT associated with the electronic game and the user;
 - receiving, from the blockchain system, an ownership validation response that indicates whether user's ownership of the game NFT is verified; and

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facilitating, via the game application, a minting of the in-app NFT when the ownership validation response indicates the user's ownership of the game NFT is verified.

2. The method of claim 1, further comprising:
 - generating a payment request in response to the in-app issuance request; and
 - receiving a payment response verifying the in-app issuance request;

wherein the ownership check is generated in response to the payment response.

3. The method of claim 1, wherein the in-app NFT has in-app NFT metadata associated therewith, wherein the in-app NFT metadata includes achievement data associated with the electronic game and the user.

4. The method of claim 1, wherein the in-app NFT has in-app NFT metadata associated therewith, wherein the in-app NFT metadata includes an application identifier associated with the electronic game.

5. The method of claim 1, wherein the in-app NFT has in-app NFT metadata associated therewith, wherein the in-app NFT metadata includes a user identifier associated with the user.

6. The method of claim 1, wherein the in-app NFT has in-app NFT metadata associated therewith, wherein the in-app NFT metadata includes an instance identifier indicating an instance of the electronic game associated with the user.

7. The method of claim 1, further comprising:
 - preventing, via the game application, the minting of the in-app NFT when the ownership validation response indicates the user's ownership of the game NFT is not verified.

8. The method of claim 1, wherein the game application is configured to determine if the user's play of the game is permitted based on ownership of the game NFT.

9. The method of claim 1, further comprising:
 - generating a payment request in response to the in-app issuance request; and

preventing, via the game application, the minting of the in-app NFT when the payment request is denied.

10. The method of claim 1, wherein the in-app issuance request is generated automatically by the gaming application.

11. A non-fungible token (NFT) platform comprises:
 - a network interface configured to communicate via a network;

at least one processor;

a non-transitory machine-readable storage medium that stores operational instructions that, when executed by the processor, cause the at least one processor to perform operations that include:

receiving, at the NFT platform and from a game application associated with an electronic game played by a user of a client device, an in-app issuance request for an in-app NFT;

generating, via the NFT platform, an ownership check; sending the ownership check to a blockchain system that stores a game NFT associated with the electronic game and the user;

receiving, from the blockchain system, an ownership validation response that indicates whether user's ownership of the game NFT is verified; and

facilitating, via the game application, a minting of the in-app NFT when the ownership validation response indicates the user's ownership of the game NFT is verified.

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12. The NFT platform of claim 11, wherein the operations further include:

generating a payment request in response to the in-app issuance request; and
receiving a payment response verifying the in-app issuance request;

wherein the ownership check is generated in response to the payment response.

13. The NFT platform of claim 11, wherein the in-app NFT has in-app NFT metadata associated therewith, wherein the in-app NFT metadata includes achievement data associated with the electronic game and the user.

14. The NFT platform of claim 11, wherein the in-app NFT has in-app NFT metadata associated therewith, wherein the in-app NFT metadata an application identifier associated with the electronic game.

15. The NFT platform of claim 11, wherein the in-app NFT has in-app NFT metadata associated therewith, wherein the in-app NFT metadata includes a user identifier associated with the user.

16. The NFT platform of claim 11, wherein the in-app NFT has in-app NFT metadata associated therewith, wherein

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the in-app NFT metadata includes an instance identifier indicating an instance of the electronic game associated with the user.

17. The NFT platform of claim 11, wherein the operations further include:

preventing, via the game application, the minting of the in-app NFT when the ownership validation response indicates the user's ownership of the game NFT is not verified.

18. The NFT platform of claim 11, wherein the game application is configured to determine if the user's play of the game is permitted based on ownership of the game NFT.

19. The NFT platform of claim 11, wherein the operations further include:

generating a payment request in response to the in-app issuance request; and

preventing, via the game application, the minting of the in-app NFT when the payment request is denied.

20. The NFT platform of claim 11, wherein the in-app issuance request is generated automatically by the gaming application.

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